

Informal Notes on Automatic Controls

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Chapter 1

Subject Matter Overview

A **control system** is an interconnection of devices that alter the natural behavior of a **plant**. A plant is itself a device we aim to manipulate. Some basic examples should help to clarify the definitions

1. Rocket During Launch

Plant: Rocket; which consists of mechanical, electrical, electronic, and chemical systems

Control system: sensors that measure the rocket's position and attitude; effectors that manipulate the rocket engine nozzle

- (a) Natural behavior (without control): the forces of thrust and gravity to align driving the rocket into the ground
- (b) Controlled behavior: align the thrust of the rocket in a direction to propel the rocket into space

2. Home Oven

Plant: Oven, which consists of electronic and chemical (the fuel) systems

Control system: thermostat, flow control effector on the fuel line

- (a) Natural behavior: heat the volume to maximum temperate possible
- (b) Controlled behavior: heat to a set temperature and maintain that temperature for cooking

3. Self-Balancing Bicycle

Plant: Bicycle; which consists of mechanical systems

Control system: sensors that monitor the tip angle of the of plant; effectors that control the steering angle of the front tire

- (a) Natural behavior: the bicycle falls over
- (b) Controlled behavior: balance the vehicle as it is propelled forward

For the three examples above, and for most systems, accurate mathematical models that represent the physical characteristics of the system and the controls devices are necessary. Note that for the rocket and the bicycle without control the natural response will lead to *detrimental results*; without proper control, the outcome results in damage to the plant and likely the control system. These systems are considered **unstable** because the uncontrolled result damages the plant and produces an unbounded result (mathematical definition) – much more on this later.

The study of control systems may be separated into three areas: **model development**, **analysis**, and **controller design**. In model development, we find new ways to mathematically model systems we want to understand and manipulate. In analysis, we investigate a system with proven techniques. Finally, for controller design we design and implement a controller to manipulate a system to achieve our controls objectives. The time to go from “idea” to final design may be weeks or, for complex problems, years. This content touches on all three aspects, but focuses on analysis and controller design.

Generally, differential equations are transformed into Laplace domain equivalent expressions where they are manipulated into standard forms such that the dynamic response can be determined without actually needing to perform time domain simulation. In ECE 4375 the models are linear time-invariant which allows for highly accurate model development. For more challenging systems, such as nonlinear and time-varying devices, advanced techniques are employed — they’re just beyond the scope of a senior level course. Modern approaches use software simulation tools, such as MATLAB, to directly observe the time domain behavior of the system.

Conversion from the time domain differential equation, such as

$$\ddot{y}(t) + a_1\dot{y}(t) + a_0y(t) - \dot{u}(t) - b_0u(t) = 0, \quad (1.1)$$

to the Laplace domain equivalent equation is typically performed with a lookup table.

With the case of (1.1), the Laplace equivalent equation is

$$s^2Y(s) - sy(0) - \dot{y}(0) + a_1sY(s) - a_1y(0) + a_0Y(s) - sU(s) + u(0) - b_0U(s) = 0 \quad (1.2)$$

where the $y(0)$ and $u(0)$ terms are the initial conditions. For many studies – including this course – the initial conditions are assumed zero. Therefore, (1.2) simplifies to

$$s^2Y(s) + a_1sY(s) + a_0Y(s) - sU(s) - b_0U(s) = 0 \quad (1.3)$$

Now, in the Laplace form, the system is manipulated into a transfer function¹. A transfer function is an equation that is the ratio of output to input. In this class, the system output will always be $y(t)$ in the time domain and $Y(s)$ in the Laplace domain. The input will be either $u(t)$, $U(s)$ or $r(t)$, $R(s)$. In the case of (1.3), the input is $U(s)$, the output is $Y(s)$. To re-write the equation into a transfer function, first move the input expressions to the right

$$s^2Y(s) + a_1sY(s) + a_0Y(s) = sU(s) + b_0U(s) \quad (1.4)$$

factor the input and output from the right and left of the equation

$$(s^2 + a_1s + a_0)Y(s) = (s + b_0)U(s)$$

¹When the end goal is a transfer function all initial conditions are zero, $y(0) = u(0) = 0$.

Next, divide both sides of the equation by the left-hand expression

$$Y(s) = \frac{(s + b_0)U(s)}{(s^2 + a_1s + a_0)}$$

Finally, divide both sides by the input signal to create the transfer function, which is a output over input ratio

$$\frac{Y(s)}{U(s)} = \frac{s + b_0}{s^2 + a_1s + a_0} \quad (1.5)$$

Now, (1.5) is the Laplace Domain transfer function of (1.1). They are equivalent equations of the same mathematical system. We may transform from the time domain to the Laplace domain and back to the time domain without loss of information, knowledge, or generality. This statement is true for all time domain, Laplace domain representations.

In industry, many devices have a datasheet with a transfer function. When this is the case, the mathematical models may be cascaded together to form an overall model of the system, which then may be analyzed to understand the performance of the system before ever physically constructing the plant. For example, a complex robot consists of electronic devices (e.g. amplifiers and sensors), mechanical members, and actuators (e.g. DC-servo motors). The systems are arranged such that the robotic device performs as designed.

In general, most systems, regardless of complexity, may be mathematically modeled. Of course, more complex the system (e.g. nonlinear and/or time-varying) the more difficult it is to establish a mathematical model.

In this course, the focus is on linear time-invariant systems with little or no noise which allows for a near perfect mathematical model. There are some linear devices that are common and reoccur in most non-trivial systems.

1.1 Basic Models

Let's take a look at a couple of math models:

1. Amplifier

Input, U: low-voltage, low-power

Output, Y: higher-voltage, higher-power

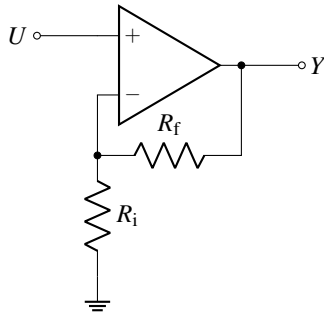
Device: operational amplifier (op amp)

The transfer function

$$\frac{Y(s)}{U(s)} = K_{oa} \quad (1.6)$$

where $-\infty < K_{oa} < \infty$.

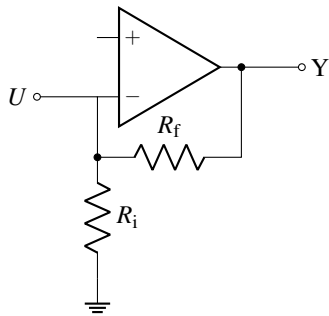
For example, if we use an op amp circuit



then

$$K_{oa} = 1 + \frac{R_f}{R_i} = \frac{R_i + R_f}{R_i}$$

If we apply the input to the inverting terminal, such as



then

$$K_{oa} = -\frac{R_f}{R_i}$$

2. Summation Junction

Input, U_i : Multiple signal sources

Output, Y: summation of signal sources

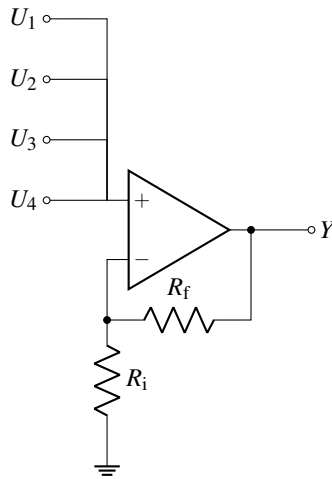
Device: operational amplifier (op amp)

Mathematical representation:

$$Y(s) = \sum_{i=1}^n U_i(s) \quad (1.7)$$

Note: the summation junction is not a transfer function

The summation junction circuit with four inputs



here

$$\begin{aligned}
 Y &= \left(1 + \frac{R_f}{R_i}\right)(U_1 + U_2 + U_3 + U_4) \\
 &= \left(\frac{R_i + R_f}{R_i}\right)(U_1 + U_2 + U_3 + U_4)
 \end{aligned}$$

1.2 Time Domain to Laplace Domain Transformations

Transfer functions are the primary modeling tool we use for this course – it's also a incredibly common in industry. When possible, we chose to work with transfer functions.

Transfer functions require a Laplace domain representation. When given a time domain equation, we use a Look Up Table, LUT, to transform between the time and Laplace domains. If the provided table is insufficient to complete a transformation, find another table. There is no expectation to use the mathematical equations to complete a transformation. However, no set of notes is complete without the equations

$$F(s) = \int_0^{\infty} f(t)e^{-st} dt$$

and

$$f(t) = \frac{1}{2\pi j} \int_{c-j\infty}^{c+j\infty} F(s)e^{st} ds$$

for $t > 0$ and where c is larger than the real parts of all singular points of $F(s)$.

1.3 Selected Functions

Some functions are more common in the field than others. Below are the most common input signals and the exponential function.

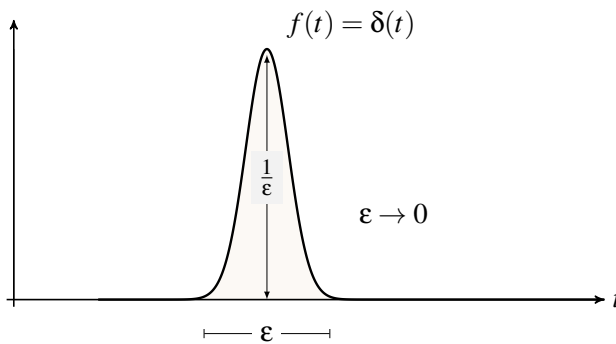
	$f(t)$	$F(s)$	Notes
1	$k \cdot f(t)$	$kF(s)$	k is a constant
2	$f_1(t) + f_2(t)$	$F_1(s) + F_2(s)$	
3	$\delta(t)$	1	impulse
4	$1(t)$	$\frac{1}{s}$	step
5	t	$\frac{1}{s^2}$	
6	$\frac{t^2}{2}$	$\frac{1}{s^3}$	
7	t^n	$\frac{n!}{s^{n+1}}$	$n = 1, 2, 3, \dots$
8	$\frac{t^{n-1}}{(n-1)!}$	$\frac{1}{s^n}$	$n = 1, 2, 3, \dots$
9	e^{-at}	$\frac{1}{s+a}$	
10	$\sin(bt)$	$\frac{b}{s^2+b^2}$	
11	$\cos(bt)$	$\frac{s}{s^2+b^2}$	
12	$e^{-at} \sin(bt)$	$\frac{b}{(s+a)^2+b^2}$	
13	$e^{-at} \cos(bt)$	$\frac{s+a}{(s+a)^2+b^2}$	
14	$t^n e^{-at}$	$\frac{n!}{(s+a)^{n+1}}$	$n = 1, 2, 3, \dots$
15	$\dot{f}(t)$	$sF(s) - f(0)$	$f(0)$ is initial condition
16	$\frac{d^n f(t)}{dt^n}$	$s^n F(s) + \text{initial conditions}$	initial conditions are zero in ECE 4375
17	$\int f(t) dt$	$\frac{F(s)}{s}$	
18	$\int \dots \int f(t) dt$	$\frac{F(s)}{s^n}$	n integrations
19	$f_1(t) * f_2(t)$	$F_1(s) \cdot F_2(s)$	convolution
20	$f_1(t) \cdot f_2(t)$	$F_1(s) * F_2(s)$	convolution
21	$\lim_{t \rightarrow 0} f(t)$	$\lim_{s \rightarrow \infty} sF(s)$	IVT
22	$\lim_{t \rightarrow \infty} f(t)$	$\lim_{s \rightarrow 0} sF(s)$	FVT

Impulse Function

While we call the impulse a function, it's technically not a function because of the infinite height and zero width. It is however a useful tool in software simulation and mathematical analysis.

$$f(t) = \delta(t)$$

which produces the plot



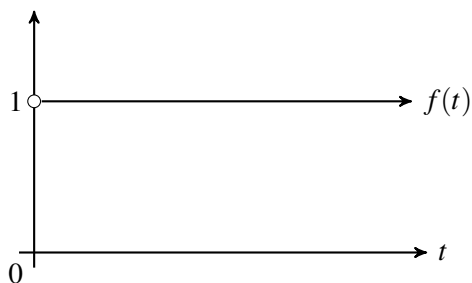
Where the area under the curve is one.

Step Function

The step function serves as a setpoint input, which unlike the impulse, is physically realizable. It is a bounded, static input

$$f(t) = \begin{cases} 1, & t > 0 \\ 0, & t \leq 0 \end{cases}$$

which produces the plot



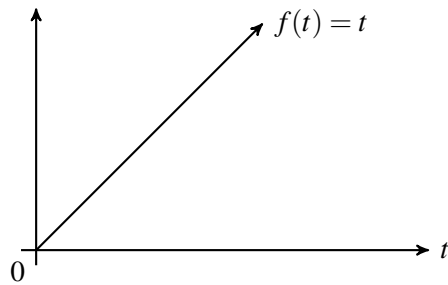
Think of the temperature setting for a home; we adjust the temperature setting on the thermostat which functions as the setpoint input into the temperature control system.

Ramp Function

The ramp function is an unbounded input

$$f(t) = t$$

used to determine the capacity of a system to follow an input with a constant velocity.

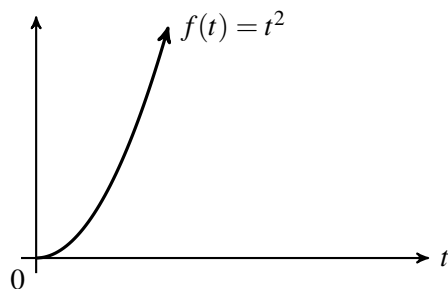


Parabola Function

The parabola function is an unbounded input

$$f(t) = t^2$$

used to determine the capacity of a system to follow an input with a constant acceleration.

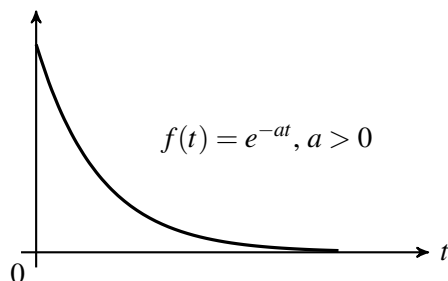


Exponential Function

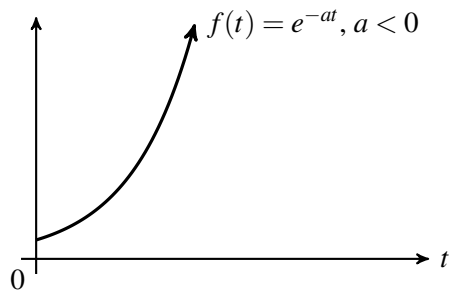
We see the exponential function not as a signal input, but as an output signal of a transfer function.

$$f(t) = e^{-at}$$

The stability of exponential function is determined by the constant **a**. When **a** is positive, the system is stable and the time-domain response is



When **a** is negative, the system is unstable and the time-domain response is

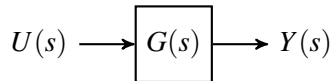


Chapter 2

Block Diagrams

Control systems generally consist of multiple devices (sensors, effectors) and a plant (electrical, mechanical, or a combination). These devices are linked together to form a system that performs a given function. The mathematical representation of each device is usually a transfer function. Devices are linked together by an input output relationship; where the output of one device becomes the input to another.

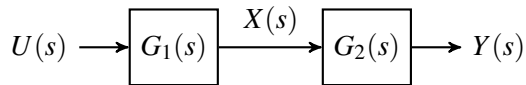
The most basic system consists of an input, output, and single transfer function



Mathematically, the input is $U(s)$ and an output is $Y(s)$. The output function $Y(s) = G(s)U(s)$, where $G(s)$ is a transfer function.

For simple problems cascaded as this example is, the algebra is simple. For more complex problems, a better representation is needed. The block diagram allows for transfer function representation and complex signal visualization without compromising accuracy.

The next example has an external input of $U(s)$ and an external output of $Y(s)$, however it is made up of two cascaded systems. The output signal of the first transfer function is the input the signal into the second transfer function. The signal between the first and second transfer functions (aka. blocks) is considered an **internal signal** while the input and output are **external signals**. When systems are cascaded together in such a way one can find the overall transfer function, $\frac{Y(s)}{U(s)}$.



To solve for the transfer function

$$Y(s) = G_2 X(s)$$

where

$$X(s) = G_1 U(s)$$

thus

$$Y(s) = G_2(s)(G_1 U(s))$$

or

$$Y(s) = G_1 G_2 U(s)$$

The transfer function is then

$$\frac{Y(s)}{U(s)} = G_1 G_2(s) \quad (2.1)$$

We use three fundamental constructs to develop complex representations of physical systems.

Transfer Functions

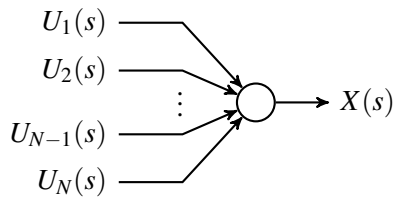
A linear, time-invariant, differential equation represented in the Laplace Domain as a ratio of output to input. to achieve a transfer function, the initial conditions must all be zero.

Summation Junctions

A function that sums all of the inputs into a single output

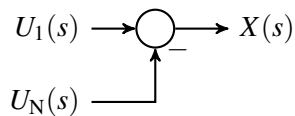
$$X(s) = \sum_{i=1}^N U_i(s) \quad (2.2)$$

where $U_i(s)$ are inputs and $X(s)$ is the output. The generalized summation diagram



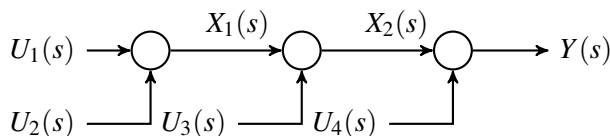
Additionally, the summation junction is used to add or subtract input signals. The sign operator is outside the summation circle near the signal to be subtracted (*Note: the absence of a sign designation implies a positive operator*).

Given the diagram

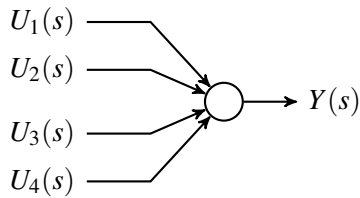


the output $X(s) = U_1(s) - U_N(s)$.

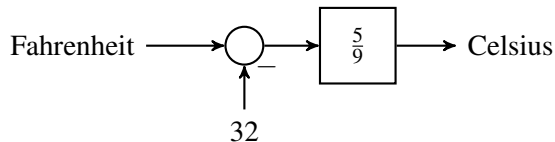
Summation junctions may be cascaded together, combined or separated. For example



has two internal signals, $X_1(s) = U_1(s) + U_2(s)$, $X_2(s) = X_1(s) + U_3(s)$, and the external signal $Y(s) = X_2(s) + U_4(s)$. The output may be represented as a summation of all of the external inputs, $Y(s) = U_1(s) + U_2(s) + U_3(s) + U_4(s)$ and as a diagram



Block diagrams can represent processes, both simple and complex. The diagram below uses a static gain and a summation junction to represent the mathematical operation to convert Fahrenheit to Celsius.

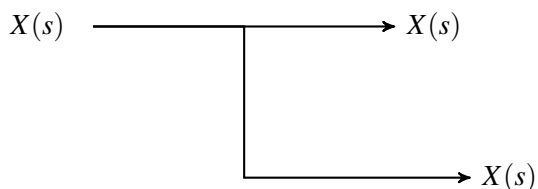


Which is

$$\text{Celsius} = \frac{5}{9}(\text{Fahrenheit} - 32)$$

Branches

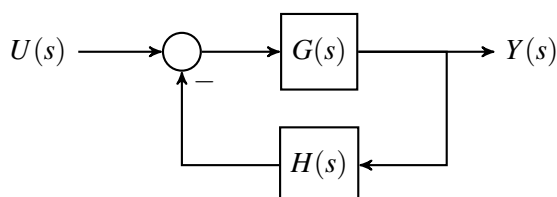
A branch is an ideal split such that the signal after the branch has two or more paths.



2.1 Manipulation

The block diagram approach is a good visualization tool that allows for quick combination of multiple devices by reducing the process to basic mathematics.

A typical model with negative feedback



where $G(s)$ is the forward path transfer function and $H(s)$ is the sensor feedback transfer function. Which produces an input/output transfer function

$$\frac{Y(s)}{U(s)} = T(s) = \frac{G(s)}{1 + GH(s)} \quad (2.3)$$

which can be written in **continued fraction** form by dividing the numerator and denominator by the numerator value

$$T(s) = \frac{G(s)}{1 + GH(s)} = \frac{\frac{G(s)}{G(s)}}{\frac{1 + GH(s)}{G(s)}} = \frac{1}{\frac{1}{G(s)} + H(s)} \quad (2.4)$$

The result in (2.1) is a useful tool for model construction.

For example, we are given a transfer function

$$T(s) = \frac{s + b_0}{s^2 + a_1s + a_0}$$

and we want to construct a block diagram with feedback. Simply find the continued fraction model of the transfer function and compare it to the general continued fraction model from (2.1).

$$T(s) = \frac{s + b_0}{s^2 + a_1s + a_0} = \frac{1}{\frac{s^2 + a_1s + a_0}{s + b_0}} = \frac{1}{\frac{s^2 + a_1s}{s + b_0} + \frac{a_0}{s + b_0}}$$

Now, compare the expression to the continued fraction model

$$T(s) = \frac{1}{\frac{s^2 + a_1s}{s + b_0} + \frac{a_0}{s + b_0}} = \frac{1}{\frac{1}{G(s)} + H(s)}$$

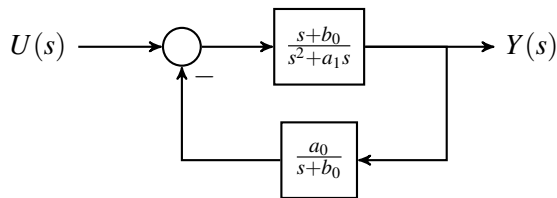
Now, one must make a choice as to the expressions for G and H . The rule of thumb: select the lowest order expression for H ¹. In this example

$$H(s) = \frac{a_0}{s + b_0}$$

and

$$\frac{1}{G(s)} = \frac{s^2 + a_1s}{s + b_0} \rightarrow G(s) = \frac{s + b_0}{s^2 + a_1s} = \frac{s + b_0}{s(s + a_1)}$$

Now, the block diagram for the transfer function $T(s)$ is

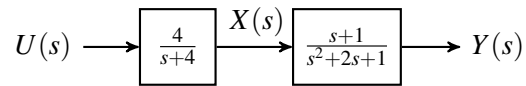


Let's consider some examples that demonstrate some of the most common block diagram manipulations

¹Selecting the wrong expression produces a model that will not work in software simulation tools

Example — Cascaded Blocks

Given:



Find: the transfer function $\frac{Y(s)}{U(s)}$

Solution:

$$Y(s) = \frac{s+1}{s^2+2s+1} \cdot X(s)$$

where

$$X(s) = \frac{4}{s+4} \cdot U(s)$$

$$Y(s) = \frac{s+1}{s^2+2s+1} \cdot \frac{4}{s+4} \cdot U(s) = \frac{4s+4}{s^3+6s^2+9s+4} U(s)$$

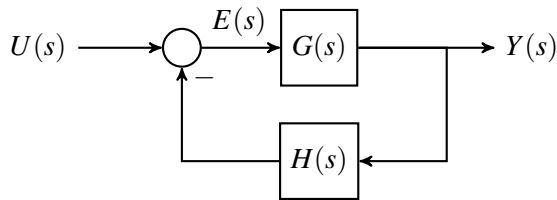
Finally

$$\frac{Y(s)}{U(s)} = \frac{4s+4}{s^3+6s^2+9s+4}$$

As is the custom, the final transfer function solution is written as a ratio of descending order polynomials.

Example — Negative Feedback

Given:



Find: the transfer function $\frac{Y(s)}{U(s)}$

Solution:

$$Y(s) = GE(s)$$

where

$$E(s) = U(s) - HY(s)$$

thus

$$Y(s) = G(s)(U(s) - HY(s))$$

$$Y(s) = GU(s) - GHY(s)$$

$$Y(s) + GHY(s) = GU(s)$$

$$(1 + GH(s))Y(s) = GU(s)$$

or

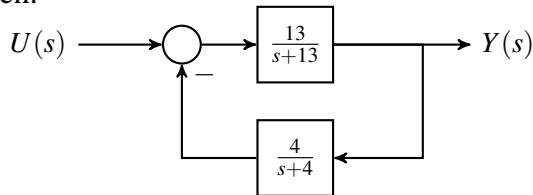
$$Y(s) = \frac{GU(s)}{1 + GH(s)}$$

Finally as a transfer function,

$$\frac{Y(s)}{U(s)} = \frac{G(s)}{1 + GH(s)} \quad (2.5)$$

Example — Another Negative Feedback

Given:



Find: the transfer function $\frac{Y(s)}{U(s)}$

Solution: The solution may be obtained using the block algebra from the previous example; here, let's solve it using direct mathematical analysis.

The output signal

$$Y(s) = \frac{13}{s+13} \left(U(s) - \frac{4}{s+4} Y(s) \right)$$

distribute the transfer function through the expression in the parentheses

$$Y(s) = \frac{13}{s+13} \cdot U(s) - \frac{52}{(s+13)(s+4)} Y(s)$$

collect the output signal Y on the lefthand side and factor out the signal

$$\left(1 + \frac{52}{(s+13)(s+4)} \right) Y(s) = \frac{13}{s+13} U(s)$$

write the expression in the parentheses as a fraction

$$\left(\frac{(s+13)(s+4) + 52}{(s+13)(s+4)} \right) Y(s) = \frac{13}{s+13} U(s)$$

isolate the output signal on the lefthand side

$$Y(s) = \left(\frac{(s+13)(s+4)}{(s+13)(s+4) + 52} \right) \frac{13}{s+13} U(s)$$

simplify

$$Y(s) = \left(\frac{13(s+4)}{(s+13)(s+4)+52} \right) U(s)$$

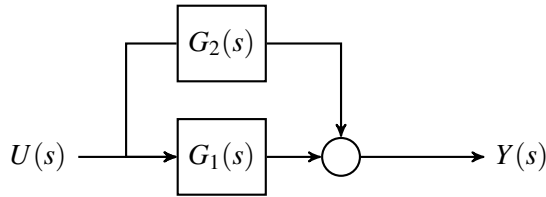
The final transfer function

$$Y(s) = \frac{13s+52}{s^2+17s+104} U(s)$$

Once again, the final transfer function solution is written as a ratio of descending order polynomials.

Example — Feedforward

Given:



Find: the transfer function $\frac{Y(s)}{U(s)}$

Solution:

$$Y(s) = G_1 U(s) + G_2 U(s)$$

thus,

$$Y(s) = (G_1(s) + G_2(s)) U(s)$$

Finally as a transfer function,

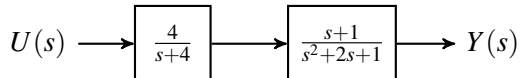
$$\frac{Y(s)}{U(s)} = G_1(s) + G_2(s) \quad (2.6)$$

2.2 Algebra

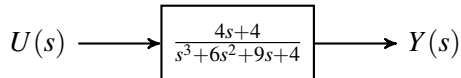
There are several algebraic approaches that allow us to manipulate block diagrams.

2.2.1 Cascaded Block Manipulation

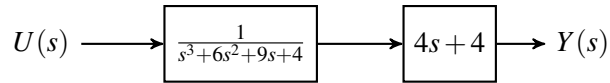
Just as above, we can manipulate cascaded blocks and form new math models. For example,



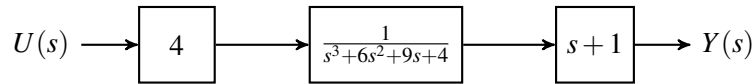
is the same as



which can be rewritten as



or as

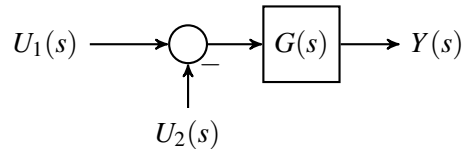


There are many potential representations – as long as the transfer function $\frac{Y(s)}{U(s)}$ is consistent among the models.

2.2.2 Moving Summation Junctions

After a Transfer function

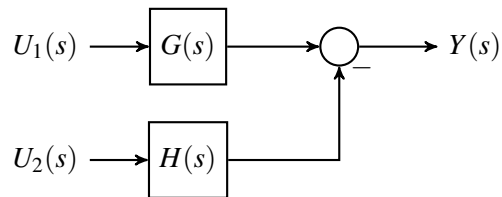
We have an original diagram



and we want to move the summation junction after the transfer function. We first consider the mathematical solution to the block diagram, which is

$$Y = G(s)(U_1(s) - U_2(s)) = GU_1(s) - GU_2(s) \quad (2.7)$$

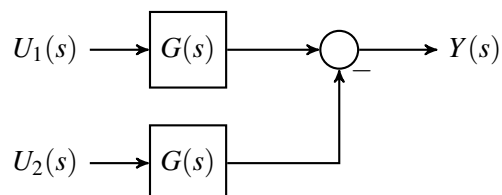
After moving the summation junction the solution **must** be the same



which produces an equation

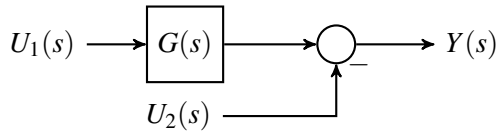
$$Y = GU_1(s) - HU_2(s) \quad (2.8)$$

Now, just compare (2.7) to (2.8) to resolve the unknown transfer function $H(s)$. Obviously, $H(s) = G(s)$, so the final block diagram is



Before a Transfer function

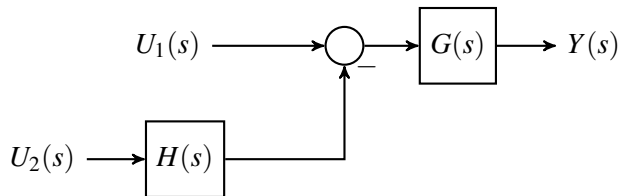
We have an original diagram



which produces an equation

$$Y(s) = GU_1(s) - U_2(s) \quad (2.9)$$

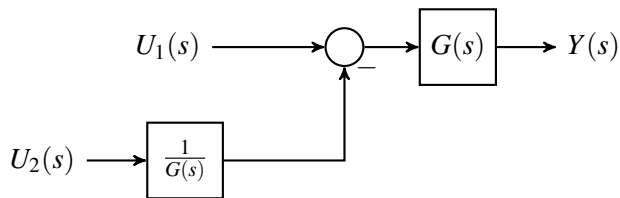
We want to move the summation junction before the transfer function to produce a block diagram



which produces the equation

$$Y(s) = G(s) \left(U_1(s) - HU_2(s) \right) = GU_1(s) - GHU_2(s) \quad (2.10)$$

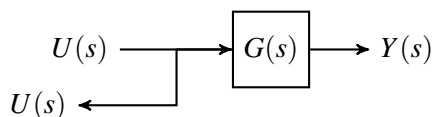
and now we just compare (2.9) to (2.10) to resolve the transfer function $H(s)$. If we set $H(s) = \frac{1}{G(s)}$ the equations will be equal and produce the final block diagram



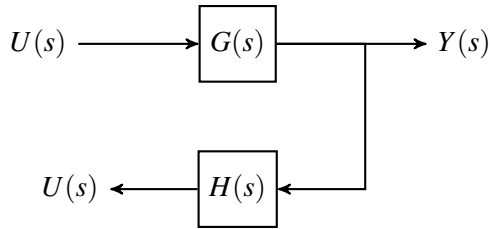
2.2.3 Moving Branches

After a Transfer function

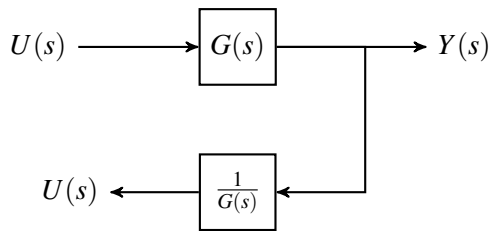
We start with a block diagram with an input signal that branches before going into a transfer function and we want to move the branch after the transfer function



Once again, after the move, we will have an unknown transfer function that we must determine

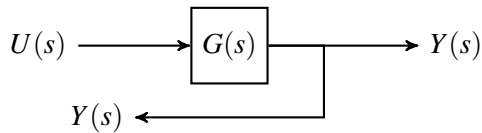


From the diagram $Y(s) = GU_1(s)$, therefore, $U_1(s) = \frac{1}{G(s)}Y(s)$, so the unknown transfer function $H(s) = \frac{1}{G(s)}$. The block diagram becomes

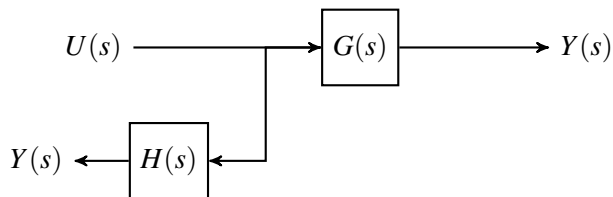


Before a Transfer function

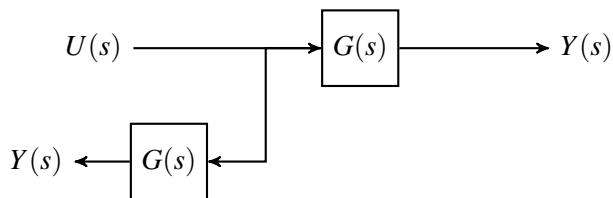
If we start with a block diagram with an output signal that branches after a transfer function and we want to move the branch before the transfer function



We repeat the process of placing an unknown transfer function in the block diagram

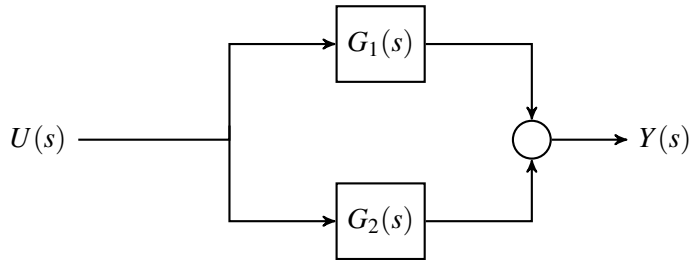


Of course, we know that $Y = GU(s)$, which means the unknown variable is simply $G(s)$. The block diagram becomes

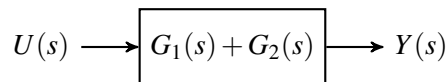


2.2.4 Combining Parallel Paths

When we have multiple transfer functions with the same input signal and their outputs are passed into a summation junction, we have parallel paths. See, Section 2.1 for an additional example.

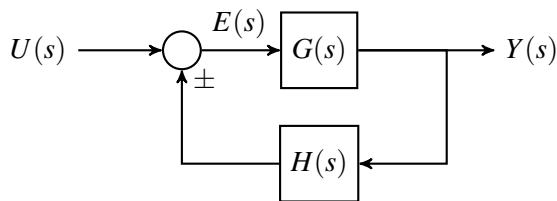


which produces an output signal $Y(s) = G_1U(s) + G_2U(s)$ and can be combined into a single transfer function



2.2.5 Eliminating a Feedback Loop

Section 2.1 is a good example of this concept, but let's formalize the process. We have a diagram with feedback (either positive or negative)



The output

$$Y(s) = G(s)E(s)$$

where

$$E(s) = U(s) \pm HY(s)$$

thus,

$$Y(s) = G(s)(U(s) \pm HY(s))$$

$$Y(s) = GU(s) \pm GHY(s)$$

$$Y(s) \mp GHY(s) = GU(s)$$

$$(1 \mp GH(s))Y(s) = GU(s)$$

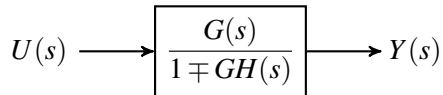
or

$$Y(s) = \frac{GU(s)}{1 \mp GH(s)}$$

which produces a transfer function

$$\frac{Y(s)}{U(s)} = \frac{G(s)}{1 \mp GH(s)}$$

and the block diagram simplifies to

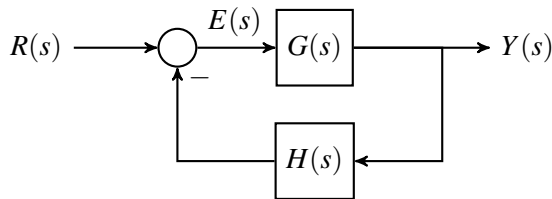


Which brings up an interesting concept and a common “conflict” in the field.

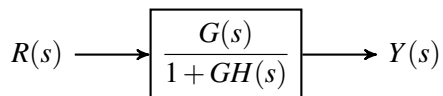
With this algebraic approach, we start with a diagram that has a feedback loop, and we simplify it to a diagram without a feedback loop. Additionally, we can reverse the process and start with a transfer function and end up with a diagram with a feedback loop.

Open-Loop vs Closed-Loop

Just like before we have a feedback system (I’m going to set the feedback to be negative to clean up the math, but the same concept applies for positive feedback as well).



which simplifies to



Visually, the first diagram has a feedback path (or loop), while the second does not. If the system has a loop then the signal can loop back on itself so it is closed-loop. Additionally, with the closed-loop system we define a **loop-gain** transfer function which is the product of all transfer functions within the loop; in this example $L(s) = GH(s)$.

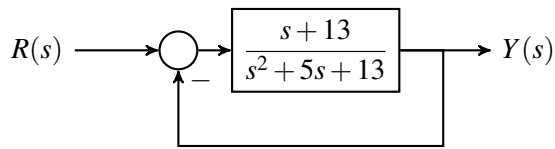
The second diagram does not loop back on itself, the signal passes from the input through the transfer to the output, so it is open-loop. I know, it should really be a “no loop system,” but that’s not the industry nomenclature – sorry.

What you won’t find in many source texts is presented in **Section 2.1**, which is the interoperability between the two models. For most systems, we can transform between the two models; each model presents unique analytical insights. The open-loop transfer function $\left(\frac{Y(s)}{R(s)} = \frac{B(s)}{A(s)}\right)$ is written as a ratio of polynomials where the denominator $A(s)$ is the **characteristic polynomial** and determines stability.

The closed-loop transfer function $\left(\frac{Y(s)}{R(s)} = \frac{G(s)}{1+L(s)}\right)$ yields the loop-gain $L(s)$ which determines the response error and frequency stability – among other things. Often, we need to consider both models when analyzing a system or designing a controller.

Example

Given:



Find: The open-loop and closed-loop representations

By observation, $\frac{Y(s)}{R(s)} = \frac{G(s)}{1+GH(s)}$ where

$$G(s) = \frac{s+13}{s^2+5s+13}$$

and

$$H(s) = 1$$

So

$$\frac{Y(s)}{R(s)} = \frac{G(s)}{1+GH(s)} = \frac{\frac{s+13}{s^2+5s+13}}{1+\frac{s+13}{s^2+5s+13}}$$

which is the closed-loop model. Now, to find the open-loop, just simplify the closed-loop to a ratio of polynomials.

Write the closed-loop model as a ratio of polynomials to achieve the open-loop representation

$$\frac{Y(s)}{R(s)} = \frac{\frac{s+13}{s^2+5s+13}}{\frac{(s^2+5s+13)+s+13}{s^2+5s+13}} = \frac{s+13}{s^2+5s+13} \cdot \frac{s^2+5s+13}{s^2+6s+26} = \frac{s+13}{s^2+6s+26}$$

Now, we have both representations (**closed-loop** and **open-loop**)

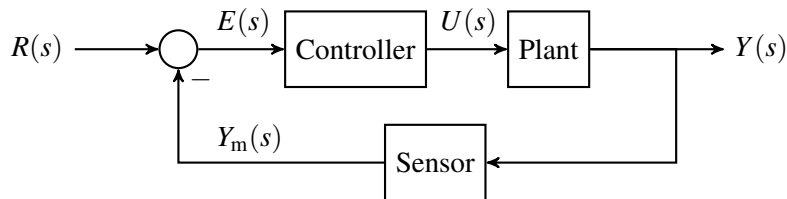
$$\frac{\frac{s+13}{s^2+5s+13}}{1+\frac{s+13}{s^2+5s+13}} \Leftrightarrow \frac{s+13}{s^2+6s+26}$$

which are equivalent, but offer different analytical insights into the system and how it will respond to various inputs.

Chapter 3

Systems with Feedback

Ideally, we simplify a plant with block diagram algebra to a single transfer function, then design a control system with negative feedback such that the controlled block diagram is of the form



where we have three internal signals

1. $E(s)$ is the error signal

- When the feedback is negative the signal is the difference between the reference input and the measured output

$$E(s) = R(s) - Y_m(s)$$

2. $U(s)$ is the control effort

- The controller output that drives the plant

$$U(s) = \text{Controller} \cdot E(s)$$

3. $Y_m(s)$ is the measured output signal, which is used for feedback

$$Y_m(s) = \text{Sensor} \cdot Y(s)$$

- Often, the difference between the output signal and the measured signal is insignificant when designing a controlled process

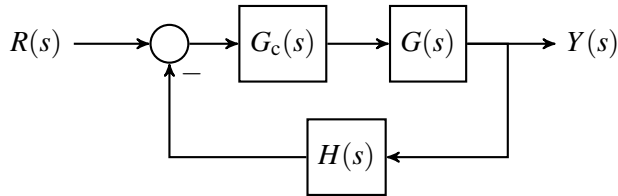
and two external signals

1. $R(s)$ is the reference input signal

2. $Y(s)$ is the controlled system output signal

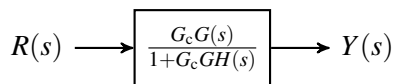
$$Y(s) = \text{Plant} \cdot U(s)$$

The above block diagram is accurate (but wordy) and is simplified to

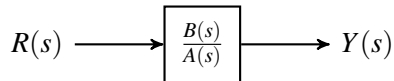


where the controller is $G_c(s)$, the plant is $G(s)$, the internal signals are the same, but not identified on the figure. The output feedback to the summation junction creates a loop in the system — this representation is a closed-loop model.

The diagram simplifies to a single block diagram



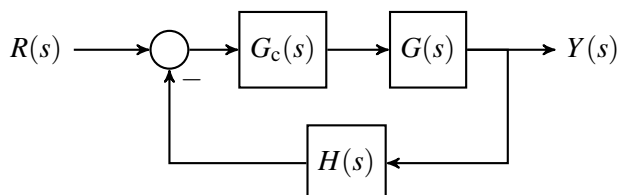
when the transfer function is simplified to a ratio of polynomials the solution is considered an open-loop representation, such as



Both models are valid representations and there's no loss of generality or information with either model, so we may transform between the two forms as needed for analysis and design. The biggest distinction between the two representations is the final form. The closed-loop form retains the forward path transfer function, which is $G_c(s)$ and $G(s)$ cascaded, and the loop-gain transfer function, which is $G_c(s)$, $G(s)$, and $H(s)$ cascaded or $L(s) = G_cGH(s)$. The primary reason we consider the closed-loop form is to analyze the loop-gain. So, if a transfer function is closed-loop the loop-gain is determined by direct observation. While the open-loop transfer function is always $\frac{Y(s)}{R(s)} = \frac{B(s)}{A(s)}$ where $B(s)$ and $A(s)$ are polynomials. Additionally, $A(s)$ has a special designation, it is the **characteristic polynomial**.

Example 1

Given:



where

$$G_c(s) = \frac{3}{s+3}$$

$$G(s) = \frac{10}{s+1}$$

and

$$H(s) = \frac{2}{s+2}$$

Find: the open-loop transformation

$$G_c G(s) = \frac{3 \cdot 10}{(s+3)(s+1)} = \frac{30}{s^2 + 4s + 3}$$

and

$$G_c GH(s) = \frac{3 \cdot 10 \cdot 2}{(s+3)(s+1)(s+2)} = \frac{60}{s^3 + 6s^2 + 11s + 6}$$

The single block diagram solution, which is $T(s) = \frac{G_c G(s)}{1 + G_c GH(s)}$ is

$$T(s) = \frac{\frac{30}{s^2 + 4s + 3}}{1 + \frac{60}{s^3 + 6s^2 + 11s + 6}} \quad (3.1)$$

which may be simplified

$$T(s) = \frac{\frac{30}{s^2 + 4s + 3}}{\frac{s^3 + 6s^2 + 11s + 6 + 60}{s^3 + 6s^2 + 11s + 6}} = \frac{30(s+2)}{s^3 + 6s^2 + 11s + 66}$$

written in polynomial form as

$$T(s) = \frac{B(s)}{A(s)} = \frac{30s + 60}{s^3 + 6s^2 + 11s + 66} \quad (3.2)$$

Now, we have two solutions (3.1) and (3.2). The first solution (3.1) still has a distinct representation for the forward path $G_c G(s)$ and the loop-gain

$$L(s) = G_c GH(s)$$

which means it is in closed-loop form. The loop-gain is

$$L(s) = \frac{60}{s^3 + 6s^2 + 11s + 6}$$

The second solution (3.2) has lost the individual representations for the transfer functions, so it is considered the open-loop solution – the open-loop solution is always a ratio of polynomials.

Example 2

Given a transfer function in open-loop form:

$$\frac{Y(s)}{R(s)} = T(s) = \frac{s+4}{s^2+s+9}$$

Find: the closed-loop transformation

Use the **continued fraction model** (Section 2.1) to determine the closed-loop model.

We begin by dividing the numerator and denominator in the open-loop transfer function by the numerator

$$T(s) = \frac{1}{\frac{s^2+s+9}{s+4}}$$

then separate the denominator fraction into two expressions. Take care to make one expression as simple as possible – a constant if possible, and the lowest order

$$T(s) = \frac{1}{\frac{s^2+s}{s+4} + \frac{9}{s+4}}$$

now set it equal to the continued fraction model of a negative feedback block diagram

$$T(s) = \frac{1}{\frac{s^2+s}{s+4} + \frac{9}{s+4}} = \frac{1}{\frac{1}{G(s)} + H(s)}$$

Select the simplest expression for the sensor transfer function and the other as the forward path transfer function $G(s)$

$$H(s) = \frac{9}{s+4}$$

and

$$G(s) = \frac{s+4}{s^2+s}$$

which produces a loop-gain

$$L(s) = GH(s) = \left(\frac{s+4}{s^2+s} \right) \left(\frac{9}{s+4} \right) = \frac{9}{s^2+s}$$

The closed-loop representation is then

$$\frac{Y(s)}{R(s)} = \frac{\frac{s+4}{s^2+s}}{1 + \frac{9}{s^2+s}}$$

Example 3

Given a transfer function in open-loop form:

$$\frac{Y(s)}{R(s)} = T(s) = \frac{4}{s+13}$$

Find: the closed-loop transformation

Just as before use the **continued fraction model** (Section 2.1) to determine the loop-gain.

$$T(s) = \frac{1}{\frac{s+13}{4}}$$

then separate the denominator fraction into two expressions.

$$T(s) = \frac{1}{\frac{s}{4} + \frac{13}{4}}$$

now set it equal to the continued fraction model of a negative feedback block diagram

Select the simplest expression for the sensor transfer function and the other as the forward path transfer function $G(s)$

$$H(s) = \frac{13}{4}$$

and

$$G(s) = \frac{4}{s}$$

which produces a loop-gain

$$L(s) = GH(s) = \frac{13}{s}$$

and a closed-loop transfer function

$$\frac{Y(s)}{R(s)} = \frac{\frac{4}{s}}{1 + \frac{13}{s}}$$

Example 4 — Biproper

Given a transfer function in open-loop form:

$$\frac{Y(s)}{R(s)} = T(s) = \frac{s^2 + 3s + 9}{s^2 - 2s - 7}$$

Find: the closed-loop transformation

The solution to this problem is not as easy as the previous problems because the open-loop transfer function is **biproper** – the order of the numerator is the same as the order of the denominator. Now, our first step is perform long division on transfer function

$$T(s) = \frac{s^2 + 3s + 9}{s^2 - 2s - 7} = 1 + \frac{5s + 16}{s^2 - 2s - 7} = 1 + \hat{T}(s)$$

Now, we have two expressions that represent $T(s)$ and we can use the **continued fraction** model to determine the loop-gain of $\hat{T}(s)$.

$$\hat{T}(s) = \frac{1}{\frac{s^2 - 2s - 7}{5s + 16}}$$

then separate the denominator fraction into two expressions.

$$\hat{T}(s) = \frac{1}{\frac{s^2 - 2s}{5s + 16} - \frac{7}{5s + 16}}$$

Select the simplest expression for the sensor transfer function and the other as the forward path transfer function $G(s)$

$$H(s) = -\frac{7}{5s + 16}$$

and

$$G(s) = \frac{5s + 16}{s^2 - 2s}$$

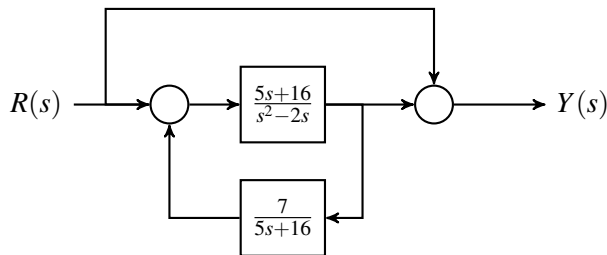
which produces a loop-gain

$$L(s) = GH(s) = -\frac{7}{s^2 - 2s}$$

and a closed-loop transformation

$$\frac{Y(s)}{R(s)} = 1 + \frac{\frac{5s + 16}{s^2 - 2s}}{1 - \frac{7}{s^2 - 2s}}$$

Let's take this example a step further by developing the block diagram. We have $G(s)$ and $H(s)$ defined for the partial expression $\hat{T}(s)$, but must also consider the constant in $T(s)$, which is a path directly from the input signal to the output signal.



Note, the feedback through $H(s)$ is positive due to the negative sign in the $H(s)$ transfer function.

Example 5 — Improper

Given a transfer function in open-loop form:

$$\frac{Y(s)}{R(s)} = T(s) = \frac{s^3 + s^2 + 9s + 9}{s^2 + 4s + 2}$$

Find: the closed-loop transformation

The denominator is higher order than the numerator, which means the open-loop transfer function is **improper**. There is no way to develop a closed-loop model for an improper open-loop transfer function.

3.1 Closed-Loop Models

When using the continued fraction model to transform from open-loop to closed-loop you are presented with a choice for $G(s)$ and $H(s)$. The choice should always be simplest expression for sensor feedback $H(s)$. Ideally, the sensor feedback should be a static gain of one — a.k.a. unity feedback. If unity feedback is not possible due to the system model, then select the simplest model possible — usually a constant over a low-order polynomial. Choosing the simplest model for $H(s)$ does two things:

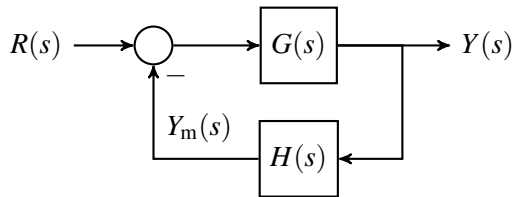
1. Minimizes the feedback circuits
 - The simpler the feedback the less lag is introduced
2. Avoids differentiation
 - Amplifies high frequency noise

Differentiation and Noise Example

Given an open-loop transfer function:

$$\frac{Y(s)}{R(s)} = \frac{1}{s + 13}$$

Find: the closed-loop block diagram



Using the continued fraction model approach we have two options

$$\frac{1}{s + 13} = \frac{1}{\frac{1}{G(s)} + H(s)}$$

Option 1

$$G(s) = \frac{1}{s}$$
$$H(s) = 13$$

which produces an integrator in the forward path.

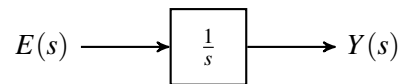
Option 2

$$G(s) = \frac{1}{13}$$
$$H(s) = s$$

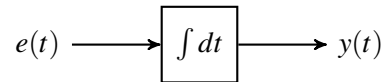
produces a differentiator in the feedback path.

Now, the mathematics tells us the two options are equivalent. However, **Option 1** has the simpler solution for $H(s)$

Let's consider the forward path of **Option 1**



and in the time-domain



where the input error signal is corrupted by noise

$$e = \text{signal} + \text{noise}$$

or

$$e(t) = \sin(t) + 0.01 \sin(10^{10}t)$$

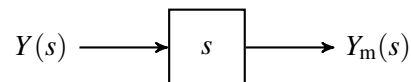
So the output

$$y(t) = \int_0^t e(t) dt = \int_0^t \sin(t) + 0.01 \sin(10^{10}t) dt = \int_0^t \sin(t) dt + \int_0^t 0.01 \sin(10^{10}t) dt$$

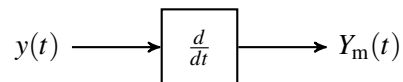
which is

$$y(t) = \text{signal} + \text{noise} = -\cos(t) - \frac{0.01}{10^{10}} \cos(10^{10}t) \approx \text{signal}$$

Now, let's consider the feedback path of **Option 2**



and in the time-domain



where the output signal is corrupted by noise

$$y = \text{signal} + \text{noise}$$

or

$$y(t) = \sin(t) + 0.01 \sin(10^{10}t)$$

So the output

$$Y_m(t) = \frac{dy}{dt} = \frac{d}{dt} \sin(t) + 0.01 \sin(10^{10}t)dt = \frac{d}{dt} \sin(t)dt + \frac{d}{dt} 0.01 \sin(10^{10}t)dt$$

which is

$$Y_m(t) = \text{signal} + \text{noise} = \cos(t) + (0.01 \cdot 10^{10}) \cos(10^{10}t) \approx \text{noise}$$

What we find is that the integrator in the presence of high-frequency noise produces a signal that is fully corrupted by the noise. In electronics, high-frequency noise is common. Switching power supplies generally have a 200 kHz to 2 MHz clock. Additionally, oscillators, time-sync circuits, and timers produce high-frequency noise as well.

3.1.1 Minimal Realization

Block diagram manipulation produce an open-loop transfer function where the zeros and poles will cancel. Always be careful in taking this step to reduce the complexity of the model with zero/pole cancelation. Only stable poles may be canceled in the open-loop transfer function. Unstable poles should *never* be canceled.

Stable Cancellation

If

$$\begin{aligned} T(s) &= \frac{s^2 + 7s + 6}{s^3 + 18s^2 + 104s + 192} \\ &= \frac{(s+1)(s+6)}{(s+8)(s+6)(s+4)} \end{aligned}$$

We can simplify to

$$\begin{aligned} T(s) &= \frac{s+6}{(s+8)(s+4)} \\ &= \frac{s+6}{s^2 + 12s + 32} \end{aligned}$$

Unstable Realization

If

$$\begin{aligned} T(s) &= \frac{s^2 - 5s - 6}{s^3 + 6s^2 - 40s - 192} \\ &= \frac{(s+1)(s-6)}{(s+8)(s-6)(s+4)} \end{aligned}$$

We adjust the model to

$$T(s) = \frac{(s+1)(s-6+\epsilon)}{(s+8)(s-6+\epsilon)(s+4)}$$

Chapter 4

Static Analysis

Generally, analysis is broken down into i) stability and ii) system response to inputs. Both areas of study are vast enough to have their own dedicated topics. This brief discussion is intended to be an overview of the subject.

When transforming a system from the time-domain to the Laplace-domain the knowledge of time dependent characteristics is lost. However, two time-domain data points can be determined from the Laplace-domain transform: i) initial value of the output and ii) final value of the output. Additionally, there is a stability constraint on the evaluation of the output: the final value analysis is only valid when the output response is bounded as $t \rightarrow \infty$. Given a general transfer function

$$\frac{Y(s)}{R(s)} = \frac{b_2s^2 + b_1s + b_0}{s^3 + a_2s^2 + a_1s + a_0}$$

the output wrt to the input is written

$$Y(s) = \frac{b_2s^2 + b_1s + b_0}{s^3 + a_2s^2 + a_1s + a_0} \cdot R(s)$$

where b_i and a_i are real valued constants. If the system is stable and the input signal, $R(s)$ is known and bounded then the initial and final values of the output signal can be determined. The initial value of the output signal is $y(t = 0)$, while the final value of the output signal is $y(t \rightarrow \infty)$.

1. The Initial Value Theorem (IVT)

$$y(t = 0) = \lim_{s \rightarrow \infty} s \cdot Y(s) \quad (4.1)$$

2. The Final Value Theorem (FVT)

$$y(t \rightarrow \infty) = \lim_{s \rightarrow 0} s \cdot Y(s) \quad (4.2)$$

Note: in both theorems there is an additional s term that must be considered when solving for the output signal.

4.1 Initial Value Theorem (IVT)

Given a problem where the output signal is defined by

$$Y(s) = \frac{4s^3 + 2s + 6}{s^3 + 5s^2 + 7s + 1} \cdot R(s)$$

and the input signal is a setpoint, $R(s) = \frac{1}{s}$. Then, the initial value is

$$y(t=0) = \lim_{s \rightarrow \infty} s \frac{4s^3 + 2s + 6}{s^3 + 5s^2 + 7s + 1} \cdot \frac{1}{s}.$$

Simplify the expression and evaluate the limit

$$y(t=0) = \lim_{s \rightarrow \infty} \frac{4 + \frac{2}{s^2} + \frac{6}{s^3}}{1 + \frac{5}{s} + \frac{7}{s^2} + \frac{1}{s^3}} = \frac{4}{1} = 4$$

4.2 Final Value Theorem (FVT)

Given the same problem where the output signal is defined by

$$Y(s) = \frac{4s^3 + 2s + 6}{s^3 + 5s^2 + 7s + 1} \cdot R(s)$$

and the input signal is a step input, $R(s) = \frac{1}{s}$. Then, the final value is

$$y(t \rightarrow \infty) = \lim_{s \rightarrow 0^+} s \frac{4s^3 + 2s + 6}{s^3 + 5s^2 + 7s + 1} \cdot \frac{1}{s}$$

Now, either evaluate with L'Hôpital's rule or simplify the expression and evaluate the limit

$$y(t \rightarrow \infty) = \lim_{s \rightarrow 0^+} \frac{4s^3 + 2s + 6}{s^3 + 5s^2 + 7s + 1} = \frac{6}{1} = 6$$

The solution is only valid when the output $Y(s)$ is stable.

Recall: L'Hôpital's Rule

If, instead of algebraically manipulating the expression one prefers to use the more rigorous approach, here's the rule use

$$\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \lim_{x \rightarrow c} \frac{f'(x)}{g'(x)}$$

4.3 Zero — Pole Determination

We must use the open-loop model to determine the zeros and poles of a system

$$T(s) = \frac{B(s)}{A(s)}$$

Set $B(s) = 0$ and solve for the variable s , the solutions are the system zeros. Set $A(s) = 0$ and solve for the variable s , the solutions are the system poles. **Poles alone determine the system stability.** The system performance is determined by both the zeros and the poles.

Example

Given:

$$T(s) = \frac{s^2 + 14s}{s^4 + 5s^3 - 14s^2 - 140s - 200}$$

Find: The zeros and poles

The zeros:

$$\begin{aligned}s^2 + 14s &= s(s + 14) = 0 \\ s &= \{-14, 0\}\end{aligned}$$

The poles:

$$\begin{aligned}s^4 + 5s^3 - 14s^2 - 140s - 200 &= 0 \\ s &= \{-2, -4 \pm 2j, 5\}\end{aligned}$$

Note, the complex solutions for the poles. Zeros and poles may be real, complex, or imaginary.

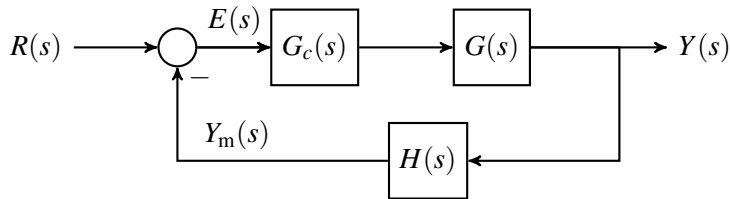
4.4 Steady State Error

Most controls problems involve the output signal tracking the input signal. Unimaginatively, we call these **tracking problems**. The typical implementation involves applying a desired or reference input to a system while designing a controller that modifies the natural behavior to the system so the output signal tracks the input signal after the transient (short term initial) response has settled.

The difference between the input and the output is the system error — at any point in time. When the system response has settled to a steady state, the error is then the steady state error, $e_{ss}(t \rightarrow \infty)$.

In the time-domain, $e(t) = r(t) - y(t)$. Where, $r(t)$ is the input **reference signal** and $y(t)$ is the ideal output signal. In the Laplace-domain, $E(s) = R(s) - Y(s)$. If the sensor feedback is poor, then the actual error as described the diagram $e(t) = r(t) - y_m(t)$ will differ from the ideal error in the above equation.

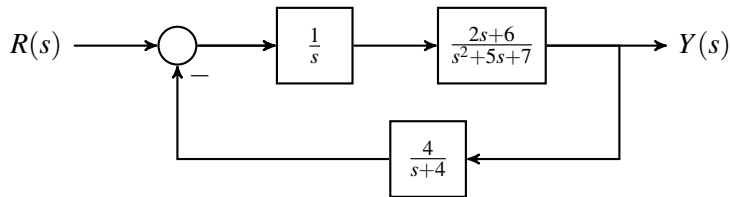
Steady state error analysis utilizes the negative feedback model, often with the controller explicitly defined in the forward path.



Predictably, there is more than one approach to solving these problems.

Option 1

Take a direct route by determining the transfer function $\frac{Y(s)}{R(s)}$ then using $E(s) = R(s) - Y(s)$.



where

$$Y(s) = \frac{\left(\frac{1}{s}\right)\left(\frac{2s+6}{s^2+5s+7}\right)}{1 + \left(\frac{1}{s}\right)\left(\frac{2s+6}{s^2+5s+7}\right)\left(\frac{4}{s+4}\right)} R(s)$$

which simplifies to

$$Y(s) = \frac{2s^2 + 14s + 24}{s^4 + 9s^3 + 27s^2 + 36s + 24} R(s)$$

So,

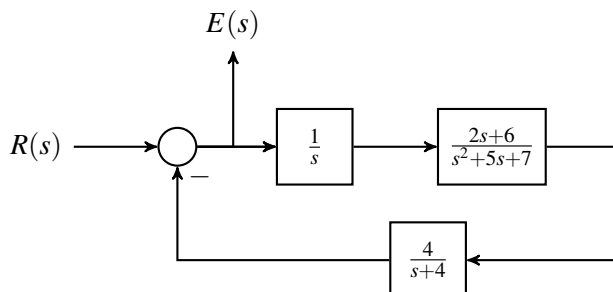
$$E(s) = R(s) - Y(s) = R(s) - \frac{2s^2 + 14s + 24}{s^4 + 9s^3 + 27s^2 + 36s + 24} R(s)$$

find the steady state response when $R(s) = \frac{1}{s}$

$$e(t \rightarrow \infty) = \lim_{s \rightarrow 0} s(R(s) - Y(s)) = \lim_{s \rightarrow 0} s\left(\frac{1}{s} - \frac{2s^2 + 14s + 24}{s^4 + 9s^3 + 27s^2 + 36s + 24} \frac{1}{s}\right) = 1 - \frac{24}{24} = 0$$

Option 2

Redefine the output to be the error signal and solve for the transfer function $\frac{E(s)}{R(s)}$.



Now,

$$E(s) = \frac{1}{1 + L(s)}$$

where the loop gain

$$L(s) = G_cGH(s) = \left(\frac{1}{s}\right)\left(\frac{2s+6}{s^2+5s+7}\right)\left(\frac{4}{s+4}\right)$$

Then

$$e(t \rightarrow \infty) = \lim_{s \rightarrow 0} sE(s) = \lim_{s \rightarrow 0} s\left(\frac{1}{1 + L(s)}\right)\frac{1}{s}$$

After simplification

$$e(t \rightarrow \infty) = \frac{1}{1 + L(0)}$$

where

$$L(0) = \lim_{s \rightarrow 0} \left(\frac{1}{s}\right)\left(\frac{2s+6}{s^2+5s+7}\right)\left(\frac{4}{s+4}\right) = \left(\frac{1}{0}\right)\left(\frac{6}{7}\right)\left(\frac{4}{4}\right) = \infty$$

which simplifies to

$$e(t \rightarrow \infty) = \frac{1}{1 + L(0)} = \frac{1}{\infty} = 0$$

Both approaches produce the correct solution, the first approach contains more algebraic manipulation than the second. Generally speaking, the second choice is the one with fewer possible pitfalls.

There is more on steady state error in Chapter 10.

Chapter 5

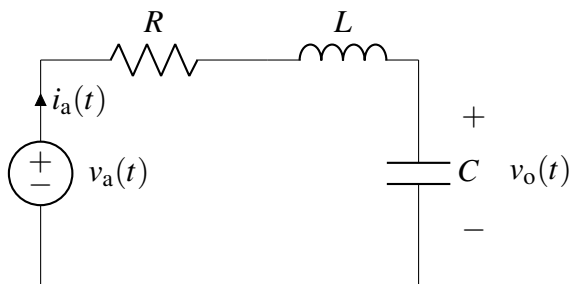
Practical Examples

Differential equations are used to mathematically model physical systems we want to control and analyze. The Laplace Domain representations are often used to model dynamic system.

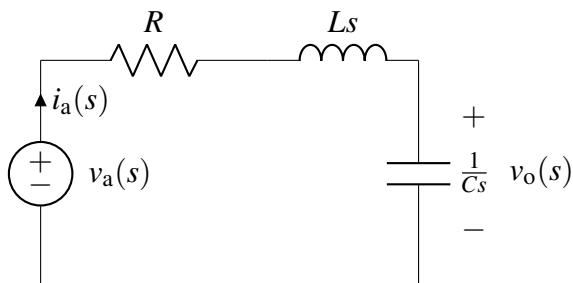
5.1 Electrical Circuit

Let's start with a problem that should be familiar.

Given the circuit diagram, find the open-loop transfer function



Let's transform the circuit into the Laplace Domain equivalent



Perform KVL around the circuit

$$-V_a(s) + Ri_a(s) + Lsi_a(s) + \frac{i_a(s)}{Cs} = 0$$

Now, define the current through the circuit with respect to the output signal using Ohm's Law

$$i_a(s) = \frac{V_o(s)}{\frac{1}{Cs}} = CsV_o(s)$$

the KVL equation simplifies to

$$-V_a(s) + RCsV_o(s) + LCs^2V_o(s) + V_o(s) = 0$$

Since a transfer function is a ratio of output to input, we need to isolate the I/O signals

$$V_a(s) = RCsV_o(s) + LCs^2V_o(s) + V_o(s)$$

$$V_a(s) = (RCs + LCs^2 + 1)V_o(s)$$

$$\frac{V_o(s)}{V_a(s)} = \frac{1}{RCs + LCs^2 + 1}$$

Finally, in proper form

$$\frac{V_o(s)}{V_a(s)} = \frac{1}{LCs^2 + RCs + 1}$$

Note: "Proper form" implies the numerator and denominator are written in descending order polynomial form.

5.2 Mechanical Diagram

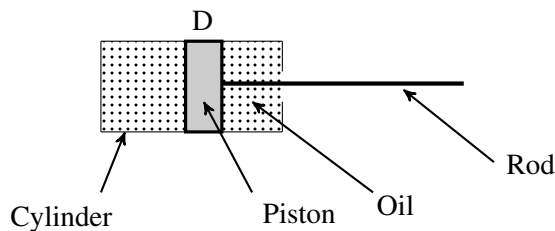
In electrical engineering we use a circuit diagram to find the Laplace domain equation, then the transfer function, block diagram, etc. In mechanical engineering we use a free body diagram (also called a force diagram) to find the Laplace domain equation, transfer function, etc. For dynamic mechanical systems, there are several common devices to consider.

Damper

The damper absorbs energy and dissipates it as heat, it does not store kinetic or potential energy.

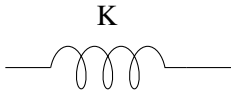
As constructed, it's a cylinder, piston, rod, and oil. The cylinder is filled with oil and the leaky piston – which is connected to the rod.

Relative motion between the rod and the cylinder forces oil to move from one side of the piston to the other. The amount of energy that is dissipated is a function of the physical characteristics, such as, the oil viscosity and the leakiness of the piston. The mathematical model uses a damper coefficient D to represent the characteristics of the device.



Spring

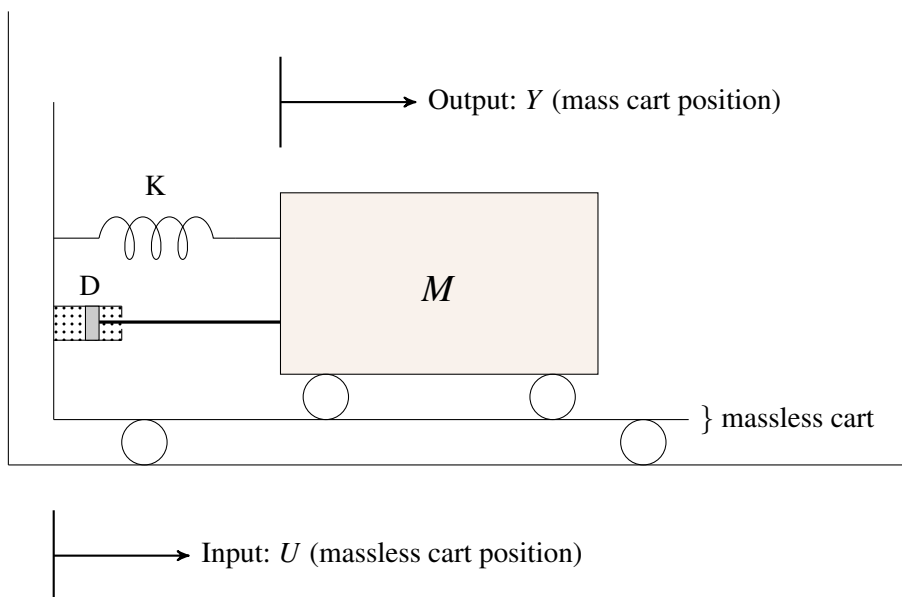
Every spring has a natural relaxed position called the free length. Compress a spring and it opposes the compression. Stretch a spring it opposes the stretching. The mathematical models uses a spring constant K .



Mechanical systems utilize a free body diagram much like electrical systems use electrical circuits.

Free Body Diagram

Given a free body diagram composed of a cart with mass M sitting on a massless cart. The two carts are connected to each other with a spring and a damper. The input to the system is the position of the massless cart and the output is the position of the mass cart.



To find the open-loop transfer function from the free body diagram, we start with Newton's second law

$$\sum F = ma \quad (5.1)$$

where $\sum F$ is the summation of all of the forces acting on the system, m is the mass of the mass cart and a is the acceleration of the mass cart. Since $y(t)$ is defined as the position of the mass cart, the acceleration of the mass cart is the second derivative of the cart's position

$$a = \ddot{y}(t)$$

and

$$\sum F = F_D + F_S$$

where F_D is the damper force

$$F_D = -D(\dot{y}(t) - \dot{u}(t))$$

and F_S is the spring force

$$F_S = -K(y(t) - u(t))$$

Newton's second law for the free body diagram is

$$m\ddot{y}(t) = -D(\dot{y}(t) - \dot{u}(t)) - K(y(t) - u(t))$$

$$m\ddot{y}(t) = -D\dot{y}(t) + D\dot{u}(t) - Ky(t) + Ku(t)$$

transform into the Laplace domain

$$ms^2Y(s) = -DsY(s) + DsU(s) - KY(s) + KU(s)$$

isolate the I/O signals

$$ms^2Y(s) + DsY(s) + KY(s) = DsU(s) + KU(s)$$

and factor them

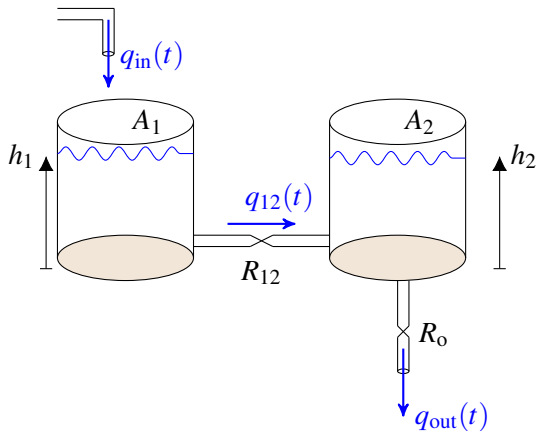
$$(ms^2 + Ds + K)Y(s) = (Ds + K)U(s)$$

Finally, write the open-loop transfer function

$$\frac{Y(s)}{U(s)} = \frac{Ds + K}{ms^2 + Ds + K}$$

5.3 Two Tank Fluid Flow Describing Diagram

Given the describing diagram for a two tank fluid flow problem



From the diagram, determine the fluid flow from input to output and construct an open-loop transfer function. First, define the differential equations. The volume of fluid in the first tank is

$$V_1(t) = A_1 \cdot h_1(t)$$

however, there is fluid flowing into the tank, $q_{in}(t)$ and fluid flowing out of the tank, $q_{12}(t)$; so we consider the rate of change of the fluid in the first tank.

$$q_{in}(t) - q_{12}(t) = \frac{dV_1(t)}{dt}$$

or written with h_1 as the dynamic variable

$$q_{in}(t) - q_{12}(t) = A_1 \cdot \frac{dh_1(t)}{dt} \quad (5.2)$$

The flow from tank 1 to tank 2 is a function of the heights of the tanks and the restriction of the flow tube.

$$q_{12}(t) = \frac{h_1(t) - h_2(t)}{R_{12}} \quad (5.3)$$

where $1 \leq R_{12} < \infty$, the larger R_{12} the greater the restriction on the flow rate.

Similar to the fluid flow in tank 1, the fluid flow and change of volume of tank 2 is

$$q_{12}(t) - q_{out}(t) = A_2 \cdot \frac{dh_2(t)}{dt} \quad (5.4)$$

Finally, the output equation is simply

$$q_{out}(t) = \frac{h_2(t)}{R_o} \quad (5.5)$$

where $1 \leq R_o < \infty$, the larger R_o the greater the restriction on the flow rate.

Now that we have defined the system from input to output we need to convert (5.2) to (5.5) into the Laplace domain equivalent equations

Starting with (5.2)

$$Q_{in}(s) - Q_{12}(s) = A_1 s H_1(s)$$

now, we rewrite it with the input $Q_{in}(s)$ and the output $H_1(s)$

$$H_1(s) = \frac{Q_{in}(s) - Q_{12}(s)}{A_1 s} \quad (5.6)$$

Moving to (5.3)

$$Q_{12}(s) = \frac{H_1(s) - H_2(s)}{R_{12}} \quad (5.7)$$

Next, (5.4)

$$Q_{in}(s) - Q_{12}(s) = A_2 s H_2(s)$$

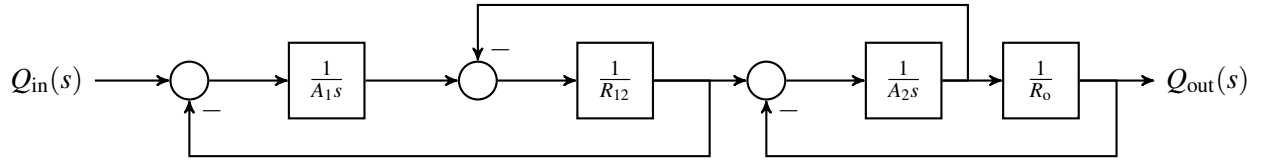
Now, we rewrite the equation with the input $Q_{in}(s)$ and the output $H_1(s)$

$$H_2(s) = \frac{Q_{12}(s) - Q_{out}(s)}{A_2 s} \quad (5.8)$$

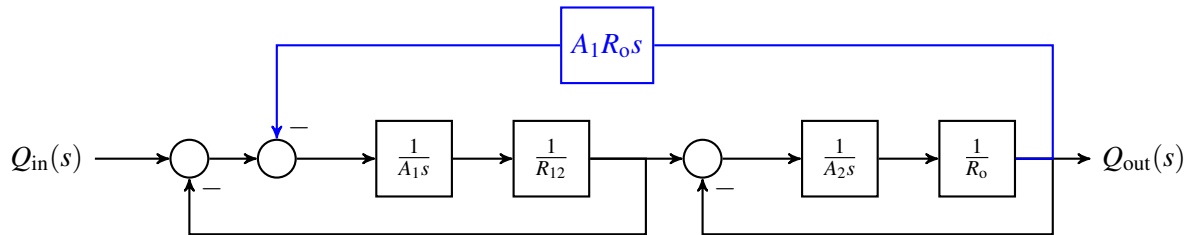
Finally (5.5)

$$Q_{out}(s) = \frac{H_2(s)}{R_o} \quad (5.9)$$

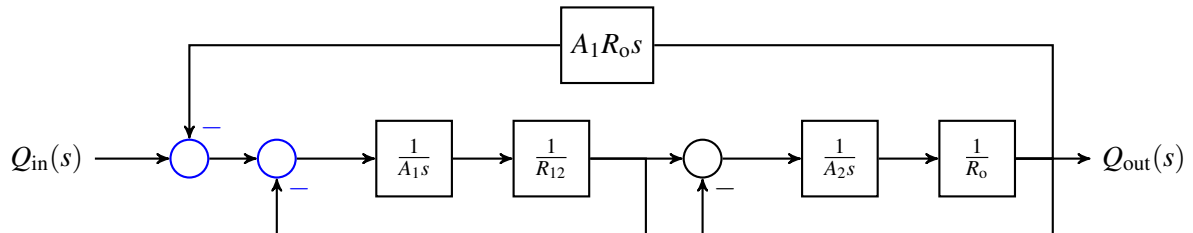
Now, we use the Laplace domain equations to construct the block diagram of the system



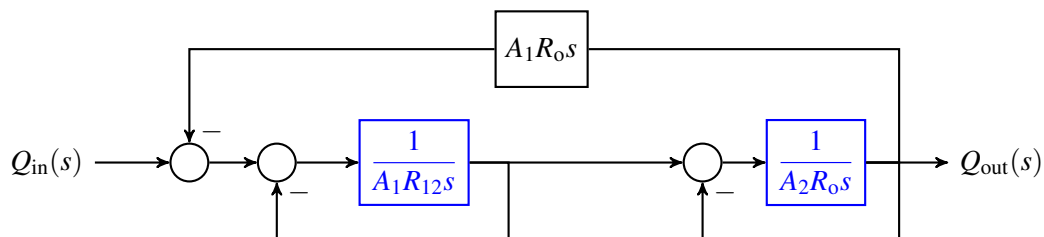
Next, we need to simplify the block diagram down to a transfer function. First move the branch from in front of the $\frac{1}{R_o}$ to after it and move the summation junction after $\frac{1}{A_1 s}$ to before it



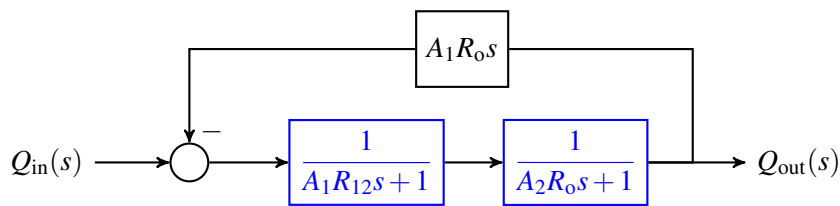
Change the order of the first two summation junctions



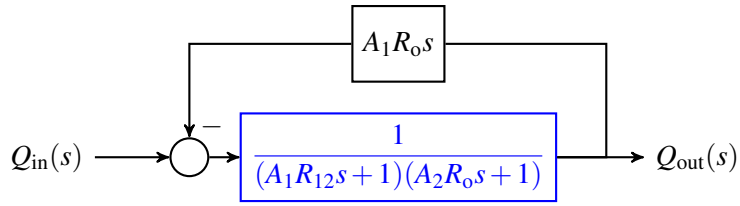
Combine cascaded blocks in the forward path



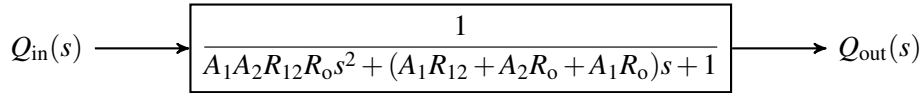
Next, eliminate the unity feedback loops



Combine the two remaining cascaded blocks in the forward path



Finally, eliminate the remaining feedback loop and reduce the diagram to produce a single open-loop transfer function block



Of course, the single open-loop block diagram reveals the input/output transfer function

$$\frac{Q_{\text{out}}(s)}{Q_{\text{in}}(s)} = \frac{1}{A_1 A_2 R_{12} R_o s^2 + (A_1 R_{12} + A_2 R_o + A_1 R_o) s + 1}$$

Chapter 6

Inverse Laplace Transforms

So far, we've taken differential equations and converted them into the Laplace Domain equivalent so we can

1. Form block diagrams
2. Perform basic analysis
 - (a) Initial Value
 - (b) Final Value
 - (c) Find the zeros and poles
3. Transform in the open-loop and closed-loop models

The transformation to the Laplace Domain can be reversed using a LUT with only a little effort in utilizing partial-fraction expansion, PFE. To directly utilize this approach, the open-loop transfer function must be strictly proper (the order of the numerator must be less than the order of the denominator). If the system is not strictly proper, perform long division once to form a mixed expression with a constant and a strictly proper remainder.

We represent the transfer function in open-loop form with the denominator represented in binomial form with real roots. Any complex roots are written as second order expressions.

For systems without complex poles

$$\frac{Y(s)}{R(s)} = \frac{B(s)}{A(s)} = \frac{k_1}{s+p_1} + \frac{k_n}{s+p_2} + \dots + \frac{k_n}{s+p_n}$$

For systems with real and complex poles

$$\frac{Y(s)}{R(s)} = \frac{B(s)}{A(s)} = \frac{k_1}{s+p_1} + \frac{k_2}{s+p_2} + \dots + \frac{k_n}{s+p_n} + \frac{k_ms + k_{m+1}}{\alpha_2 s^2 + \alpha_1 s + \alpha_0}$$

And of course we may have repeated poles — both real and complex. For a system such as that the generic equation is

$$\frac{Y(s)}{R(s)} = \frac{B(s)}{A(s)} = \frac{k_1}{(s+p_1)^2} + \frac{k_2}{s+p_1} + \frac{k_3}{s+p_2} + \dots + \frac{k_n}{s+p_n} + \frac{k_ms + k_{m+1}}{(\alpha_2 s^2 + \alpha_1 s + \alpha_0)^2} + \frac{k_{m+2}s + k_{m+3}}{\alpha_2 s^2 + \alpha_1 s + \alpha_0}$$

PFE can be difficult for more complex systems. Additionally, the solution begins to be less accurate with more poles. Systems with many repeated poles or even poles close to one another being to degrade the solution. Mathematically, those are **ill-conditioned** systems. Most physical systems may be accurately solved with PFE.

Let's take a look at some examples.

Example — Distinct Real Poles and an impulse input

Given:

$$\frac{Y(s)}{R(s)} = \frac{3}{(s+6)(s+2)}$$

Find: the $y(t)$ when $r(t) = \delta(t)$

Represent the output as a sum of partials

$$Y(s) = \frac{3}{(s+6)(s+2)} = \frac{k_1}{s+6} + \frac{k_2}{s+2}$$

To solve for the residue k_1 , multiply both sides by the divisor $s+6$ and then solve the equation for $s = -6$

$$\frac{3(s+6)}{(s+6)(s+2)} = \frac{k_1(s+6)}{s+6} + \frac{k_2(s+6)}{s+2}, s = -6$$

$$\frac{3}{(s+2)} = k_1 + \frac{k_2(s+6)}{s+2}, s = -6$$

$$\frac{3}{(-4)} = k_1 + \frac{k_2(0)}{-4}$$

$$k_1 = \frac{-3}{4}$$

Now, repeat the process for k_2

$$\frac{3(s+2)}{(s+6)(s+2)} = \frac{k_1(s+2)}{s+6} + \frac{k_2(s+2)}{s+2}, s = -2$$

$$\frac{3}{(s+6)} = \frac{k_1(s+2)}{s+6} + k_2, s = -2$$

$$\frac{3}{4} = \frac{k_1(0)}{4} + k_2$$

$$k_2 = \frac{3}{4}$$

So

$$Y(s) = \frac{\frac{-3}{4}}{s+6} + \frac{\frac{3}{4}}{s+2}$$

Now, use a LUT on the expressions to find the final solution

$$y(t) = \frac{-3}{4}e^{-6t} + \frac{3}{4}e^{-2t}$$

In this form, $y(t)$ may be evaluated for any value $0 \leq t < \infty$.

Example — Biproper with Distinct Real Poles and an impulse input

Given:

$$\frac{Y(s)}{R(s)} = \frac{s^2 + 13s + 6}{s^2 + 9s + 8}$$

Find: $y(t)$ when $r(t) = \delta$

After long division

$$Y(s) = 1 + \frac{4s - 2}{s^2 + 9s + 8}$$

Then, use PFE on the denominator polynomial term

$$\frac{4s - 2}{s^2 + 9s + 8} = \frac{4s - 2}{(s + 8)(s + 1)} = \frac{k_1}{s + 8} + \frac{k_2}{s + 1}$$

So

$$k_1 = \frac{4s - 2}{s + 1}, s = -8$$

substitute $s = -8$ and simplify

$$k_1 = \frac{4(-8) - 2}{(-8) + 1} = \frac{-34}{-7} = \frac{34}{7}$$

and

$$k_2 = \frac{4s - 2}{s + 8}, s = -1$$

substitute $s = -1$ and simplify

$$k_2 = \frac{4(-1) - 2}{-1 + 8} = \frac{-6}{7}$$

Now

$$Y(s) = 1 + \frac{\frac{34}{7}}{s + 8} + \frac{\frac{-6}{7}}{s + 1}$$

And using a LUT

$$y(t) = \delta(t) + \frac{34}{7}e^{-8t} - \frac{6}{7}e^{-t}$$

Example — Repeated Real Poles and a Setpoint Input

Given:

$$\frac{Y(s)}{R(s)} = \frac{s + 72}{s^3 + 14s^2 + 57s + 72}$$

Find: $y(t)$ when $r(t) = 1$

Write $Y(s)$ as a sum of partials

$$Y(s) = \frac{s + 72}{s(s + 3)(s + 3)(s + 8)} = \frac{k_1}{s} + \frac{k_2}{(s + 3)^2} + \frac{k_3}{s + 3} + \frac{k_4}{s + 8}$$

Note the repeated residues k_2 and k_3 .

Residues k_1 , k_2 , and k_4 are solved as before. The results: $k_1 = 1$, $k_2 = -4.6$, and $k_4 = -0.32$. The only residue that cannot be solved as before is k_3 ; to solve it

$$k_3 = \left\{ \frac{d}{ds} \left[(s+3)^2 \frac{s+72}{s(s+3)^2(s+8)} \right] \right\}, s = -3$$

which simplifies to

$$k_3 = \left\{ \left[\frac{d}{ds} \frac{s+72}{s(s+8)} \right] \right\}, s = -3$$

Use the Quotient Rule to take the derivative

$$k_3 = \frac{(1)(s^2+8s) - (s+72)(2s+8)}{(s^2+8s)^2}, s = -3$$

$$k_3 = \frac{((-3)^2 - 24) - (72 - 3)(2(-3) + 8)}{((-3)^2 - 24)^2}$$

$$k_3 = \frac{(9 - 24) - (72 - 3)(8 - 6)}{(9 - 24)^2}$$

$$k_3 = \frac{-15 - (2 \cdot 69)}{225}$$

$$k_3 = \frac{-15 - 138}{225}$$

$$k_3 = \frac{-153}{225} = -0.68$$

Now that all the residues are determined

$$Y(s) = \frac{1}{s} - \frac{4.6}{(s+3)^2} - \frac{0.68}{s+3} - \frac{0.32}{s+8}$$

Take the inverse Laplace transform for the final solution

$$y(t) = 1 - 4.6te^{-3t} - 0.68e^{-3t} - 0.32e^{-8t}$$

Quotient Rule

When

$$f(x) = \frac{g(x)}{h(x)}$$

then

$$f'(x) = \frac{g'(x)h(x) - g(x)h'(x)}{(h(x))^2}$$

6.1 MATLAB

The `residue` function in MATLAB is used to recompose a function into its partials.

For the previous example

$$\frac{Y(s)}{R(s)} = \frac{s+72}{s^3+14s^2+57s+72}$$

we define $B = [1, 72]$ and $A = [1, 14, 57, 72, 0]$ (note: setpoint input is distributed in A) then the function call

```
[r, p, k] = residue(B,A)
```

produces the residues r , poles p , and gain k that match the calculated values from the example

```
>> [r,p,k] = residue(B,A)
```

```
r =
```

```
-0.3200  
-0.6800  
-4.6000  
1.0000
```

```
p =
```

```
-8.0000  
-3.0000  
-3.0000  
0
```

```
k =
```

```
[]
```

Chapter 7

Bounded Input Bounded Output Stability

Stability is a deep concept and is broken-down into several sections.

For a stable system, a bounded input (e.g. step and impulse) produces a bounded output. Therefore, this is considered bounded-input bounded-output stability or BIBO stability. A signal is bounded when there is a finite value that the signal magnitude never exceeds.

BIBO stability is analyzed with the open-loop representation of a transfer function by determining the roots of the characteristic polynomial (poles).

When the real value of a pole is negative, then the pole is stable. When all poles of a characteristic polynomial are stable, then the system that produces the signal is stable. If a characteristic polynomial has even one unstable pole then the system is unstable.

Let's look at a few examples.

Example — Stable System

Given:

$$\frac{Y(s)}{R(s)} = \frac{s+4}{s^2+3s+2}$$

Find: The BIBO system stability

A **time domain** approach:

Apply a bounded input, $r(t) = \delta(t)$ and determine the output signal using partial fraction expansion

$$Y(s) = \frac{s+4}{s^2+3s+2} = \frac{s+4}{(s+1)(s+2)} = \frac{k_1}{s+1} + \frac{k_2}{s+2}$$

where k_1 and k_2 are residues — non-zero real numbers in this example.

Then,

$$y(t) = k_1 e^{-t} + k_2 e^{-2t}$$

therefore

$$y(\infty) = k_1 e^{-\infty} + k_2 e^{-\infty} = 0$$

is a bounded output.

Let's choose a different bounded input, set $r(t) = 1$. Then

$$Y(s) = \frac{s+4}{s^2+3s+2} \left(\frac{1}{s} \right) = \frac{s+4}{s(s+1)(s+2)}$$

and as a summation of partials

$$Y(s) = \frac{s+4}{s(s+1)(s+2)} = \frac{k_1}{s} + \frac{k_2}{s+1} + \frac{k_3}{s+2}$$

which produces

$$y(t) = k_1 + k_2 e^{-t} + k_3 e^{-2t}$$

where k_1 , k_2 , and k_3 are residues — non-zero real numbers in this example. Therefore

$$y(\infty) = k_1 + k_2 e^{-\infty} + k_3 e^{-\infty} = k_1$$

which is non-zero, but bounded output.

Example — Unstable System

Given:

$$\frac{Y(s)}{R(s)} = \frac{s+4}{s^2+3s-2}$$

Find: The BIBO system stability

Apply a bounded input, $r(t) = \delta(t)$ and determine the output signal using partial fraction expansion

$$Y(s) = \frac{s+4}{s^2+3s-2} = \frac{s+4}{(s+1)(s-2)} = \frac{k_1}{s+1} + \frac{k_2}{s-2}$$

where k_1 and k_2 are residues — non-zero real numbers in this example. Then,

$$y(t) = k_1 e^{-t} + k_2 e^{2t}$$

which leads to

$$y(\infty) = k_1 e^{-\infty} + k_2 e^{\infty} \rightarrow \infty$$

Now, the first expression $k_1 e^{-\infty} \rightarrow 0$, is a function of a stable pole and by itself is bounded; however, the second expression, $k_2 e^{\infty} \rightarrow \infty$, is a function of an unstable pole and is unbounded. The end result is an unbounded output signal $y(t)$.

Note, in all three examples above there is no need to go into the time domain to determine BIBO stability, one must just observe the poles of the system.

From the first example,

$$\frac{Y(s)}{R(s)} = \frac{B(s)}{A(s)} = \frac{s+4}{s^2+3s+2}$$

Set the characteristic polynomial to zero and solve for the roots

$$s^2 + 3s + 2 = (s + 1)(s + 2) = 0$$

The poles, $s = -1, -2$ are both stable; therefore, the system is BIBO stable.

From the second example,

$$\frac{Y(s)}{R(s)} = \frac{B(s)}{A(s)} = \frac{s + 4}{s^2 + 3s - 2}$$

Set the characteristic polynomial to zero and solve for the roots

$$s^2 + 3s - 2 = 0$$

The second pole, $s = -1, 2$ is unstable; therefore, the system is BIBO is unstable.

When considering an open-loop transfer function

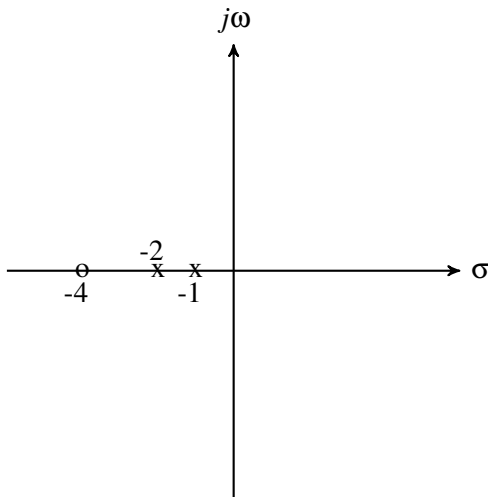
$$\frac{Y(s)}{R(s)} = \frac{B(s)}{A(s)}$$

The roots (zeros) for $B(s)$ shape the dynamic system response to an input, but do not affect stability. The roots (poles) for $A(s)$ shape the dynamic system response and determine the stability of the system. Typically, one plots the zeros and poles in the s-plane. In the plot, zeros are always identified with an “o” symbol and poles with an “x” symbol.

From the stable example above

$$\frac{Y(s)}{R(s)} = \frac{s + 4}{(s + 1)(s + 2)} = \frac{s + 4}{s^2 + 3s + 2}$$

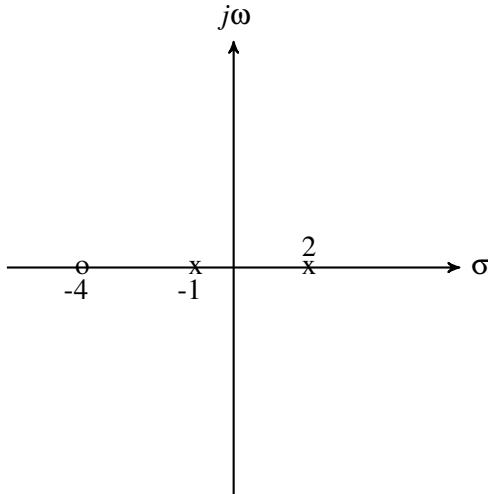
we have an s-plane plot of the zeros and poles



Likewise, from the unstable example above

$$\frac{Y(s)}{R(s)} = \frac{s + 4}{(s + 1)(s - 2)} = \frac{s + 4}{s^2 + 3s - 2}$$

we have an s-plane plot of the zeros and poles



Example — Stable System with Complex Poles

Given:

$$\frac{Y(s)}{R(s)} = \frac{s+4}{s^2+6s+10}$$

Find: The BIBO system stability From the **time domain**, set $r(t) = \delta(t)$

$$Y(s) = \frac{s+4}{s^2+6s+10} = \frac{k_1}{s+3+j} + \frac{k_2}{s+3-j}$$

$$k_1 = \frac{(s+3+j)(s+4)}{(s+3+j)(s+3-j)}, s = -3-j$$

$$k_1 = \frac{(s+4)}{(s+3-j)}, s = -3-j$$

$$k_1 = \frac{(-3-j+4)}{(-3-j+3-j)} = \frac{1-j}{-2j} = \frac{(1-j)2j}{(-2j)2j}$$

$$k_1 = \frac{2+2j}{4} = \frac{1+j}{2}$$

While

$$k_2 = \frac{(s+4)}{(s+3+j)}, s = -3+j$$

$$k_2 = \frac{(-3+j+4)}{(-3+j+3+j)} = \frac{1+j}{2j} = \frac{(1+j)(-2j)}{(2j)(-2j)}$$

$$k_2 = \frac{(2-2j)}{4} = \frac{1-j}{2}$$

Note, $k_1 = k_2^*$. This is always the case for complex numbers, their residues are always complex conjugate pairs.

$$Y(s) = \frac{s+4}{s^2+6s+10} = \frac{k_1}{s+3+j} + \frac{k_1^*}{s+3-j}$$

where the inverse Laplace

$$\mathcal{L}^{-}\{Y(s)\} = k_1 \mathcal{L}^{-}\left\{\frac{1}{s+3+j}\right\} + k_1^* \mathcal{L}^{-}\left\{\frac{1}{s+3-j}\right\}$$

Consider the expressions independently

$$k_1 \mathcal{L}^{-}\left\{\frac{1}{s+3+j}\right\} = k_1 (e^{(-3-j)t}) = k_1 (e^{(-3t)} e^{(-tj)})$$

and

$$k_1^* \mathcal{L}^{-}\left\{\frac{1}{s+3-j}\right\} = k_1^* (e^{(-3+j)t}) = k_1^* (e^{(-3t)} e^{(jt)})$$

where

$$e^{-jt} = \cos(-t) + j \sin(-t) = \cos(t) - j \sin(t)$$

$$e^{jt} = \cos(t) + j \sin(t) = \cos(t) + j \sin(t)$$

The inverse Laplace then simplifies to

$$y(t) = k_1 e^{-3t} (\cos(t) - j \sin(t)) + k_1^* e^{-3t} (\cos(t) + j \sin(t))$$

factor the common exponential function and distribute the residues

$$y(t) = e^{-3t} (k_1 \cos(t) - k_1 j \sin(t) + k_1^* \cos(t) + k_1^* j \sin(t))$$

now, collect like terms

$$y(t) = e^{-3t} ((k_1 + k_1^*) \cos(t)) + (k_1^* - k_1) j \sin(t)$$

which simplifies to

$$y(t) = e^{-3t} \left((k_1 + k_1^*) \cos(t) \right) = e^{-3t} \left(\left(\frac{1+j}{2} + \frac{1-j}{2} \right) \cos(t) \right) = e^{-3t} \cos(t)$$

7.1 Marginal Stability

Stable and unstable poles are straight forward and easily explained using the PFE model and the time domain. Marginally stable poles are *somewhat* more difficult to explain and grasp — but only by a little.

We start with a transfer function with stable poles and a pair of complex conjugate roots represented as a PFE

$$\frac{Y(s)}{R(s)} = \frac{k_1}{s+p_1} + \dots + \frac{k_n}{s+p_n} + \frac{b}{s^2+b^2}$$

The binomial expressions are from the stable poles, the last expression is from the complex roots on the imaginary axis. Set $r(t) = 1$ and solve for $y(t)$

$$Y(s) = \frac{k_0}{s} + \frac{k_1}{s+p_1} + \dots + \frac{k_n}{s+p_n} + \frac{b}{s^2+b^2}$$

Taking the inverse Laplace transform

$$y(t) = k_0 + k_1 e^{-p_1 t} + \dots + k_n e^{-p_n t} + \sin(bt)$$

As $t \rightarrow \infty$

$$y(\infty) = k_0 + \sin(bt)$$

because all of the expressions from the stable poles go to zero as $t \rightarrow \infty$. We can expand this concept to as many complex roots on the imaginary axis as we want. Each pair will generate an additional oscillation expression in the final output, but the output will still be stable.

$$Y(s) = \frac{k_0}{s} + \frac{k_1}{s+p_1} + \dots + \frac{k_n}{s+p_n} + \frac{b_1}{s^2+b_1^2} + \frac{b_2}{s^2+b_2^2}$$

Taking the inverse Laplace transform

$$y(t) = k_0 + k_1 e^{-p_1 t} + \dots + k_n e^{-p_n t} + \sin(b_1 t) + \sin(b_2 t)$$

As $t \rightarrow \infty$

$$y(\infty) = k_0 + \sin(b_1 t) + \sin(b_2 t)$$

Now, we consider a stable system with a single pole at the origin

$$\frac{Y(s)}{R(s)} = \frac{k_1}{s} + \frac{k_2}{s+p_2} + \dots + \frac{k_n}{s+p_n}$$

Let's set the input to an impulse $R(s) = 1$

$$Y(s) = \frac{k_1}{s} + \frac{k_2}{s+p_2} + \dots + \frac{k_n}{s+p_n}$$

The time domain equation

$$y(t) = k_1 + k_2 e^{-p_2 t} + \dots + k_n e^{-p_n t}$$

As $t \rightarrow \infty$

$$y(\infty) = k_1$$

which is a BIBO stable response.

Repeat for a setpoint input

$$Y(s) = \frac{k_1}{s} + \frac{k_2}{s+p_2} + \dots + \frac{k_n}{s+p_n} + \frac{k_m}{s^2}$$

The time domain equation

$$y(t) = k_1 + k_2 e^{-p_2 t} + \dots + k_n e^{-p_n t} + k_m t$$

As $t \rightarrow \infty$

$$\begin{aligned} y(\infty) &= k_1 + k_m(\infty) \\ &= \infty \end{aligned}$$

which is unbounded.

BIBO stable systems with marginally poles:

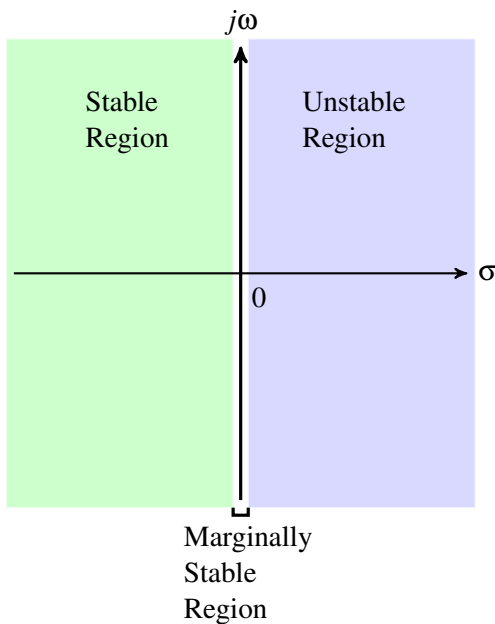
1. Are stable if there is no root at the origin. Complex roots on the $j\omega$ axis produce a oscillating bounded output
2. Are dependent on the input for stability if there is a root at the origin. An impulsive input produces a bounded output; while a setpoint input produces an unstable output
3. Are unstable if there are repeated roots at the origin; regardless of the input

7.2 BIBO Stability Summary

Stability analysis decomposes the s-plane in three regions of stability, as determined by the real axis. A pole is defined as $s = \sigma + j\omega$, where σ is the real part and $j\omega$ is the imaginary part. The three regions of stability are defined as

1. Stable region, $\sigma < 0$
2. Unstable region, $\sigma > 0$
3. Marginally stable region, $\sigma = 0$

The complex component produces the oscillations as observed about as the cosine operator in the previous example, however, the exponential function from the real part drives the solution to zero as $t \rightarrow \infty$.



7.3 Routh-Hurwitz Stability Criterion

Allows us to evaluate the BIBO stability of a system without directly calculating the poles of the system. The process creates an array (more like a table, really) that is then analyzed to determine the BIBO stability based off of the first column. The array is colloquially called the “Routh Array”.

When $A(s)$ is of degree n , the Routh Array will have $n + 1$ rows. The elements of the first two rows have elements composed of the coefficients in $A(s)$.

If $A(s) = a_n s^n + a_{n-1} s^{n-1} + \dots + a_0$ then the first two rows of the array are

Row Label	Col 1	Col 2	Col 3	...
s^n	a_n	a_{n-2}	a_{n-4}	...
s^{n-1}	a_1	a_{n-3}	a_{n-5}	...

the remaining rows ($s^{n-2}, \dots, s^0$) must be calculated.

A complete Routh Array takes on the form

Row Label	Col 1	Col 2	Col 3	...
s^n	a_n	a_{n-2}	a_{n-4}	...
s^{n-1}	a_1	a_{n-3}	a_{n-5}	...
s^{n-2}	x_{31}	x_{32}	x_{33}	...
s^{n-3}	x_{41}	x_{42}	x_{43}	...
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
s^0	x_{n1}			

where each x_{ij} is an element in the i^{th} row and j^{th} column.

Note: after the first two rows, each successive row has one fewer calculated column than the previous row. Therefore, the array shrinks by one column each row we travel down the array, until the final row has only a single column.

For example

Example — Generic 1

Given: $A(s) = a_3 s^3 + a_2 s^2 + a_1 s + a_0$

Find the Routh Array

s^3	a_3	a_1
s^2	a_2	a_0
s^1	x_{31}	x_{32}
s^0	x_{41}	

Now for the calculated elements x_{ij} , construct a sub-matrix for each unknown element based on the elements in the two previous rows. Always use the elements in the first column and elements in the column to the right of the unknown element to be calculated.

For element x_{ij} the sub-matrix is

$$\begin{bmatrix} x_{(i-2)1} & x_{(i-2)(j+1)} \\ x_{(i-1)1} & x_{(i-1)(j+1)} \end{bmatrix}$$

If $i = 3$ and $j = 1$, the sub-matrix is

$$\begin{bmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{bmatrix}$$

note, in this example, $x_{11} = a_n$, $x_{12} = a_{n-2}$, $x_{21} = a_{n-1}$, and $x_{22} = a_{n-3}$.

Finally, if $i = 7$ and $j = 4$, the sub-matrix is

$$\begin{bmatrix} x_{51} & x_{55} \\ x_{61} & x_{65} \end{bmatrix}$$

Once the sub-matrix is determined, the element to be calculated is simply the negative of the determinant divided by the element $x_{(i-1)1}$

$$x_{ij} = \frac{- \begin{vmatrix} x_{(i-2)1} & x_{(i-2)(j+1)} \\ x_{(i-1)1} & x_{(i-1)(j+1)} \end{vmatrix}}{x_{(i-1)1}}$$

where

$$\begin{vmatrix} x_{(i-2)1} & x_{(i-2)(j+1)} \\ x_{(i-1)1} & x_{(i-1)(j+1)} \end{vmatrix} = x_{(i-2)1} \cdot x_{(i-1)(j+1)} - x_{(i-2)(j+1)} \cdot x_{(i-1)1}$$

so

$$x_{ij} = \frac{x_{(i-2)(j+1)} \cdot x_{(i-1)1} - x_{(i-2)1} \cdot x_{(i-1)(j+1)}}{x_{(i-1)1}}$$

Once again, the math looks more complicated than it really is. Let's consider a few examples

Example — Generic 1, Reboot

For this example, we'll finish the previous one.

Given: $A(s) = a_3s^3 + a_2s^2 + a_1s + a_0$

Find: The stability of the system

The Routh Array

s^3	a_3	a_1
s^2	a_2	a_0
s^1	x_{31}	x_{32}
s^0	x_{41}	

where

$$x_{31} = \frac{- \begin{vmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{vmatrix}}{x_{21}} = \frac{- \begin{vmatrix} a_3 & a_1 \\ a_2 & a_0 \end{vmatrix}}{a_2} = \frac{a_2 \cdot a_1 - a_3 \cdot a_0}{a_2}$$

$$x_{32} = \frac{-\begin{vmatrix} x_{11} & x_{13} \\ x_{21} & x_{23} \end{vmatrix}}{x_{21}} = \frac{-\begin{vmatrix} a_3 & 0 \\ a_2 & 0 \end{vmatrix}}{a_2} = \frac{a_2 \cdot a_1 - a_3 \cdot a_0}{a_2} = 0$$

Note: Pad the array with zeros during the x_{ij} calculations.

$$x_{41} = \frac{-\begin{vmatrix} x_{21} & x_{22} \\ x_{31} & x_{32} \end{vmatrix}}{x_{31}} = \frac{-\begin{vmatrix} a_2 & a_0 \\ x_{31} & x_{32} \end{vmatrix}}{x_{31}} = \frac{x_{31} \cdot a_0 - a_2 \cdot x_{32}}{x_{31}} = \frac{x_{31} \cdot a_0 - a_2 \cdot 0}{x_{31}} = a_0$$

The final Routh array

s^3	a_3	a_1
s^2	a_2	a_0
s^1	x_{31}	0
s^0	a_0	

Now, for the stability analysis: inspect the elements in the first column. Starting with the first row s^n moving down, compare the signs (positive or negative). The number of sign changes in the first column is equal to the number of unstable poles.

Example — Second Order

Given: $A(s) = s^2 - 3s - 4$

Find: The stability of the system with the Routh Array

s^2	1	-4
s^1	-3	0
s^0	x_{31}	

Note: element x_{22} is not defined from the characteristic polynomial, but will be needed for later calculations so it's set to zero.

Only x_{31} needs to be calculated to determine stability

$$x_{31} = \frac{-\begin{vmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{vmatrix}}{x_{21}} = \frac{-\begin{vmatrix} 1 & -4 \\ -3 & 0 \end{vmatrix}}{-3} = \frac{12}{-3} = -4$$

The first column

$$\begin{bmatrix} +1 \\ -3 \\ -4 \end{bmatrix}$$

There is only one sign change in the first column; therefore, there is one unstable pole.

Example — Second Order

Given: $A(s) = s^2 + 5s + 4$

Find: The stability of the system with the Routh Array

s^2	1	4
s^1	5	0
s^0	x_{31}	

Only x_{31} needs to be calculated to determine stability

$$x_{31} = -\frac{\begin{vmatrix} 1 & 4 \\ 5 & 0 \end{vmatrix}}{5} = \frac{20}{5} = 4$$

So, the first column

$$\begin{bmatrix} +1 \\ +5 \\ +4 \end{bmatrix}$$

There are no sign changes in the first column; therefore, there are no unstable poles.

Example — Second Order

Given: $A(s) = s^2 - 5s + 4$

Find: the stability of the system with the Routh Array

s^2	1	4
s^1	-5	0
s^0	x_{31}	

Again, only x_{31} needs to be calculated to determine stability

$$x_{31} = -\frac{\begin{vmatrix} 1 & 4 \\ -5 & 0 \end{vmatrix}}{-5} = \frac{-20}{-5} = 4$$

So, the first column

$$\begin{bmatrix} +1 \\ -5 \\ +4 \end{bmatrix}$$

Now, there are two sign changes in the first column; therefore, there are two unstable poles. Since this is a second order system, all poles are unstable.

Example — Tunable Parameter

Given: $A(s) = s^3 + 9s^2 + Ks + 12$, where $-\infty < K < \infty$

Find: The stability of the system

Find: The range of K such that the system is stable

s^3	1	k
s^2	9	12
s^1	x_{31}	x_{32}
s^0	x_{41}	

Again, only x_{31} needs to be calculated to determine stability

$$x_{31} = \frac{-\begin{vmatrix} 1 & K \\ 9 & 12 \end{vmatrix}}{9} = \frac{9K - 12}{9}$$

$$x_{32} = \frac{-\begin{vmatrix} 1 & 0 \\ 9 & 0 \end{vmatrix}}{9} = 0$$

$$x_{41} = \frac{-\begin{vmatrix} 9 & 12 \\ x_{31} & 0 \end{vmatrix}}{x_{31}} = \frac{12x_{31}}{x_{31}} = 12$$

So, the first column

$$\begin{bmatrix} +1 \\ +9 \\ x_{31} \\ +12 \end{bmatrix}$$

For the system to be stable $x_{31} > 0$

$$x_{31} = \frac{9K - 12}{9} > 0$$

Therefore,

$$9K - 12 > 0$$

or

$$K > \frac{12}{9} = 1\frac{1}{3}$$

The range of stability is then

$$1\frac{1}{3} < K < \infty$$

Any value of K outside of this range produces an unstable system.

7.3.1 Advanced Analysis

There are four cases the Routh array analysis will fall into

1. Nominal

- This is the case for all previous Routh examples — basically, everything works as expected

2. Zero element in the first column

- Either from the characteristic polynomial or from a calculated element, some $x_{i1} = 0$, with another element in the row being non-zero

3. A row with all zeros

- There exists a row with all zeros for elements.

4. Indeterminate array

- There is an array that cannot be solved. The Routh array is **sufficient** to determine stability, but not **necessary**.

Nominal

The complete Routh array is formed and the first column may be analyzed directly.

Example — Nominal

Given: $A(s) = s^4 + 16s^3 + 70s^2 + 196s - 816$

Find: The system stability with the Routh Array

s^4	1	70	-816
s^3	16	196	0
s^2	x_{31}	x_{32}	x_{33}
s^1	x_{41}	x_{42}	
s^0	x_{51}		

$$x_{31} = \frac{- \begin{vmatrix} 1 & 70 \\ 16 & 196 \end{vmatrix}}{16} = \frac{924}{16} = 57.75$$

$$x_{32} = \frac{- \begin{vmatrix} 1 & -816 \\ 16 & 0 \end{vmatrix}}{16} = \frac{-13056}{16} = -816$$

$$x_{33} = \frac{- \begin{vmatrix} 1 & 0 \\ 16 & 0 \end{vmatrix}}{16} = \frac{0}{16} = 0$$

$$x_{41} = \frac{- \begin{vmatrix} 16 & 196 \\ 57.75 & -816 \end{vmatrix}}{57.75} = \frac{24375}{57.75} = 422.08$$

$$x_{42} = \frac{- \begin{vmatrix} 16 & 0 \\ 57.75 & 0 \end{vmatrix}}{57.75} = \frac{0}{57.75} = 0$$

$$x_{51} = \frac{- \begin{vmatrix} 57.75 & -816 \\ 422.08 & 0 \end{vmatrix}}{422.08} = \frac{-344420}{422.08} = -816$$

The final Routh Array

s^4	1	70	-816
s^3	16	196	0
s^2	57.75	-816	0
s^1	422.08	0	
s^0	-816		

There is a single sign change in the first column; therefore, the system has one unstable pole (among four poles) and is unstable.

Zero Element in the First Column

Initially, the Routh Array cannot be completed because a zero in the first column produces subsequent infinite elements.

Example — Zero Element

Given: $A(s) = s^5 + 2s^4 + 2s^3 + 4s^2 + 11s + 10$

Find: The system stability using the Routh Array

s^5	1	2	11
s^4	2	4	10
s^3	x_{31}	x_{32}	x_{33}
s^2	x_{41}	x_{42}	x_{43}
s^1	x_{51}	x_{52}	x_{53}
s^0	x_{61}	x_{62}	x_{63}

$$x_{31} = \frac{- \begin{vmatrix} 1 & 2 \\ 2 & 4 \end{vmatrix}}{2} = \frac{0}{2} = 0$$

We have encountered a zero in the first row; however, we need to calculate more elements to determine if the entire row is zero or just the first element.

$$x_{32} = \frac{- \begin{vmatrix} 1 & 11 \\ 2 & 10 \end{vmatrix}}{2} = \frac{12}{2} = 6$$

Since $x_{32} \neq 0$ we have a zero element in the first column only case. The process is simple, set the zero element equal to a variable, ϵ , complete the array and then evaluate the elements in the first column as $\epsilon \rightarrow 0^+$. Update the Routh Array with the ϵ variable and the x_{32} element.

s^5	1	2	11
s^4	2	4	10
s^3	ϵ	6	x_{33}
s^2	x_{41}	x_{42}	x_{43}
s^1	x_{51}	x_{52}	x_{53}
s^0	x_{61}	x_{32}	x_{63}

Moving on

$$x_{33} = 0$$

by observation

$$x_{41} = \frac{-\begin{vmatrix} 2 & 4 \\ \epsilon & 6 \end{vmatrix}}{\epsilon} = \frac{4\epsilon - 12}{\epsilon}$$

$$x_{42} = \frac{-\begin{vmatrix} 2 & 10 \\ \epsilon & 0 \end{vmatrix}}{\epsilon} = \frac{10\epsilon}{\epsilon} = 10$$

$$x_{43} = \frac{-\begin{vmatrix} 2 & 0 \\ \epsilon & 0 \end{vmatrix}}{\epsilon} = 0$$

$$x_{51} = \frac{-\begin{vmatrix} \epsilon & 6 \\ x_{41} & 10 \end{vmatrix}}{x_{41}} = \frac{6x_{41} - 10\epsilon}{x_{41}}$$

$$x_{52} = \frac{-\begin{vmatrix} \epsilon & 0 \\ x_{41} & 0 \end{vmatrix}}{x_{41}} = 0$$

$$x_{61} = \frac{-\begin{vmatrix} x_{41} & 10 \\ x_{51} & 0 \end{vmatrix}}{x_{51}} = 10$$

Write the Routh Array once more with the calculated element that don't require limit analysis to determine their values

s^5	1	2	11
s^4	2	4	10
s^3	ϵ	6	0
s^2	x_{41}	10	0
s^1	x_{51}	0	
s^0	10		

The final evaluation

$$x_{31} = \lim_{\epsilon \rightarrow 0^+} \epsilon = +0$$

Since ϵ approaches zero from the positive side x_{31} is a small positive number.

$$x_{41} = \lim_{\epsilon \rightarrow 0^+} \frac{4\epsilon - 12}{\epsilon}$$

Let's divide numerator and denominator by ϵ and evaluate the limit

$$x_{41} = \lim_{\epsilon \rightarrow 0^+} \frac{4 - \frac{12}{\epsilon}}{1} = 4 - \frac{12}{0} \rightarrow -\infty$$

Then, finally

$$x_{51} \frac{6x_{41} - 10\epsilon}{x_{41}}$$

Divide numerator and denominator by x_{41} then evaluate

$$x_{51} \frac{6 - \frac{10\epsilon}{x_{41}}}{1} = 6 - \frac{10\epsilon}{x_{41}} = 6$$

the complete Routh Array

s^5	+1	2	11
s^4	+2	4	10
s^3	+0	6	0
s^2	$-\infty$	10	0
s^1	+6	0	
s^0	+10		

The first column

$$\begin{bmatrix} +1 \\ +2 \\ +0 \\ -\infty \\ +6 \\ +10 \end{bmatrix}$$

has two sign changes; therefore, the system is unstable with two poles (of five) in the unstable region.

7.3.2 Row of Zeros

This case occurs when there are symmetric poles about the origin in the S-plane. The approach is to replace the elements in the zero row with coefficients from an **auxiliary** equation. The auxiliary equation is derived from element in the row above the zero row.

Example — Row of Zeros

Given: $A(s) = s^4 + 12s^3 + 36s^2 + 48s + 128$

Find: The system stability with the Routh Array

s^4	1	36	128
s^3	12	48	0
s^2	32	128	0
s^1	0	0	
s^0			

Analysis stops after calculating row s^1 because all elements are zero. Look to the previous row for the coefficient of an auxiliary equation. Additionally, the row label will determine the order of the equation.

$$P(s) = 32s^2 + 128$$

Note: the coefficient for $P(s)$ (the auxiliary equation) are directly from the elements in the previous row, plus the highest order of $P(s)$ is the row label. The next element is blueuced by two.

Now, to determine the replacement element, take the firs derivative of $P(s)$

$$\frac{dP(s)}{ds} = 64s + 0$$

so, $x_{41} = 64$ and $x_{42} = 0$. We replace the zeros in the array and continue with completing the array.

s^4	1	36	128
s^3	12	48	0
s^2	32	128	0
s^1	64	0	
s^0			

which leaves on x_{51} to calculate

$$x_{51} = -\frac{\begin{vmatrix} 32 & 128 \\ 64 & 0 \end{vmatrix}}{64} = 128$$

The final Routh Array

s^4	1	36	128
s^3	12	48	0
s^2	32	128	0
s^1	64	0	
s^0	128		

with the first column

$$\begin{bmatrix} +1 \\ +12 \\ +32 \\ +64 \\ +128 \end{bmatrix}$$

Therefore, the system is stable with no sign changes in the first column. There is additional information we can learn from the auxiliary equation.

Solve for the roots of $P(s)$ to determine the symmetric poles.

$$P(s) = 32s^2 + 128 = 0$$

$$32s^2 = -128$$

$$s^2 = \pm \sqrt{-\frac{128}{32}} = \pm \sqrt{-4} = \pm 2j$$

Of the four poles in the charactering polynomial two are in the imaginary axis at $\pm 2j$. The remaining two poles are in the stable region.

7.3.3 Indeterminate Array

As previously stated, the Routh Array is sufficient to prove stability, but not necessary. If we encounter a characteristic polynomial that we are unable to produce a complete array, we cannot use the failure to the Routh analysis to make a claim as to the stability of the system.

7.3.4 Solutions with Multiple Approaches

Example — Special Case

Given: $A(s) = s^3 - 9s$

Find: The system stability with the Routh Array

For this problem, we have three options

Option 1 — Direct with Auxiliary Equation

Calculate the array of $A(s) = s^3 + 0s^2 - 9s + 0$

Determine the elements in the array as before

s^3	1	-9
s^2	0	0
s^1		
s^0		

At s^2 we have encountered a row of all zeros, so we to calculate the auxiliary equation and its derivative

$$P(s) = s^3 - 9s$$

and

$$\frac{dP(s)}{ds} = 3s^2 - 9$$

Now

s^3	1	-9
s^2	3	-9
s^1	x_{31}	0
s^0	x_{41}	

$$x_{31} = \frac{-\begin{vmatrix} 1 & -9 \\ 3 & -9 \end{vmatrix}}{3} = \frac{-27 + 9}{3} = -6$$

$$x_{41} = \frac{-\begin{vmatrix} 3 & -9 \\ -6 & 0 \end{vmatrix}}{-6} = -9$$

Then evaluation of the first column of the final array

s^3	1	-9
s^2	3	-9
s^1	-6	0
s^0	-9	

reveals one unstable pole in the system.

Option 2 — Direct with Modified Characteristic Polynomial & Limit Analysis

Given: $A(s) = s^3 - 9s$

Factor the common s from the characteristic polynomial such that

$$A(s) = s(s^2 - 9)$$

and apply the Routh array only to the expression within the parentheses

$$\hat{A}(s) = s^2 - 9 = s^2 + 0s - 9$$

s^2	1	-9
s^1	0	0
s^0		

in the s^1 row we have all zeros, so we find the auxiliary equation

$$P(s) = s^2 - 9$$

and its derivative

$$\frac{dP(s)}{ds} = 2s + 0$$

Replacing the zero row with the auxiliary equation coefficients produces a final array

s^2	1	-9
s^1	2	0
s^0	-9	

Evaluating the first column, we find the system $\hat{A}(s)$ is unstable. Since $A(s) = s(\hat{A}(s))$ it is unstable as well. Additionally, we know the original characteristic equation has one marginally stable pole at the origin. Our analysis tells us the original characteristic equation has one pole at the origin, and two pole symmetric about the origin. The symmetric pole can be found by solving $P(s) = 0$ for s . The solution is $s = \pm 3$.

There exists some special occurrences where either the limit analysis (zero in the first row) and the auxiliary equation analysis (a row with all zeros) can be used to evaluate the stability. Moreover, as seen above, we can perform the analysis on modified versions of the characteristic equation. The trick is to be consistent, follow the process, and remember what you've done.

Chapter 8

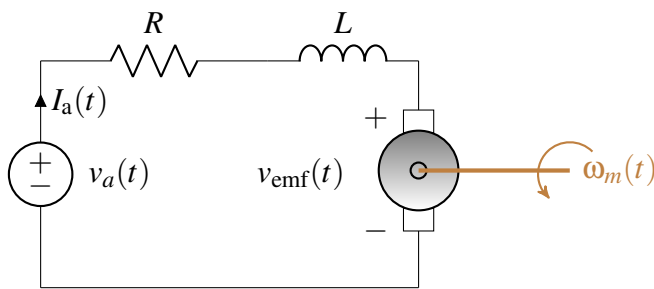
DC Servo Motor

Out of the box a DC Servo motor has a voltage input to the armature, which is $V_a(t)$ and produces an output rotational rate of the motor shaft, namely, $\omega_m(t)$.

Practical applications include using the shaft rotation to:

1. Propel a cart forward by mechanically connecting the shaft to a drivetrain
2. Provide thrust for an aerial vehicle by mechanically connecting the shaft to a fan
3. Produce mechanical energy for conveyor belts

To construct a block diagram and transfer function for the system, we first begin with a circuit diagram for a DC servo



KVL about the circuit in the Laplace domain

$$-V_a(s) + RI_a(s) + LsI_a(s) + V_{\text{emf}}(s) = 0$$

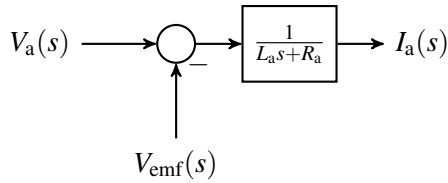
Rewritten so the output is $I_a(s)$

$$(Ls + R)I_a(s) = V_a(s) - V_{\text{emf}}(s)$$

or

$$I_a(s) = \frac{V_a(s) - V_{\text{emf}}(s)}{Ls + R}$$

Let's draw a block diagram for this equation



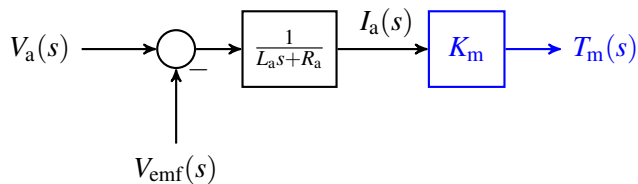
Now, the current $I_a(s)$ through the armature produces a motor torque. The ratio of torque to current is a constant, K_m .

$$K_m = \frac{T_m(s)}{I_a(s)}$$

or

$$T_m(s) = K_m I_a(s)$$

Add this to the equation to the block diagram we are building



For the torque produced in the motor to rotate the rod it must overcome inertia and friction. The inertia is that of the motor itself and the load the motor is driving. Fortunately, the total inertia is simply the summation of the motor and the load

$$J_t = J_m + J_L$$

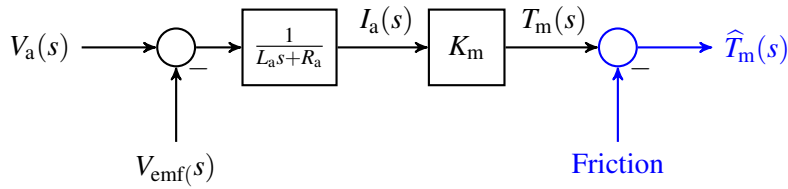
Notes on inertia:

1. For most applications the load inertia is an order of magnitude or greater than the motor inertia.
2. Inertia is the resistance of any physical object to changes in velocity.

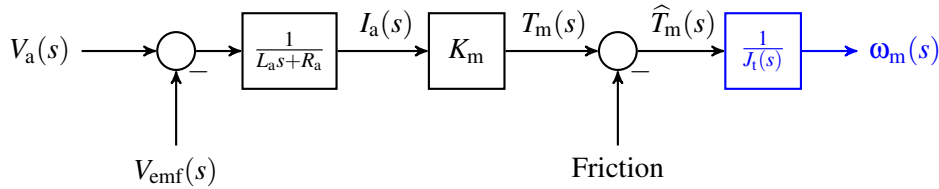
Friction is a fundamental force that resists the motion of an object sliding or rolling over another. The study of friction is in depth and is only touched on here. There are four types of friction

1. Static — occurs between surfaces that are not moving
2. Kinetic — occurs between surfaces that are sliding against one another
3. Rolling — occurs when an object rolls over a surface
4. Fluid — occurs when an object moves through a fluid

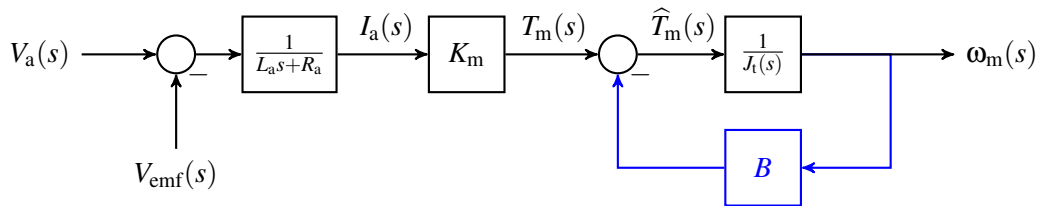
For many controls approaches only kinetic or rolling friction is considered. The frictional coefficient is considered constant, B .



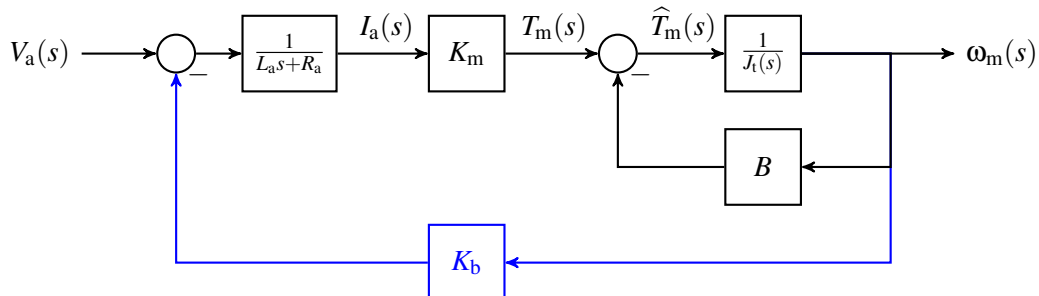
Now, let's add the inertial component to the diagram



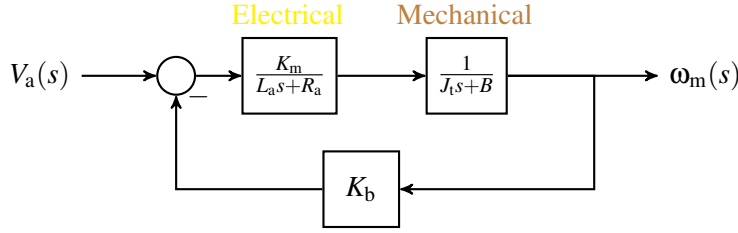
Now, we need to complete the feedback paths. The output is the shaft rotational rate, the rotation produces either kinetic or rolling friction by causing surfaces to slide or roll across one another



Not only does the shaft rotational rate produce the frictional feedback, but it also produces the V_{emf} — function of the physical properties of the motor (e.g. magnet strength, wiring size and proximity to the magnets). The coefficient K_b a measured value and provided by the motor manufacturer.



Simplify the diagram by eliminating the friction feedback and combining the first two transfer functions. This produces a block diagram with the electrical components in one block and the mechanical components in the other



where

1. L_a is the armature inductance
2. R_a is the armature resistance
3. J_t is the total inertia of the system, which is motor inertia plus load inertia, $J_t = J_m + J_L$
4. B is the frictional coefficient
5. K_b is the EMF coefficient
6. $V_a(s)$ is the armature input
7. $\omega_m(s)$ is the shaft rotational rate

Solving the block diagram produces the motor transfer function

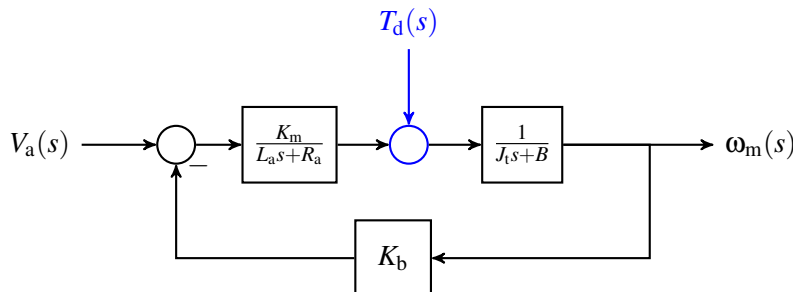
$$\frac{\omega_m(s)}{V_a(s)} = G_m(s) = \frac{K_m}{L_a J_t s^2 + (L_a B + R_a J_t) s + R_a B + K_m K_b} \quad (8.1)$$

Often, the second order model is more precise than is needed in the design, we reduce the complexity of the model by setting $L_a = 0$, which produces a **first order** DC servo model

$$G_m(s) = \frac{K_m}{R_a J_t s + R_a B + K_m K_b} \quad (8.2)$$

8.1 External Disturbances

For practical applications, we often must consider external disturbances. Most devices that move will encounter environmental effects in the form of fluid flow (wind, water) and/or environmental variations (uneven ground, wheel slippage). The best practice is to consider the disturbance as an additional input into the nominal system. Systems with multi-input / multi-output use the principle of super-position to determine the transfer function.



First, the disturbance input is set to zero, $T_d(s) = 0$, and the output with respect to the input (wrt) is the same as above in (8.1). Then, we find $\frac{\omega_m(s)}{T_d(s)}$ by setting $V_a(s) = 0$ and resolving the block diagram.

$$\begin{aligned}
\frac{\omega_m(s)}{T_d(s)} &= \frac{\frac{1}{J_t s + B}}{1 + \frac{K_m K_b}{(L_a s + R_a)(J_t s + B)}} \\
&= \frac{\frac{1}{J_t s + B}}{\frac{(L_a s + R_a)(J_t s + B) + K_m K_b}{(L_a s + R_a)(J_t s + B)}} \\
&= \frac{L_a s + R_a}{(L_a s + R_a)(J_t s + B) + K_m K_b} \\
\frac{\omega_m(s)}{T_d(s)} = \hat{G}_m(s) &= \frac{L_a s + R_a}{L_a J_t s^2 + (L_a B + R_a J_t)s + R_a B + K_m K_b}
\end{aligned} \tag{8.3}$$

Then, the output of the disturbed system is

$$\begin{aligned}
\omega_m(s) &= G_m V_a(s) + \hat{G}_m T_d(s) \\
\omega_m(s) &= \frac{K_m V_a(s)}{L_a J_t s^2 + (L_a B + R_a J_t)s + R_a B + K_m K_b} + \frac{(L_a s + R_a) T_d(s)}{L_a J_t s^2 + (L_a B + R_a J_t)s + R_a B + K_m K_b}
\end{aligned} \tag{8.4}$$

Note: the characteristic polynomial is the same for $G_m(s)$ and $\hat{G}_m(s)$.

As above, If we want to reduce the model to a first order, assume the armature inductance is negligible $L_a = 0$

$$\omega_m(s) = \frac{K_m V_a(s) + R_a T_d(s)}{R_a J_t s + R_a B + K_m K_b} \tag{8.5}$$

For either model, **basic** disturbance rejection occurs when $R_a \rightarrow 0$ and $K_m \gg 0$. However, there are times that selecting a motor with ideal parameters is not enough to reject the external disturbance; when that is the case, we must design a disturbance rejecting controller — usually a PID.

Example — Basic Disturbance Rejection

Given: A DC Servo motor system with a large inductance — too big to be ignored; determine the disturbance rejection for two different DC motors.

Motor #1 has the parameters

Parameter	Value
L_a	0.005
R_a	5.3
K_b	0.15
K_m	2.6
B	0.075
J_t	11
B	0.075

which produces a **normalized** output from (8.4) as

$$\omega_m(s) = \frac{47.27V_a(s)}{s^2 + 1060s + 14.32} + \frac{(0.09s + 96.36)T_d(s)}{s^2 + 1060s + 14.32}$$

assuming both the input and the disturbance source are bounded (for the disturbance, think of a constant headwind for a rolling cart) then

$$\begin{aligned}\omega_m(t \rightarrow \infty) &= \frac{47.27}{14.32} \cdot \lim_{s \rightarrow 0} \{sV_a(s)\} + \frac{96.36}{14.32} \cdot \lim_{s \rightarrow 0} \{sT_d(s)\} \\ &= 3.3 \cdot \lim_{s \rightarrow 0} \{sV_a(s)\} + 6.73 \cdot \lim_{s \rightarrow 0} \{sT_d(s)\}\end{aligned}$$

For complete analysis, the two input expressions need to be evaluated (FVT), but even with this incomplete setup, we can see the disturbance input is scaled significantly more than the armature voltage input. This may produce an output overly influenced by the disturbance.

On to **motor #2** which has a lower armature resistance and a larger motor torque gain than the first motor

Parameter	Value
L_a	0.005
R_a	2.3
K_b	0.15
K_m	8.5
J_t	11
B	0.075

then

$$\omega_m(s) = \frac{152V_a(s)}{s^2 + 460s + 26.32} + \frac{(0.09s + 41.82)T_d(s)}{s^2 + 460s + 26.32}$$

assume bounded signals for the inputs

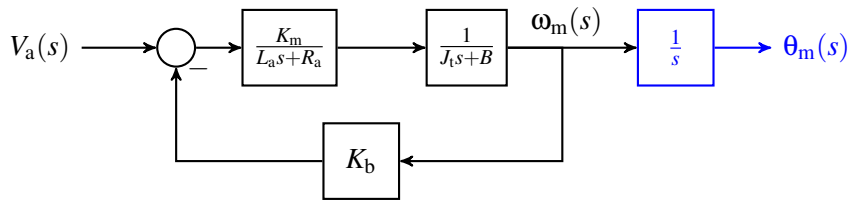
$$\begin{aligned}\omega_m(t \rightarrow \infty) &= \frac{152}{26.32} \cdot \lim_{s \rightarrow 0} \{sV_a(s)\} + \frac{41.82}{26.32} \cdot \lim_{s \rightarrow 0} \{sT_d(s)\} \\ &= 5.78 \cdot \lim_{s \rightarrow 0} \{sV_a(s)\} + 1.58 \cdot \lim_{s \rightarrow 0} \{sT_d(s)\}\end{aligned}$$

When comparing the results to the previous motor, the armature voltage input is upscaled significantly more than the disturbance input. It is likely that the output is more a function of the armature

voltage than the disturbance; however, even with this example the disturbance may contribute more to the output than is desired — or acceptable. For systems that require more disturbance rejection than what is provided by motor selection, we can implement a controller to further suppress the noise.

8.2 Position Tracking

Often the desired output of a DC Servo system is shaft position (degrees or radians) and not shaft rotational rate — especially in robotic and aerospace applications. Mathematically, all one must do is take the output $\omega_m(s)$ and pass it into an integrator. Then, the output of the integrator is the shaft position, $\theta_m(t)$ or $\theta_m(s)$

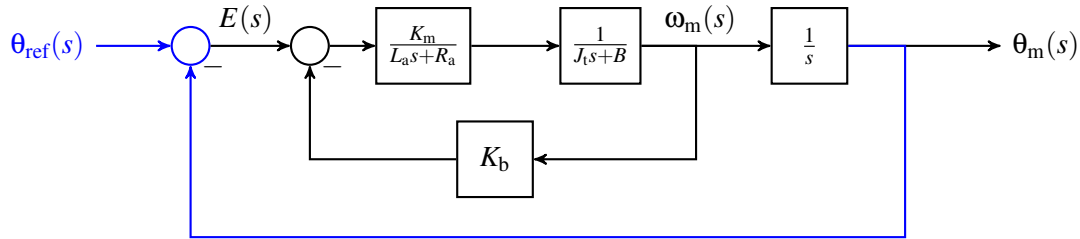


Now, considering the block diagram, the transfer function is

$$\begin{aligned} \frac{\theta_m(s)}{V_a(s)} &= \frac{K_m}{s \left(L_a J_t s^2 + (L_a B + R_a J_t) s + R_a B + K_m K_b \right)} \\ &= \frac{K_m}{L_a J_t s^3 + (L_a B + R_a J_t) s^2 + (R_a B + K_m K_b) s} \end{aligned}$$

We find a system with a pole at the origin (0,0) and two poles in the stable region. Having a pole at the origin produces a marginally stable system that depends in the input signal for BIBO stability.

Normally, we want to direct the output to a given position. Considering a robot arm, we need a joint to move to a set angle and stay there. How do we resolve our desires, verses the model? Simple, apply feedback — negative feedback.



The internal signal $E(s)$ is the difference between the reference input and the actual output. A true error signal that is driving the armature voltage.

The open-loop transfer function of the block diagram is determined to be $\frac{\theta_m(s)}{\theta_{ref}(s)}$ where $\theta_{ref}(s)$ is an input voltage comparable to a shaft angle. The transfer function is found by analyzing the block diagram

$$\frac{\theta_m(s)}{\theta_{\text{ref}}(s)} = \frac{K_m}{L_a J_t s^3 + (L_a B + R_a J_t) s^2 + (R_a B + K_m K_b) s + K_m} \quad (8.6)$$

Now, with (8.6) we have a complete third order model where the input is a *reference* angle and the output is a shaft angle, $\theta_m(s)$. Just as above, we may reduce the order of the model by assuming the armature inductance is zero, $L_a = 0$. When this is the case, the model reduces to a second order representation

$$\frac{\theta_m(s)}{\theta_{\text{ref}}(s)} = \frac{K_m}{R_a J_t s^2 + (R_a B + K_m K_b) s + K_m} \quad (8.7)$$

FVT for the position tracking system, regardless of the model order, when the input is a setpoint

$$\theta_m(t \rightarrow \infty) = \lim_{s \rightarrow 0} s \theta_m(s) = \frac{K_m}{K_m} \rightarrow 1$$

Thus, for a setpoint (unit step) input the output matches the input. The time it takes for the shaft to reach and settle to the final value is determined by the time constants (poles) of the characteristic polynomial. Also, before trusting the FVT we need to verify BIBO stability.

Note: from a practical perspective, we do not use an integrator in the system. The shaft position is achieved by attaching an encoder to the shaft that provides the position. The math model requires an integrator to correctly represent the motor with a position encoder, but we do not need to design an integrator to achieve shaft position output.

Example — Position Tracking

Take the general transfer function (8.6); then using MATLAB apply a setpoint input and observe the output when

Parameter	Value
L_a	0.0001
R_a	1.3
K_b	0.25
K_m	4
J_t	1
B	0.25

Then, the transfer function is

$$\frac{\theta_m(s)}{\theta_{\text{ref}}(s)} = \frac{4}{0.0001s^3 + 1.3s^2 + 1.325s + 4}$$

The poles to the characteristic equation are $-13000, -0.51 \pm 1.68j$ which confirms the system is stable with a complex pair of poles and a real pole. Note the two complex poles are near the $j\omega$ -axis and the real pole is very far to the left.

The two complex conjugate pair poles with real parts at -0.51 are the **dominate** poles in the system and the pole at -13000 is a **transient** pole that only contributes to the output for about five time constants. Dominate poles are the primary driver in the signal response to an input; while the transient poles only contribute the the output for a very short period of timer early in the response.

Let's consider the time constant of the real pole

$$\tau_r = \frac{1}{13000} = 0.000077$$

while the two complex poles have a time constant

$$\tau_c = \frac{1}{0.51} = 1.96$$

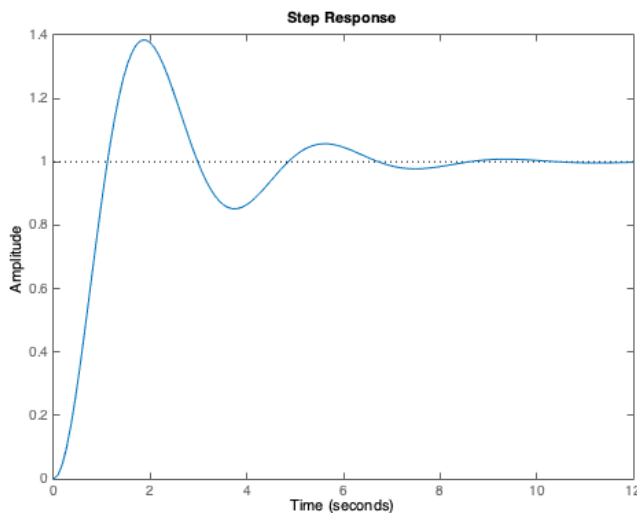
The **rule of thumb for dynamic response to a pole**: the dynamic contribution to the signal only lasts for about five time constants of a stable expression. This means the real pole far to the left is a transient signal and contributes to the dynamic output response for

$$5 \cdot \tau_r = 0.00038\text{sec}$$

while the dominant poles near the origin contribute to the dynamic output for

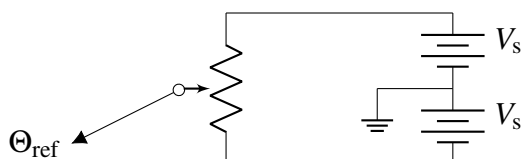
$$5 \cdot \tau_c = 9.8\text{sec}$$

So, the contribution from the transient pole is inconsequential to the dynamic response of the system, as seen here



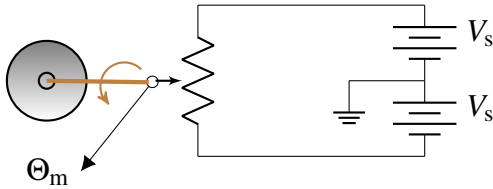
8.2.1 Rotary Position Tracking System

We use two power supplies connected in tandem and a potentiometer to produce the reference voltage



The user rotates the rheostat to produce a voltage between $+V_s$ and $-V_s$ by tuning the reference to the desired voltage.

Then, the position of the motor shaft is determined in a similar manner, but the potentiometer is directly connected to the motor shaft so that when the shaft rotates, the wiper on the potentiometer moves along the coils of the rheostat producing a voltage for the shaft position.

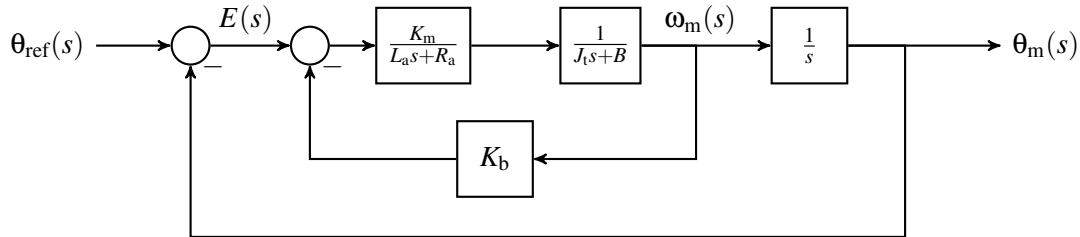


The potentiometer at the motor determines the range of rotation. A single-turn pot that has a 270° total angle of deflection will limit the movement of the shaft to 135° for $+V_s$ and -135° for $-V_s$. The motion between the limits will be linear with 0° at zero volts.

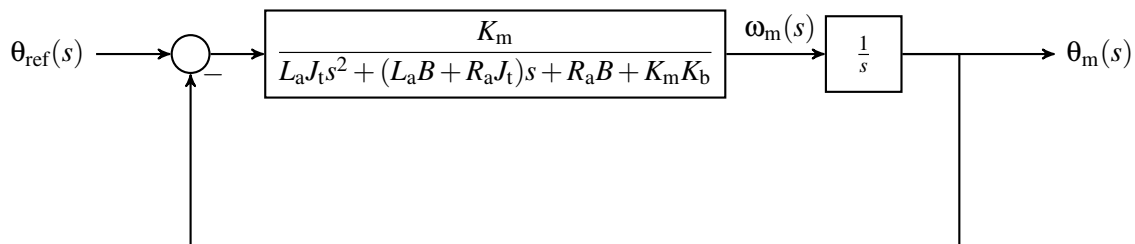
8.2.2 Tunable Gains with Tachometer Feedback

Many motors with embedded controllers have readily available tunable gain parameters — a proportional gain in the forward path with $K > 0$ and an internal feedback gain parameter called tachometer feedback $K_f \geq 0$.

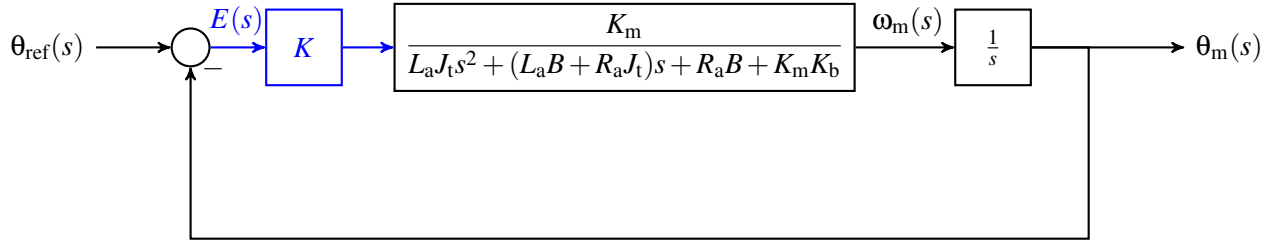
Let's begin with the expanded model



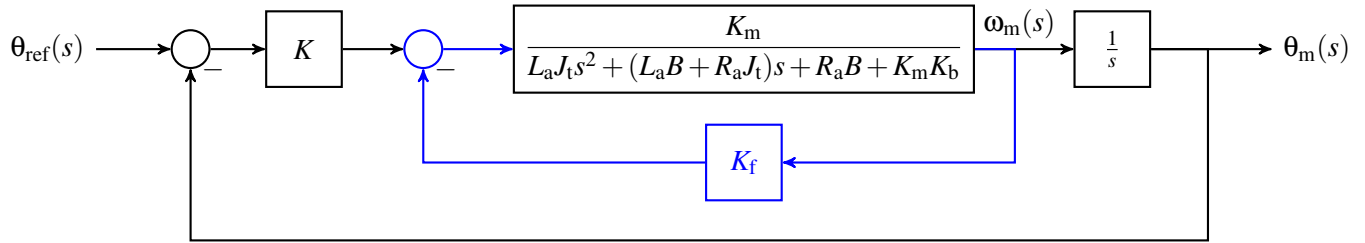
now, simplify the inner feedback loop



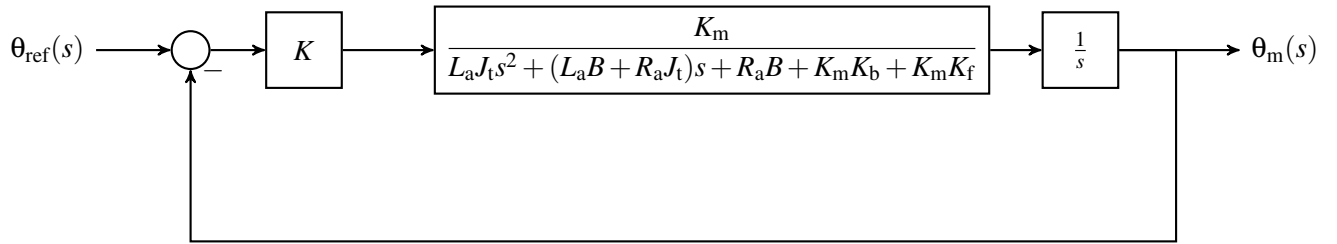
The proportional gain is between the summation junction and the first motor block so that the gain K modifies the error signal.



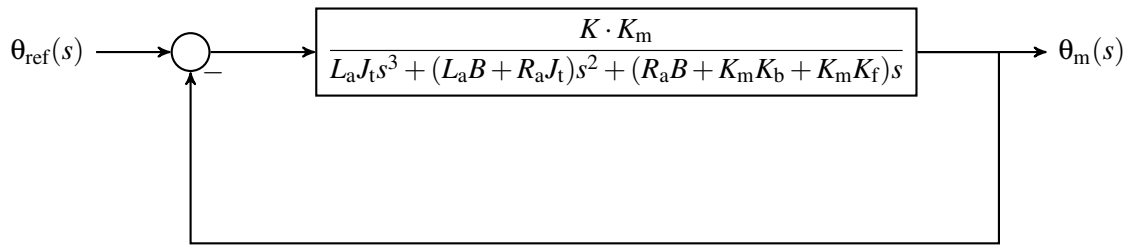
The tachometer feedback is about the motor transfer function



Now that we have a complete block diagram, let's simplify to a transfer function



Combine the three cascaded blocks in the forward path



and finally eliminate the remaining feedback loop we have a transfer function with the two tunable parameters — the proportional gain K and the tachometer feedback K_f .

$$\frac{\theta_m(s)}{\theta_{ref}(s)} = \frac{K \cdot K_m}{L_a J_t s^3 + (L_a B + R_a J_t)s^2 + (R_a B + K_m K_b + K_m K_f)s + K \cdot K_m} \quad (8.8)$$

Additionally, we can develop a simpler model by setting the armature inductance to zero

$$\frac{\theta_m(s)}{\theta_{ref}(s)} = \frac{K \cdot K_m}{R_a J_t s^2 + (R_a B + K_m K_b + K_m K_f)s + K \cdot K_m} \quad (8.9)$$

From here, we can use all of the other techniques taught in class to analyze the system and tune the response with the variables at our disposal.

Example — Basic Position Tracking with K & K_f

Given the same motor from a previous example

Parameter	Value
L_a	0.005
R_a	2.3
K_b	0.15
K_m	8.5
J_t	11
B	0.075

we find the third order transfer function

$$\frac{\theta_m(s)}{\theta_{ref}(s)} = \frac{8.5K}{0.06s^3 + 25.3s^2 + (1.45 + 8.5K_f)s + 8.5K}$$

Now, let us take a look at the poles for different tunings

K	K_f	Poles
1	0	$-459.95, -0.03 \pm 0.58j$
10	0	$-459.96, -0.03 \pm 1.83j$
10	1	$-459.62, -0.19 \pm 1.82j$
10	5	$-458.27, -0.87 \pm 1.62j$
10	10	$-456.57, -1.72 \pm 1.66j$
3	5	$-458.27, -0.87 \pm 0.50j$

One can look at the poles for the various tunings and empirically choose $K = 3$ and $K_f = 5$ as the best possible tuning on the list. Simply because the dominate poles (the ones near the vertical axis) have a greater ratio of real to imaginary magnitudes. This means the output oscillations are less than the other choices. Notice the transient pole is mostly immune to changes in the tunings and is far from the vertical axis. This is the pole that is eliminated when reducing the order of the model by setting the armature inductance to zero. Let's take a brief look at the poles for the reduced order model

K	K_f	Poles
10	1	$-0.19 \pm 1.82j$
10	5	$-0.87 \pm 1.61j$
3	5	$-0.87 \pm 0.50j$

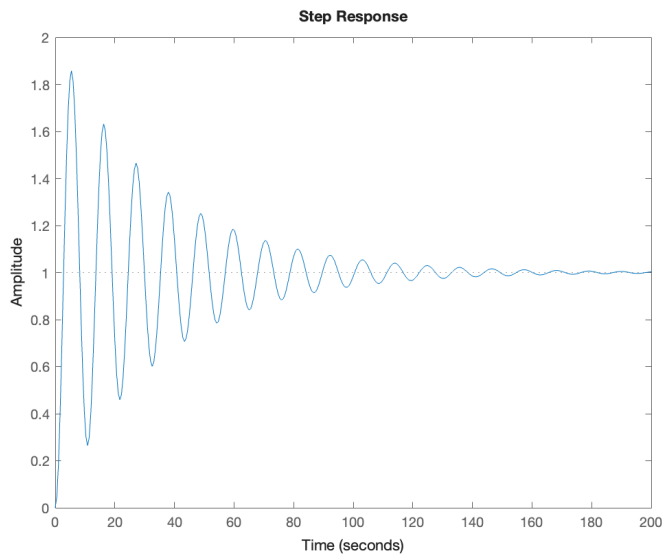
Rounding to the hundredths place, the results for the dominant poles are the same.

There are times and systems where precision is needed and the full-model is required. Other times, the reduced order model is sufficient.

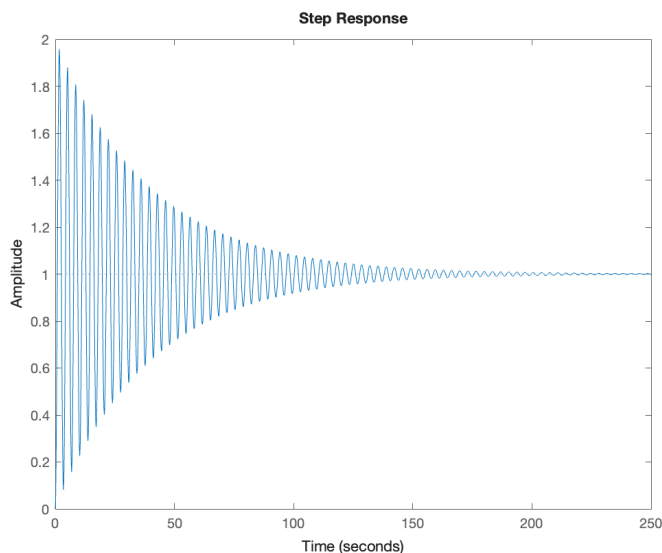
Let's take a look at the full-order model output response to a step input using MATLAB and compare to the poles. How will the selection of $K = 3$, K_f hold up to the modeling results?

Output response to a setpoint input for the various tunings

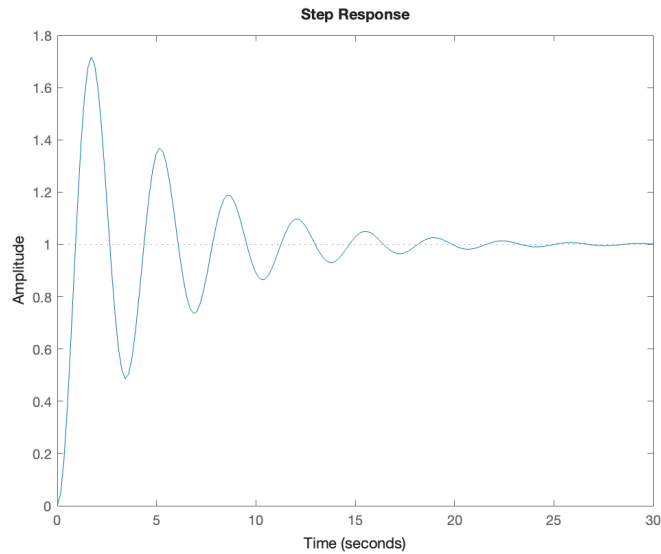
Set the parameters to $K = 1$, $K_f = 0$ and run the model. We see significant output oscillations and a settling time of about 180 seconds.



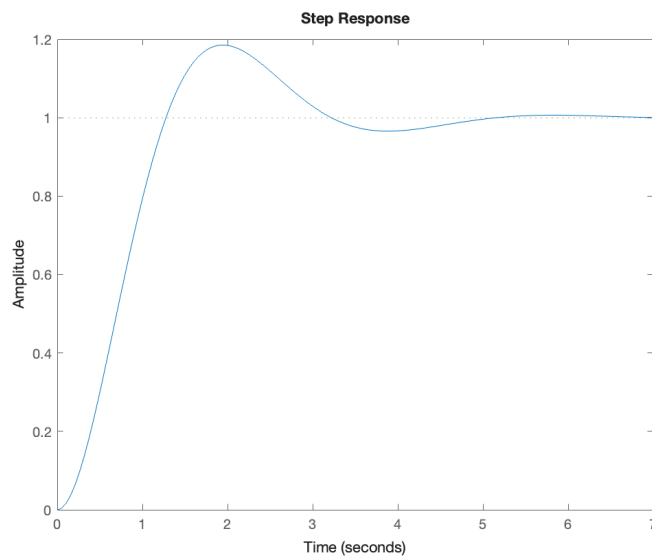
Now, set the parameters to $K = 10$, $K_f = 0$ and repeat the run. Now, the oscillations are greater in magnitude and frequency than before and the sell looks to have increased slightly — about 200 seconds.



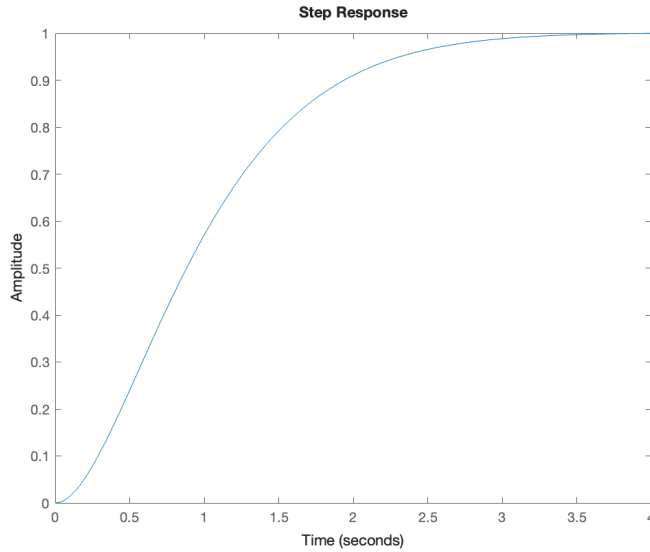
$K = 10$, $K_f = 1$ and repeat the run. Now, with the addition of tachometer feedback, the output oscillations are significantly reduced in frequency and the settling time is greatly reduced — to about 25 seconds.



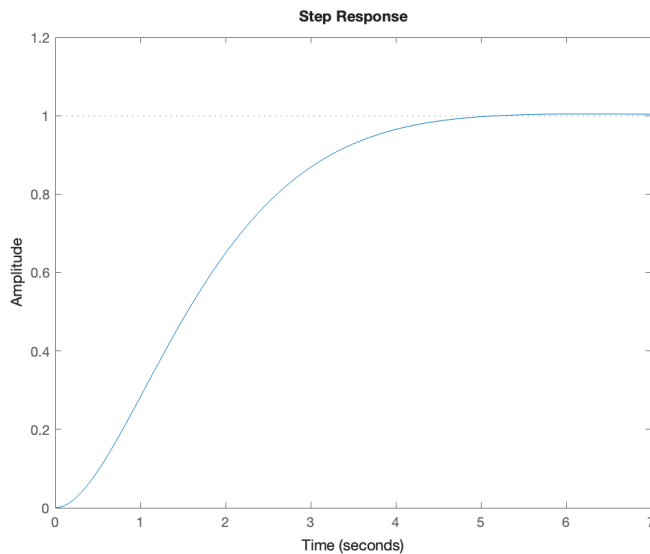
$K = 10$, $K_f = 5$. Now, the oscillations are still present, but there are only two peaks before steady state at about six seconds.



$K = 10$, $K_f = 10$ The result here appears to have no oscillations and goes to steady state in about 3.5 seconds.



$K = 3, K_f = 5$. The final result produces an output without oscillation and a settling time of about five seconds.



When considering the output response to a setpoint input only the last two tunings produce an output response with minimal oscillations. Setting $K = K_f = 10$ produces the shortest settling time; about four seconds. While $K = 3$ and $K_f = 5$ produces a settling time slightly longer; about four and a half seconds. Either produce a nice shaped response at the output. Generally, selecting the results with the smaller gain is preferable just to minimize the energy necessary to produce the desired result — think of minimizing the armature current when possible.

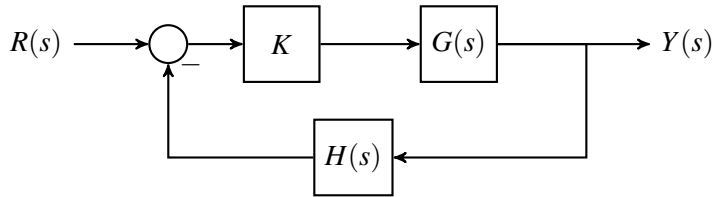
Chapter 9

Root Locus Analysis

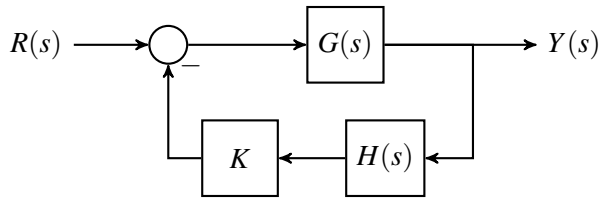
The Root Locus technique is used to place the poles of a negative feedback system by investigating the loop-gain zeros and poles of the system to be controlled with a single tunable parameter

$$0 \leq K < \infty$$

The process works when the controller is in the feedforward path, **Case 1**



or in the feedback path, **Case 2**



Note: The tedious nature of the algebraic manipulation in this section is better served by dropping the Laplace domain identifier variable “(s)” after the transfer functions and signals. Assume,

$$G = \frac{N_g}{D_g}$$

and

$$H = \frac{N_h}{D_h}$$

Case 1

K in the forward path, the open-loop transfer function

$$\frac{Y}{R} = \frac{K \cdot N_g \cdot D_h}{D_g \cdot D_h + K \cdot N_g \cdot N_h} \quad (9.1)$$

Case 2

K in the feedback loop, The open-loop transfer function

$$\frac{Y}{R} = \frac{N_g \cdot D_h}{D_g \cdot D_h + K \cdot N_g \cdot N_h} \quad (9.2)$$

The characteristic polynomial for either case is the same

$$A = D_g D_h + K N_g N_h \quad (9.3)$$

and the loop-gain for each representation is the same

$$L = KGH = \frac{K \cdot N_g N_h}{D_g D_h} = \frac{K \cdot N_L}{D_L} \quad (9.4)$$

where N_L is the loop-gain numerator and D_L is the loop-gain denominator.

Equation 9.3 is rewritten with the loop-gain expressions

$$A = D_L + K \cdot N_L \quad (9.5)$$

We have is a system where the characteristic equation is a function of the loop-gain zeros, loop-gain poles, and the variable parameter K .

From 9.5, when K is zero

$$A = D_L$$

This is a profound result: for **small values of K the solutions to the characteristic equation are near the loop-gain poles.**

To explore the solutions where $K \rightarrow \infty$, we first take 9.1 and divide the numerator and denominator by the tunable parameter K

$$\frac{Y}{R} = \frac{\frac{N_g \cdot D_h}{K}}{\frac{D_g \cdot D_h}{K} + N_g \cdot N_h} \quad (9.6)$$

In this form, the characteristic polynomial

$$A = \frac{D_g \cdot D_h}{K} + N_g \cdot N_h$$

Now, as $K \rightarrow \infty$ the characteristic polynomial $A \rightarrow N_g \cdot N_h = N_L$. Again, a profound result: for **very large values of K , the solutions to the characteristic equation are near the loop-gain zeros.** The solutions to the characteristic equation move from the loop-gain poles to the loop-gain zeros as K goes from zero to infinity.

9.1 Guidelines for Rapid Plotting

When plotting, the finite loop-gain poles are identified as “x” and the finite loop-gain zeros as “o”. The plot is in the S-Plane, so the horizontal axis is the real axis, the vertical axis is the imaginary axis.

1. Starting points ($K = 0$) are the loop-gain poles. Ending points ($K \rightarrow \infty$) are the loop-gain zeros.
2. The solutions to the characteristic equation ($A = 0$) are **continuous paths from the starting points to the ending points** as K goes from $0 \rightarrow \infty$. The continuous paths are loci (plural) or locus (singular).
3. The loci (or locus) on the real axis only exist to the left of an odd number of zero/pole crossings — that’s just how the math works.
4. A locus on the real axis only departs if it collides with another locus. The collision occurs at a breakaway point, BA. **Corollary:** a locus in the complex plane only settles on the real axis when it collides with another locus; at a break-in point, BI. Collectively, we call these **inflection points**.
5. As $K \rightarrow \infty$ each locus must settle to a loop-gain zero. However, if there are more loop-gain poles than loop-gain zeros some loci will settle to **virtual** zeros at infinity. Any locus without a loop-gain zero to settle to will approach infinity asymptotically along a straight line. The origin of the line is on the real axis at the “center of gravity” which is found using

$$\sigma_i = \frac{LG_p - LG_z}{n - m} \quad (9.7)$$

where $LG_p = \sum$ loop-gain poles, $LG_z = \sum$ loop-gain zeros, n = number of loop-gain poles, and m = number of finite loop-gain zeros. *Note:* since all complex poles and zeros have a complex conjugate pair, only the real parts of the poles and zeros need to be considered when calculating the center of gravity.

The straight line angle off the real axis is

$$\Theta_k = \frac{(2k+1)\Pi}{|n-m|} \quad (9.8)$$

where k is an integer sequence

$$k = \{0, 1, 2, \dots, |n-m| - 1\}$$

6. To find the inflection points where loci collide on the real axis, set the first derivative of the loop-gain to zero and solve for s .

$$\frac{dL}{ds} = 0 \quad (9.9)$$

The solutions will contain the inflection points, however other invalid points may also be present in the solution. One must use the Root Locus sketch to determine the true inflection points.

The gain at the inflection points is calculated by

$$K_{BX} = \frac{-D_L}{N_L}, s = BX \quad (9.10)$$

or, if $N'_L \neq 0$

$$K_{BX} = \frac{-D'_L}{N'_L}, s = BX$$

where BX is either BA or BI.

Inflection points — Breakaway and Break-in

Equation (9.9) solves for the inflection points because the first derivative of the characteristic polynomial is zero only for repeated roots

The characteristic equation

$$A = D_L + KN_L = 0$$

so the first derivative

$$\frac{dA}{ds} = D'_L + KN'_L = 0$$

only when $A = 0$ has repeated roots.

Solving for K, we get

$$K = \frac{-D'_L}{N'_L}$$

which we substitute back into the characteristic equation

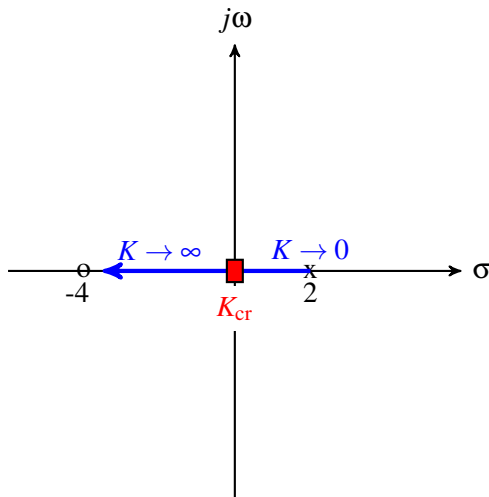
$$A = D_L + \left(\frac{-D'_L}{N'_L} \right) N_L = 0$$

and simplify to

$$A = \frac{D_L N'_L - D'_L N_L}{N'_L} = 0$$

which is $\frac{dL}{ds} = 0$. Once the inflection points are identified, the gain at the inflection points may be easily calculated.

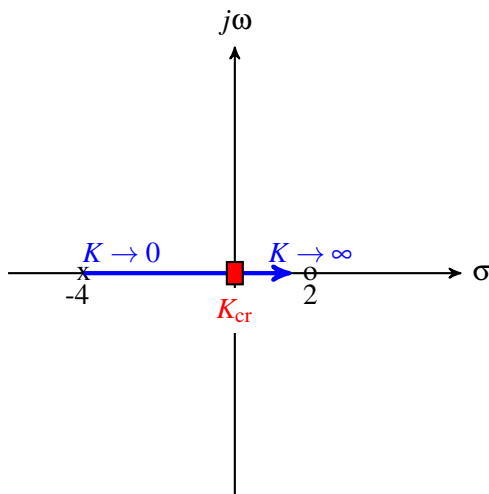
Example — Real Root and Real Zero



From the sketch:

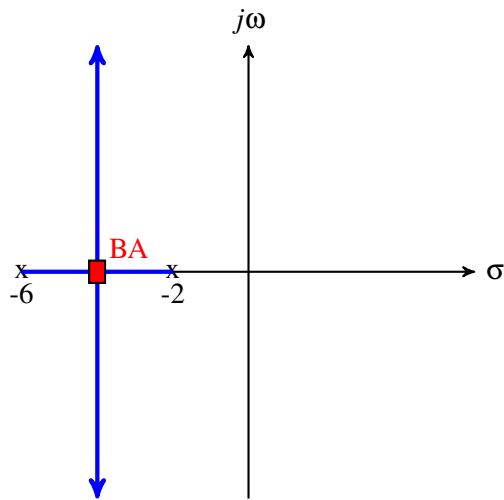
1. When the gain is small, the solution to the characteristic equation is to the left of and near the pole at positive two.
2. When the gain increases, the solution moves to the left toward the zero at negative four. It only reaches negative four when $K = \infty$.
3. There is a point in the movement where pole is at the origin. The gain at this critical point is the critical gain, K_{cr} . When $K > K_{cr}$ the system is BIBO stable; $K < K_{cr}$, BIBO unstable; and $K = K_{cr}$, marginally stable. The solution to K_{cr} can be found with the Routh Array.

Example — Real Root and Real Zero



When the gain is small, the solution to the characteristic equation is to the left of and near two. When the gain increases, the solution moves to the left toward the zero at negative four. However, it only reaches negative four when $K = \infty$. With the movement of the poles in this problem there is again a critical gain. However, for this system the stable region for K is $0 \leq K < K_{cr}$, as observed in the sketch.

Example — Real Roots



The point where the loci leave the real axis is the breakaway, BA, point. There's a gain at that point, K_{BA} . For $K \leq K_{BA}$ the poles are real for $K_{BA} < K$ the poles are complex.

Let's find the breakaway inflection point From the loop-gain roots

$$L = \frac{1}{(s+6)(s+2)} = \frac{1}{s^2 + 8s + 12} = \frac{N_L}{D_L}$$

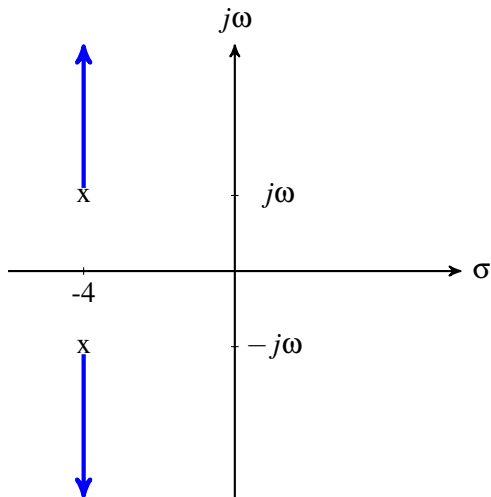
then

$$\begin{aligned} D_L N_L' - D_L' N_L &= 0 \\ (s^2 + 8s + 12)(0) - (2s + 8)(1) &= 0 \\ -2s - 8 &= 0 \\ s &= -4 \\ \text{BA} &= -4 \end{aligned}$$

Now, the gain at BA

$$\begin{aligned} K_{BA} &= -(s^2 + 8s + 12), s = \text{BA} \\ &= 4 \end{aligned}$$

Example — Complex Poles



From the Root Locus sketch:

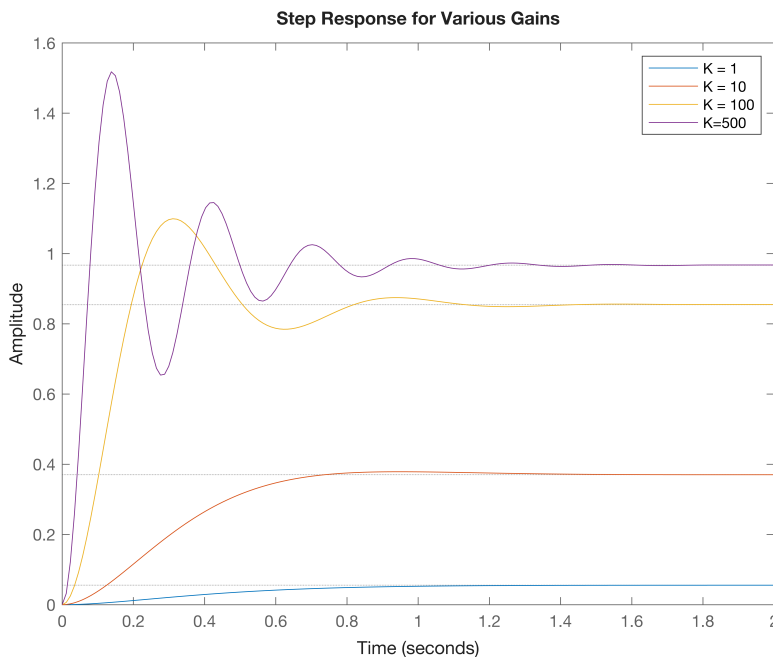
1. The system is always BIBO stable and the output always oscillates
2. The oscillation frequency increases as the gain K increases, while the settling time remains unchanged.

Let's consider the setpoint response to the open-loop transfer function for various gains.

Recall:

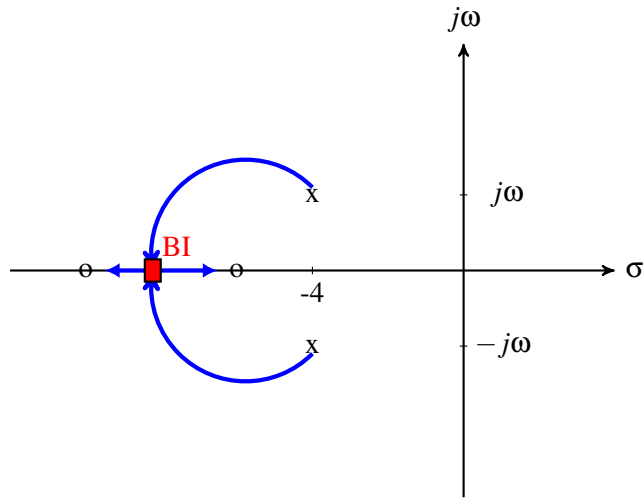
$$T = \frac{G}{1 + GH}$$

For the model, we assume $H = 1$, so $L = GH = G$. Then, we can easily construct the open-loop transfer function from the loop-gain and plot it in MATLAB.



From the open-loop transfer function, we observe the increase in frequency as K increases; while the settling time remains the same.

Example — Complex Poles and Real Zeros



From the RL sketch:

1. The system is always BIBO stable.
2. When the gain is less than K_{BI} then the output signal oscillates. When the gain is greater than K_{BI} the output no longer oscillates.

Example — Real and Complex Poles With Real Zeros

Given:

$$L(s) = \frac{s + 6}{(s + 1)(s^2 + 10s + 29)}$$

The loop-gain zero $LG_z = -6$ and the loop-gain poles $LG_p = -5 \pm 2j, -1$. So

$$n = 3$$

$$m = 1$$

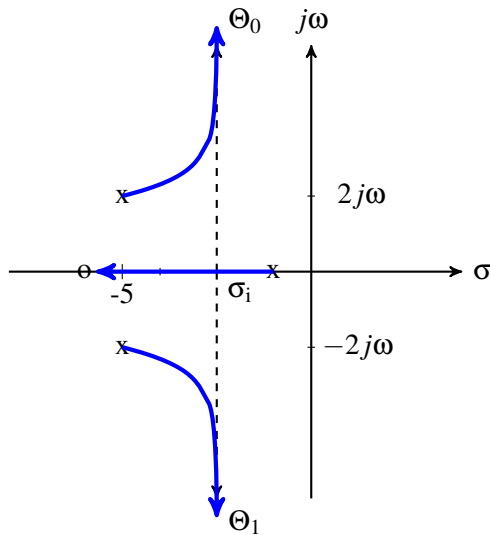
$$\sigma_i = \frac{(-5 - 5 - 1) - (-6)}{3 - 1} = \frac{-5}{2} = -2.5$$

$$k = \{0, 1\}$$

$$\Theta_0 = \frac{\Pi}{2}$$

$$\Theta_1 = \frac{3\Pi}{2}$$

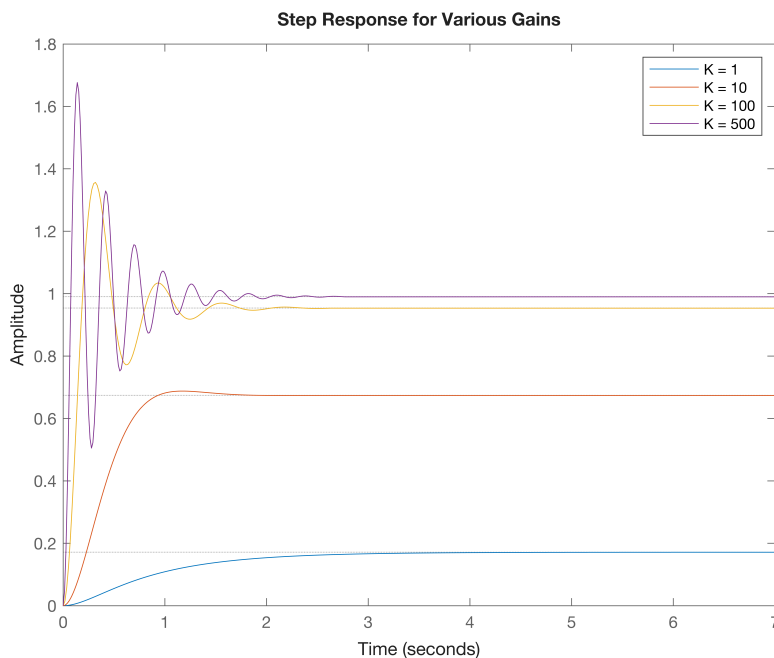
Taking the information we have gathered and calculated we can determine the sketch



From the sketch, we can determine several things;

1. The BIBO response it is always stable
2. The output will oscillate at higher frequencies as the gain increases.
3. For small values of K , the real pole is dominate, for large values of K the complex poles are dominate.

Once again, we can easily construct the open-loop transfer function from the loop-gain and plot it in MATLAB, as we did above.



Just as before, we observe the increase in frequency as K increases; while the settling time remains the same.

Example — Real and Complex Zeros and Poles

Taking the previous loop-gain and adding a complex pair of loop-gain zeros at $-8 \pm 3j$. Given:

$$L(s) = \frac{(s+6)(s^2 + 16s + 73)}{(s+1)(s^2 + 10s + 29)}$$

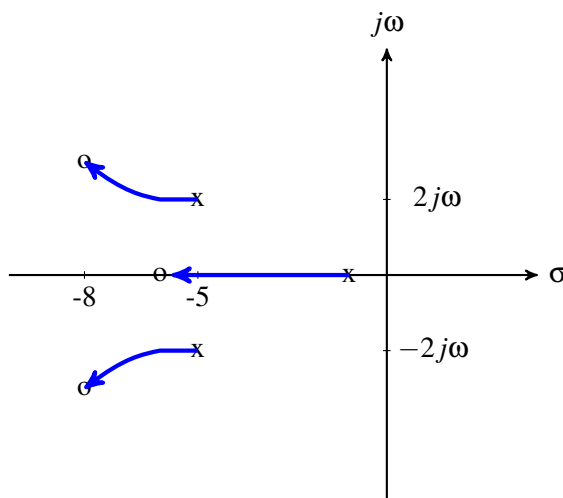
The loop-gain zeros $LG_z = -8 \pm 3j$, -6 and the loop-gain poles $LG_p = -5 \pm 2j$, -1 . So

$$n = 3$$

$$m = 3$$

Since $n = m$ there are no poles that go to infinity, so there is no need to calculate σ_i or Θ_k .

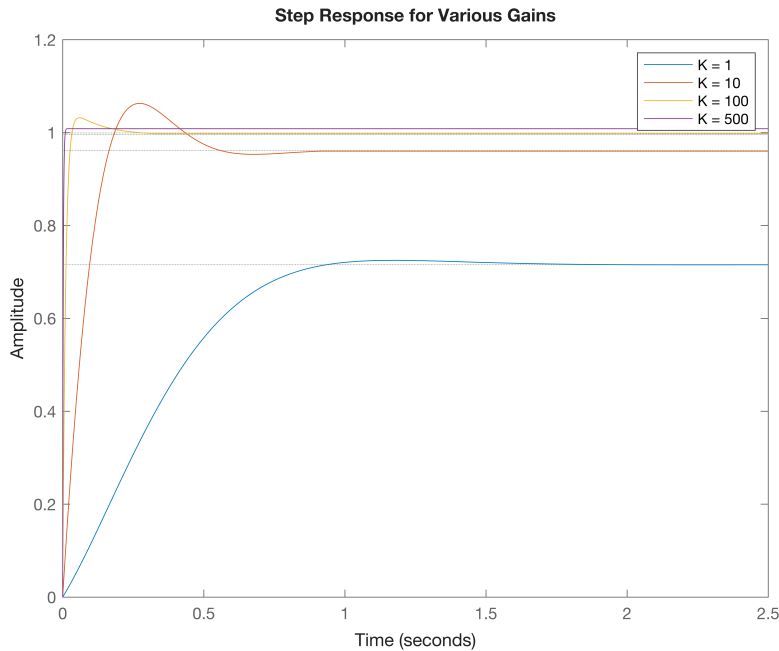
Taking the information we have gathered we can draw the sketch



From the sketch, we can determine several things:

1. The BIBO response is always stable and the output will oscillate at a higher frequency with more gain
2. The oscillations will diminish faster with increased gain because the poles are moving to the left, thus, reducing the settling time.
3. All three poles settle faster as K increases. While the frequency of the complex poles increase with K , as well, it does not increase at a comparable rate as the real parts.

Once again, we can easily construct the open-loop transfer function from the loop-gain and plot it in MATLAB, as we did above.



We observe the increase in K reduces the settling time of the system. For small values and large values of K the system response has minimal overshoot, but there appears to be a window for K such that the overshoot is maximized. Note the large overshoot for $K = 10$ and also a visible overshoot for $K = 100$. If overshoot is a concern, we would need to investigate further.

Example — Real Zeros and Poles

Given:

$$L(s) = \frac{(s+6)(s+8)}{s(s+3)} = \frac{N_L}{D_L}$$

The loop-gain zeros $LG_z = -8, -6$ and the loop-gain poles $LG_p = -3, 0$.

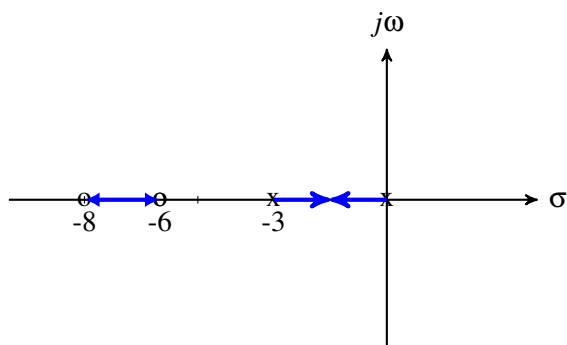
So

$$n = 3$$

$$m = 3$$

Since $n = m$ there are no poles that go to infinity, so there is no need to calculate σ_i or Θ_k .

A quick sketch of the zeros, poles, and loci on the real axis



reveals that we will breakaway and break-in points. Let's find points

$$L(s) = \frac{s+6(s+8)}{s(s+3)} = \frac{s^2+14s+48}{s^2+3s} = \frac{N_L}{D_L}$$

$$\frac{dL}{ds} = \frac{(s^2+3s)(2s+14) - (2s+3)(s^2+14s+48)}{(s^2+3s)^2}$$

$$= \frac{(s^3+17s^2+42s) - (s^3+31s^2+138s+144)}{(s^2+3s)^2}$$

$$= \frac{-14s^2-96s-144}{(s^2+3s)^2} = 0$$

The result

$$s = -6.8, -1.92$$

to determine which is the breakaway and which is the break-in, we have to look at the partial sketch from above.

$$BI = -6.8$$

because it is between the two loop-gain zeros.

$$BA = -1.92$$

because it is between the two loop-gain poles. Additionally, we find the gain at the inflection points

$$K_{BA} = \frac{-D_L}{N_L}, s = BA$$

$$= \frac{-(s^2+3s)}{s^2+14s+48}, s = -1.92$$

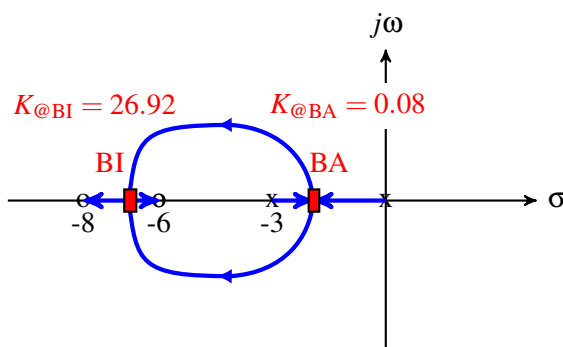
$$= 0.08$$

$$K_{BI} = \frac{-D_L}{N_L}, s = BI$$

$$= \frac{-D_L}{N_L}, s = -6.8$$

$$= 26.92$$

Completing the sketch



From the sketch we have three distinct regions — all are stable — that offer different dynamic performance:

1. $K \leq K_{BA}$

- The output does not oscillate, but the settling time is the slowest of all three regions. With one pole very near the imaginary axis. *Note:* the breakaway gain is incredibly small.

2. $K_{BA} < K < K_{BI}$

- The output oscillates and the settling time decreases as K increases.

3. $K \geq K_{BI}$

- The output stops oscillating. The increase of gain beyond K_{BI} only slightly changes the dynamic response; however, it alters the frequency response (flash forward to a later Chapter) significantly.

Example — Repeated Poles

Given:

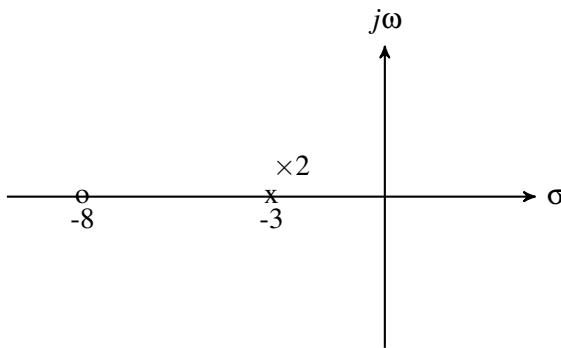
$$L(s) = \frac{s+8}{(s+3)^2}$$

The loop-gain zero $LG_z = -8$ and there are two loop-gain poles $LG_p = -3 \times 2$. So

$$n = 2$$

$$m = 1$$

A quick sketch of the zero, poles on the real axis



Using the Guidelines we observe that there will be a breakaway point at -3 and $K_{BA} = 0$. Additionally, there will be a break-in point to the left of the zero at -8. Let's find the inflection points

$$L(s) = \frac{s+8}{(s+3)^2} = \frac{s+8}{s^2+6s+9} = \frac{N_L}{D_L}$$

$$\begin{aligned} \frac{dL}{ds} &= (s^2+6s+9)(1) - (s+8)(2s+6) \\ &= (s^2+6s+9) - (2s^2+22s+48) \\ &= -s^2 - 16s - 39 = 0 \end{aligned}$$

The result

$$s = -13, -3$$

$$\text{BI} = -13$$

$$\text{BA} = -3$$

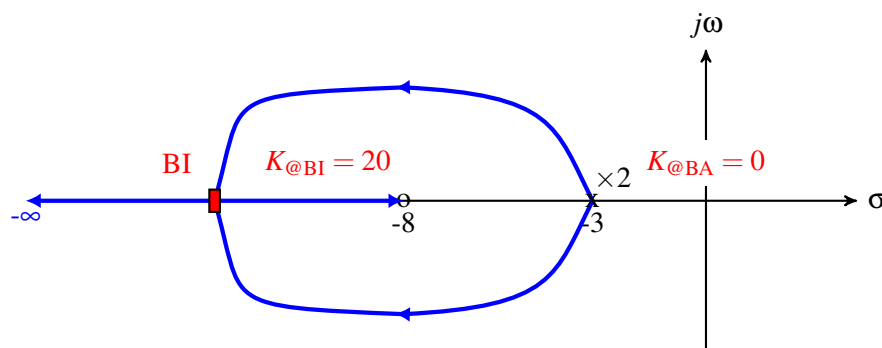
Next, we find K at the inflection points

$$\begin{aligned} K_{\text{BI}} &= \frac{-(s^2 + 6s + 9)}{s + 8}, s = -13 \\ &= 20 \end{aligned}$$

$$\begin{aligned} K_{\text{BA}} &= \frac{-(s^2 + 6s + 9)}{s + 8}, s = -3 \\ &= 0 \end{aligned}$$

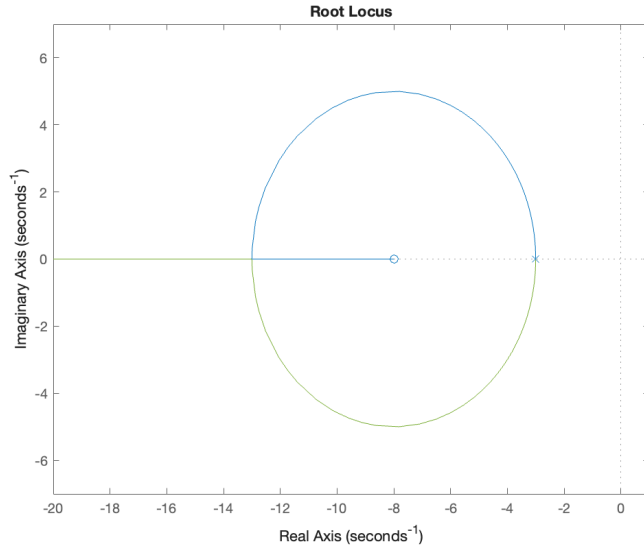
Note: From observation of the first sketch, we had already surmised the gain at the breakaway point.

Now, the sketch



Now, the actual paths from BA to BI are unknown; we know they are continuous paths that mirror one another about the real axis. *Note:* there are additional steps to understand the paths, but with software simulation tools these tedious approaches are obsolete. If an exact path is necessary, plot the Root Locus in MATLAB or another tool.

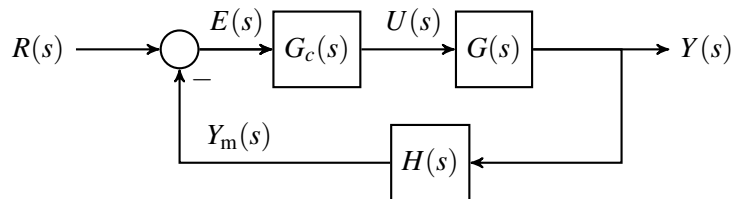
Using the `rlocus` function in MATLAB



The basics of the sketch agree with the loci plot from MATLAB. We observe that as $K \rightarrow \infty$ there is a locus that goes to $-\infty$ along the real axis. We can solve for the asymptote angle; however, it may be found thru observation. Namely, $\Theta_0 = \pi$.

Chapter 10

System Typing (Adv Steady State Error Analysis)



The analysis in this chapter is based upon the Final Value Theorem, so the system itself must be BIBO stable, in addition there are constraints applied to the feedback sensor

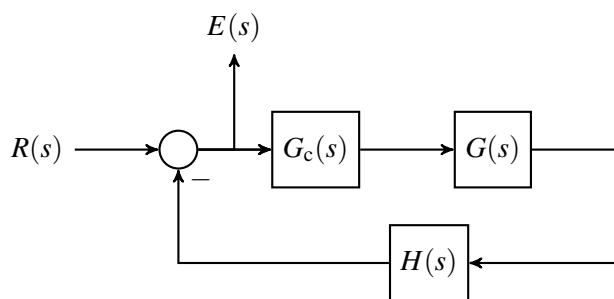
1. $H(s)$ must be stable

- Obviously, if the sensor feedback is unstable then the measurement error signal is invalid

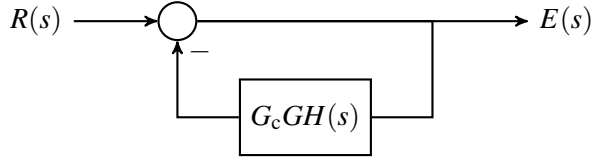
2. $H(s \rightarrow 0) = 1$

- This means that the steady state measurement error goes to zero. In other words, $y_m(t \rightarrow \infty) = y(t \rightarrow \infty)$

As in previous chapters, for steady state error analysis we redefine the output to be the error



Let's redraw the diagram to more closely match the convention



Now, the open-loop transfer function is

$$\frac{E(s)}{R(s)} = \frac{1}{1 + L(s)}$$

where the loop-gain $L(s) = G_cGH(s)$.

We consider three inputs into the system

1. Setpoint, $R(s) = \frac{1}{s}$
 - Is used to represent a fixed position.
2. Ramp, $R(s) = \frac{1}{s^2}$
 - Is used to represent a constant velocity
3. Parabola, $R(s) = \frac{1}{s^3}$
 - Is used to represent a constant acceleration

For any input the time domain steady state output error is

$$\begin{aligned} e(t \rightarrow \infty) &= \lim_{s \rightarrow 0} s \cdot E(s) \\ &= \lim_{s \rightarrow 0} s \frac{1}{1 + L(s)} R(s) \end{aligned}$$

To simplify the process and to compare relative error performance we establish three figures of merit; one for each input. The larger the figure of merit, the smaller the error. If the figure of merit is zero, the response is infinite steady state error. When the figure of merit is infinite the steady state error is zero.

Let's expound on the figures of merit by considering the error when the input is a setpoint

$$\begin{aligned} e(t \rightarrow \infty) &= \lim_{s \rightarrow 0} s \frac{1}{1 + L(s)} \frac{1}{s} \\ &= \lim_{s \rightarrow 0} \frac{1}{1 + L(s)} \\ &= \frac{1}{1 + L(0)} \end{aligned}$$

For the setpoint input, the figure of merit

$$KP = L(0) \tag{10.1}$$

Now, for a ramp input

$$\begin{aligned}
 e(t \rightarrow \infty) &= \lim_{s \rightarrow 0} s \frac{1}{1 + L(s)} \frac{1}{s^2} \\
 &= \lim_{s \rightarrow 0} \frac{1}{1 + L(s)} \frac{1}{s} \\
 &= \lim_{s \rightarrow 0} \frac{1}{s + sL(s)} \\
 &= \lim_{s \rightarrow 0} \frac{1}{sL(s)}
 \end{aligned}$$

For the ramp input, the figure of merit

$$\text{KV} = \lim_{s \rightarrow 0} \{sL(s)\} \quad (10.2)$$

Finally, for a parabolic input

$$\begin{aligned}
 e(t \rightarrow \infty) &= \lim_{s \rightarrow 0} s \frac{1}{1 + L(s)} \frac{1}{s^3} \\
 &= \lim_{s \rightarrow 0} \frac{1}{1 + L(s)} \frac{1}{s^2} \\
 &= \lim_{s \rightarrow 0} \frac{1}{s^2 + s^2L(s)} \\
 &= \lim_{s \rightarrow 0} \frac{1}{s^2L(s)}
 \end{aligned}$$

For the parabolic input, the figure of merit

$$\text{KA} = \lim_{s \rightarrow 0} \{s^2L(s)\} \quad (10.3)$$

The steady state error is a function of the loop-gain and the input. For system typing analysis we consider the loop-gain function in binomial form.

$$\begin{aligned}
 L(s) &= \frac{b_1s + b_0}{a_2s^2 + a_1s + a_0} \\
 &= \frac{K_g(s + \beta_1)}{s^N(s + \alpha_1)(s + \alpha_2)}
 \end{aligned}$$

where N is a positive (including zero) integer and K_g is the factored gain. Simplify $L(s)$ with all zero pole cancellations.

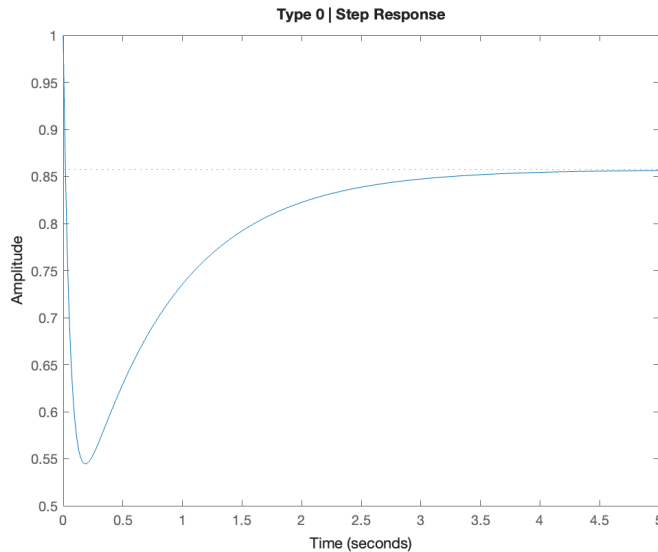
Example — Type 0 System

We have a loop-gain

$$\begin{aligned}
 L(s) &= \frac{18s + 6}{2s^2 + 18s + 36} \\
 &= \frac{18(s + 0.33)}{2(s^2 + 9s + 18)} \\
 &= \frac{9(s + 0.33)}{s^0(s + 6)(s + 3)}
 \end{aligned}$$

after normalization, $N = 0$ which means this system is a **type 0**. For this analysis, we are not concerned with the other parameters within the loop-gain, only N .

Using MATLAB, the system error for a step input



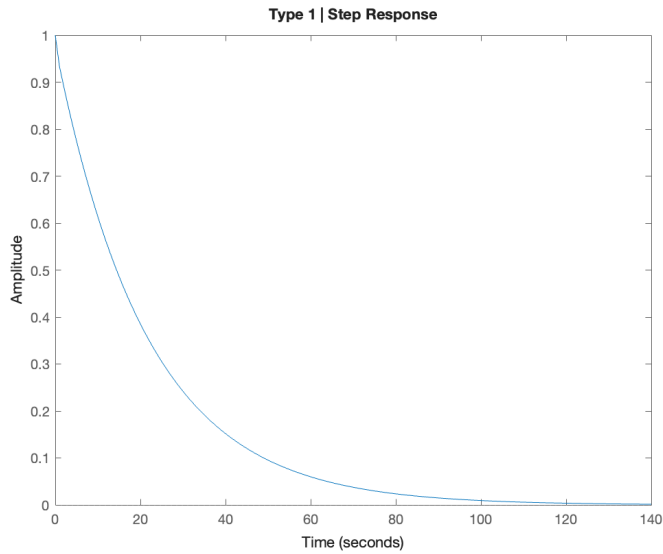
Example — Type 1 System

We have a loop-gain

$$\begin{aligned} L(s) &= \frac{s+2}{s^2+42s} \\ &= \frac{(s+2)}{s(s+42)} \end{aligned}$$

after normalization, $N = 1$ which means this system is a **type 1**.

Using MATLAB, the system error for a step input

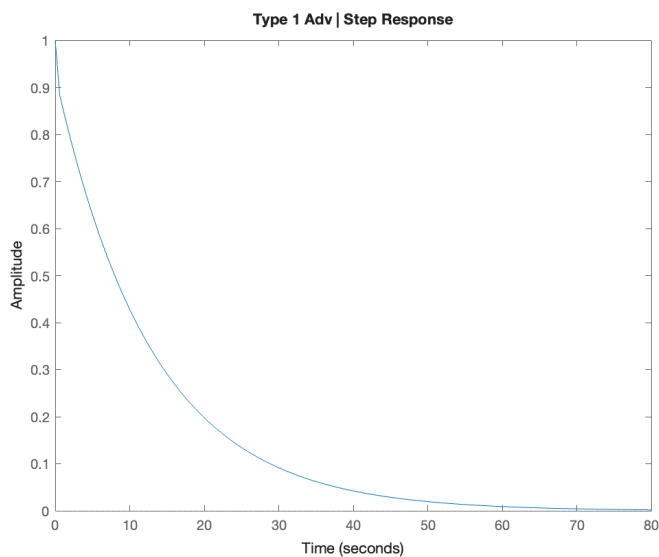


Example — Adv Type 1 System

We have a loop-gain

$$\begin{aligned}
 L(s) &= \frac{2s^2 + 2s}{s^3 + 24s^2} \\
 &= \frac{2s(s + 1)}{s^2(s + 24)} \\
 &= \frac{2(s + 1)}{s(s + 24)}
 \end{aligned}$$

after normalization, $N = 1$ so this system is a **type 1**, as well.
Using MATLAB, the system error for a step input



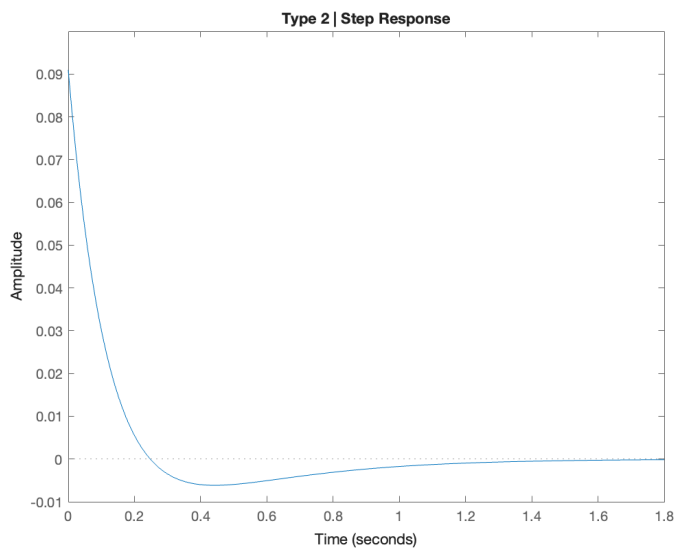
Example — Type 2 System

We have a loop-gain

$$\begin{aligned} L(s) &= \frac{10(s^2 + 12s + 27)}{s^2 + 1s} \\ &= \frac{10(s + 3)(s + 9)}{s^2(s + 1)} \end{aligned}$$

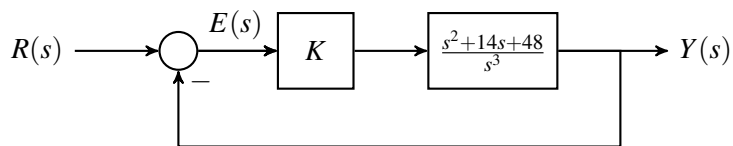
after normalization, $N = 2$ so this is a **type 2** system.

Using MATLAB, the system error for a step input



Example — Design with a Type 3 System

Given: A system

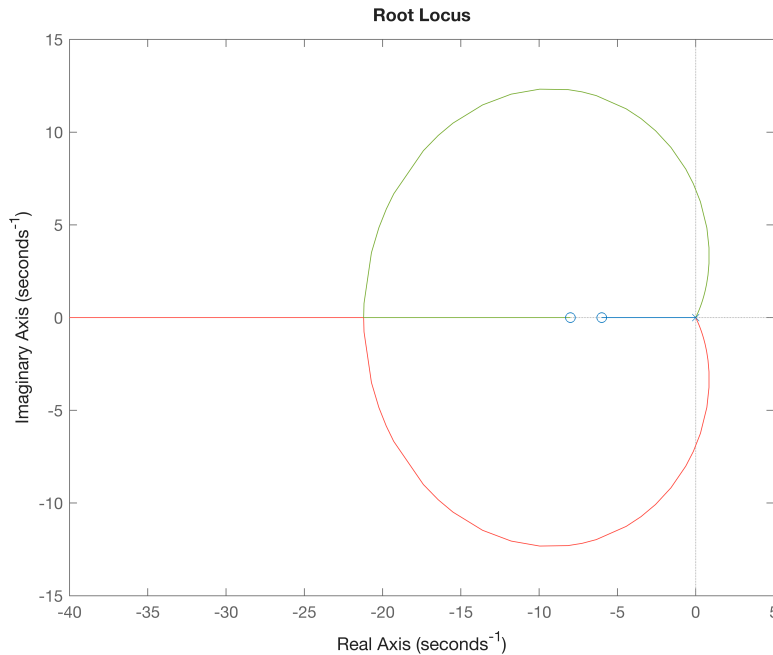


We have a loop-gain

$$L(s) = \frac{K(s^2 + 14s + 48)}{s^3}$$

which is a type 3 system.

To analyze the Root Locus of $L(s)$, set $K = 1$ and evaluate either with a sketch or with simulation software. For this problem, let's use MATLAB to obtain the Root Locus diagram

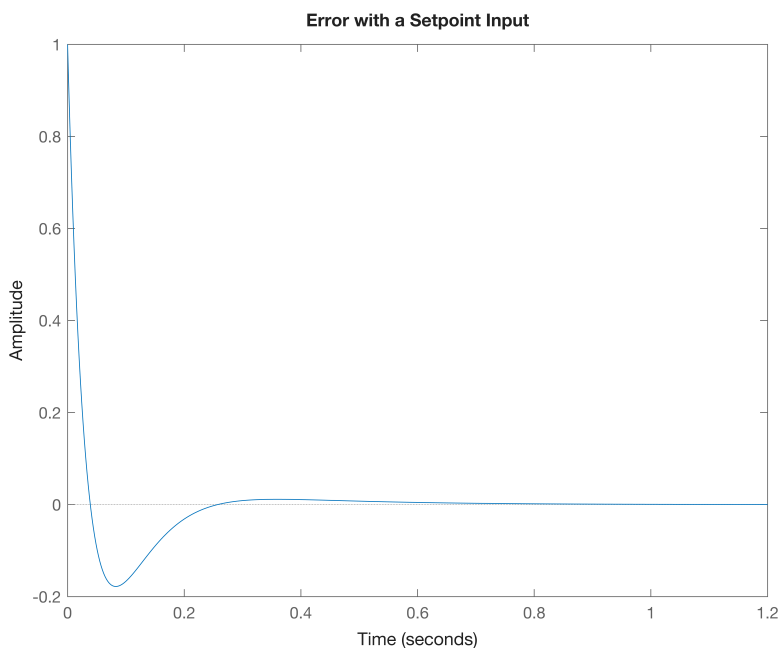


We can see from the diagram, there are values for K where the system is unstable. So, we need to be aware of the system stability. Using the Root Locus diagram, we can observe the range of stability to be $3.5 \geq K < \infty$. Additionally, for the region where the system does not oscillate is $47.5 \geq K < \infty$. For implementation purposes, let's select $K = 47.5$; which produces

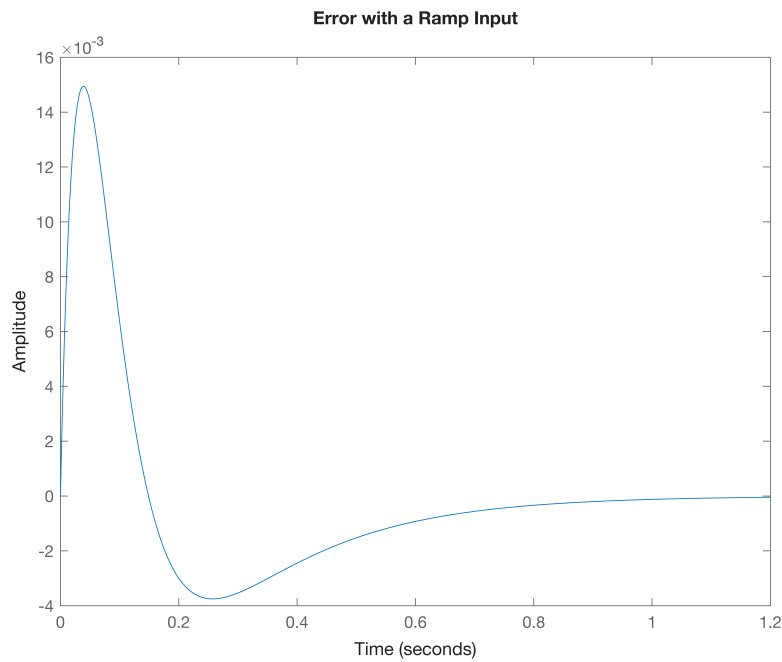
$$\frac{E(s)}{R(s)} = \frac{s^3}{s^3 + 47.5s^2 + 665s + 2280}$$

There are three zeros $s = 0 \times 3$ and three poles $s = -21.59, -20.84, -5.07$.

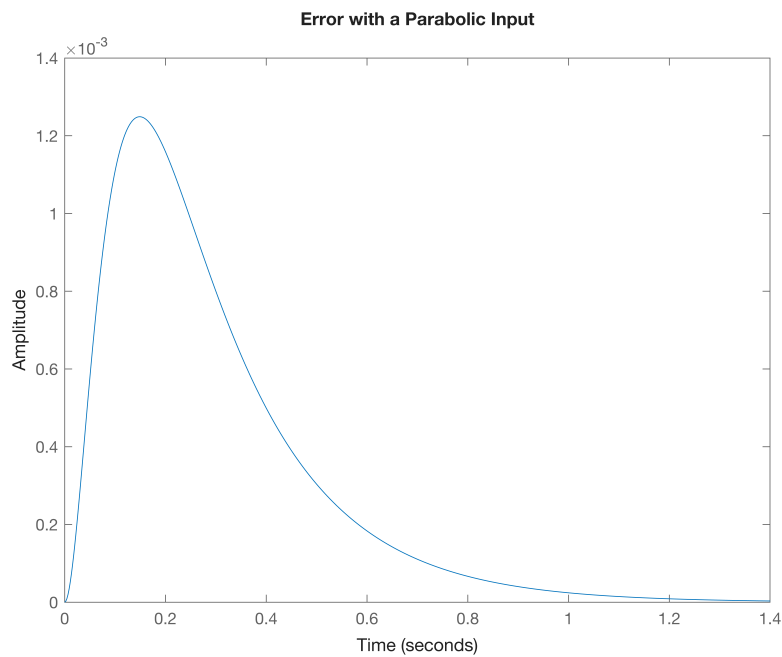
The error for a setpoint input



The error for a ramp input



The error for a parabolic input



We observe that the type 3 system has zero error for all three inputs. So, why don't we always design a type 3 system? The answer is simple, the higher the order the more complex the controller and the analysis — so we only implement the minimal system needed for the job.

10.1 Conclusion

From the system type, we may determine the steady state error without performing any calculations — just use this error response look up table.

The calculated cells are the steady state error.

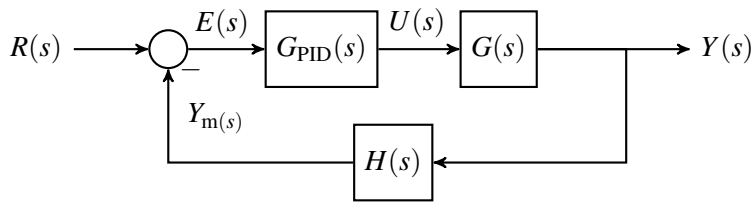
Type	Errors from input		
	Setpoint	Ramp	Parabola
0	$\frac{1}{1+KP}$	∞	∞
1	0	$\frac{1}{KV}$	∞
2	0	0	$\frac{1}{KA}$
3+	0	0	0

Before using the table to determine the steady-state error, we first must verify that the system is BIBO stable that the sensor feedback meets the criteria at the beginning of the chapter.

Chapter 11

Proportional, Integral, Differential (PID)

The PID controller is incredibly common in industry because it is flexible and easy to implement. Industrial PID controllers may be adjusted in the field using basic tuning rules. Three of the more common rules are discussed here. Additionally, there are multiple implementations of the controller, only four are presented within the chapter.



In the figure, the PID controller, $G_{PID}(s)$, can take on multiple configurations; the four most common are considered here, but there many different ways to build-up a PID.

1. Proportional:

$$G_{PID}(s) = G_P(s) = K_p \quad (11.1)$$

where $K_p > 0$, a single tunable parameter.

Strengths:

- (a) Easy to implement
- (b) A Root Locus system
- (c) Control effort scales proportionally wrt to error

Weaknesses:

- (a) Single tunable parameter (K_p) limits the location the closed-loop poles can be placed
- (b) Does not increase the system type

2. Proportional, Integral: with the gain paths as separate expressions

$$\begin{aligned} G_{PID}(s) = G_{PI}(s) &= K_p + \frac{K_i}{s} \\ &= \frac{K_p s + K_i}{s} \end{aligned} \quad (11.2)$$

and finally as a Root Locus controller

$$G_{PI}(s) = \frac{K_p(s + \hat{K}_i)}{s} \quad (11.3)$$

where, $\hat{K}_i = \frac{K_i}{K_p}$, $K_p > 0$, and $K_i > 0$.

Strengths:

- (a) The presence of the integrator in the forward path increases system type; $\therefore$ eliminates steady state tracking error for an additional input $e(\infty) = r(\infty) - y_m(\infty)$
- (b) The configuration in (11.3) is simply a Root Locus problem with an additional loop-gain pole at the origin and an tunable loop-gain zero at $-\frac{K_i}{K_p}$
- (c) Increases the system type by one

Weaknesses:

- (a) Implementation of the integrator can be a challenge with electronics. Additionally, electronic integrators have a hard limit based on component limitations
- (b) The integration control tends to produces output oscillations

Notes:

3. Proportional, Differential:

$$G_{PID}(s) = G_{PD}(s) = K_d s + K_p \quad (11.4)$$

where, $K_p > 0$ and $K_d > 0$. As a Root Locus controller,

$$G_{PD}(s) = K_d(s + \hat{K}_p) \quad (11.5)$$

where $\hat{K}_p = \frac{K_p}{K_d}$.

Strengths:

- (a) In systems that oscillate, the presence of the differential gain reduces overshoot and oscillations. Often, proper tuning can completely eliminate both.
- (b) Treated as a Root Locus problem with a tunable loop-gain zero at $-\frac{K_p}{K_d}$.

Weaknesses:

- (a) Slows the system response
- (b) Does not increase the system type

4. Proportional, Integral, Differential:

$$\begin{aligned} G_{PID}(s) &= K_p + \frac{K_i}{s} + K_d s \\ &= \frac{K_d s^2 + K_p s + K_i}{s} \end{aligned} \quad (11.6)$$

where $K_p > 0$, $K_i > 0$, and $K_d > 0$, or as a Root Locus controller,

$$\begin{aligned} G_{\text{PID}}(s) &= \frac{K_d(s^2 + \hat{K}_p s + \hat{K}_i)}{s} \\ &= \frac{K_d(s + \beta_1)(s + \beta_2)}{s} \end{aligned} \quad (11.7)$$

where $\hat{K}_p = \frac{K_p}{K_d}$ and $\hat{K}_i = \frac{K_i}{K_d}$. While $-\beta_1$ and $-\beta_2$ are user selected loop-gain zeros which produces a Root Locus problem with user set zeros.

Strengths:

- (a) Three tunable parameters allows for improved pole placement
- (b) Increases system type
- (c) Eliminates steady state tracking error, $e(\infty) = r(\infty) - y_m(\infty)$
- (d) Adds two loop-gain zeros at user selected locations, and a loop-gain pole at the origin

Weaknesses:

- (a) Can be difficult to tune to the best possible gain selection

Given the block diagram structure, where the controller and plant transfer functions are generally represented as,

$$\begin{aligned} G_{\text{PID}}(s) &= \frac{N_{\text{gc}}}{D_{\text{gc}}} \\ G(s) &= \frac{N_g}{D_g} \\ H(s) &= \frac{N_h}{D_h} \end{aligned}$$

Then, the transfer function is

$$\begin{aligned} \frac{Y(s)}{R(s)} &= \frac{G_{\text{PID}}G(s)}{1 + G_{\text{PID}}GH(s)} \\ \frac{Y(s)}{R(s)} &= \frac{\left(\frac{N_{\text{gc}}}{D_{\text{gc}}}\right)\left(\frac{N_g}{D_g}\right)}{1 + \left(\frac{N_{\text{gc}}}{D_{\text{gc}}}\right)\left(\frac{N_g}{D_g}\right)\left(\frac{N_h}{D_h}\right)} \end{aligned} \quad (11.8)$$

Finally,

$$\begin{aligned} \frac{Y(s)}{R(s)} &= T(s) \\ &= \frac{N_{\text{gc}}N_gD_h}{D_{\text{gc}}D_gD_h + N_{\text{gc}}N_gN_h} \\ &= \frac{B(s)}{A(s)} \end{aligned} \quad (11.9)$$

We can see from the analysis that controller affects the numerator and the denominator in the open-loop transfer function.

11.1 Tuning

There are several approaches to tuning a PID controller. If we have a very good math model for the system, we may use a **Root Locus** approach. If the model is not available, then a **Ziegler-Nichols Method** can be used for a set of systems. Finally, we may always tune a PID with a **heuristic** approach. Using the solution in (11.9), where the controller numerator (N_{gc}) and controller denominator (D_{gc}) vary depending on the configuration chosen (P, PI, PD, or PID).

11.1.1 Root Locus Approach

Each of the previously mentioned controllers may be analyzed from the perspective of Root Locus analysis.

Proportional

A proportional only controller is tuned as a typical Root Locus controller, see Root Locus chapter and notes. It is an ideal Root Locus system with single tunable gain.

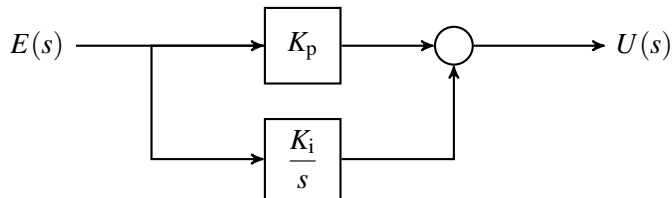
Proportional, Integral, Controller

When

$$\begin{aligned} G_{PI}(s) &= K_p + \frac{K_i}{s} \\ &= \frac{K_p s + K_i}{s} \\ &= \frac{K_p(s + \hat{K}_i)}{s} \end{aligned} \tag{11.10}$$

where we set the loop-gain zero from $\hat{K}_i$ to a desired location and then tune K_p as we would with a Root Locus system.

The control block is



The controller allows us to set a loop-gain zero on the real axis with

$$\hat{K}_i(s) = \frac{K_i}{K_p}$$

and the nature of the integrator in the controller increases the system type by one — which works to reduce the steady state error.

For the Root Locus analysis:

1. The variable gain: $0 < K_p < \infty$
2. We have have the system loop-gain zeros plus the user set zero at $-\hat{K}_i(s)$ from the controller
3. The system loop-gain poles plus the additional pole at the origin from the controller
4. The controller increases the system type

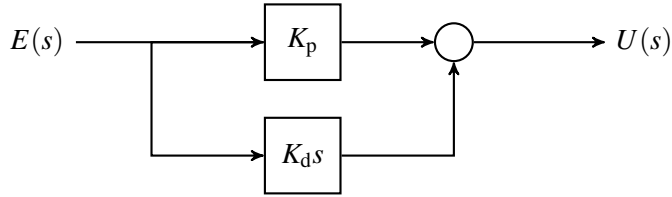
Proportional, Differential Controller

When

$$\begin{aligned} G_{PD}(s) &= K_d s + K_p \\ &= K_d(s + \hat{K}_p) \end{aligned} \quad (11.11)$$

where we set the loop-gain zero from $\hat{K}_p$ to a desired location and then tune K_d as we would with a Root Locus system.

The control block is



The controller allows us to set a loop-gain zero on the real axis with

$$\hat{K}_p = \frac{K_p}{K_d}$$

For the Root Locus analysis:

1. The variable gain: $0 < K_d < \infty$
2. We have have the system loop-gain zeros plus the user set zero at $-\hat{K}_p(s)$ from the controller

Proportional, Integral, Differential Controller

When

$$\begin{aligned} G_{PID}(s) &= K_p + \frac{K_i}{s} + K_d s \\ &= \frac{K_d s^2 + K_p s + K_i}{s} \\ &= \frac{K_d(s^2 + \hat{K}_p s + \hat{K}_i)}{s} \\ &= \frac{K_d(s + \beta_1)(s + \beta_2)}{s} \end{aligned} \quad (11.12)$$

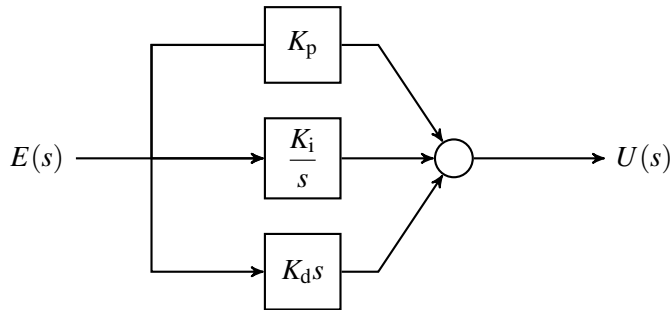
where

$$\hat{K}_p = \frac{K_p}{K_d}$$

$$\hat{K}_i = \frac{K_i}{K_d}$$

and β_1 and β_2 are user set loop-gain zeros. Now, for this controller we can set two loop-gain zeros in the s-plane — real valued or complex — there is a lot of versatility with the PID controller. Additionally, there is a loop-gain pole at the origin — once again increasing the system type and reducing the steady state error.

The control block is



For the Root Locus analysis:

1. The variable gain: $0 < K_d < \infty$
2. We have the system loop-gain zeros plus the user set zeros at $-\beta_1$ and $-\beta_2$ from the controller
3. The system loop-gain poles plus the additional pole at the origin from the controller
4. The controller increases the system type

11.1.2 Zeigler-Nichols Approach

These are a modelless methods for tuning a PID controller based on the output response to a setpoint input. They work for specific classes of systems that produce an certain output response.

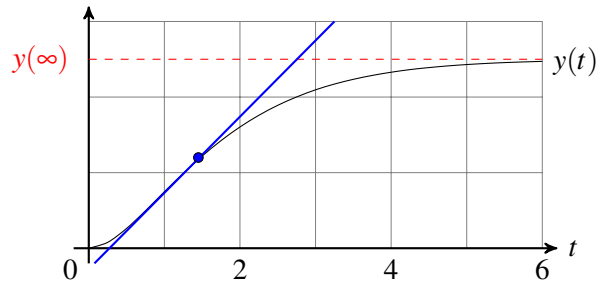
First Method

In this approach, the assumptions are:

1. The plant — the system to be controlled — is stable to a setpoint input
2. The plant can be interrogated without control and without feedback in the path
3. The output response to a setpoint input forms an “S”

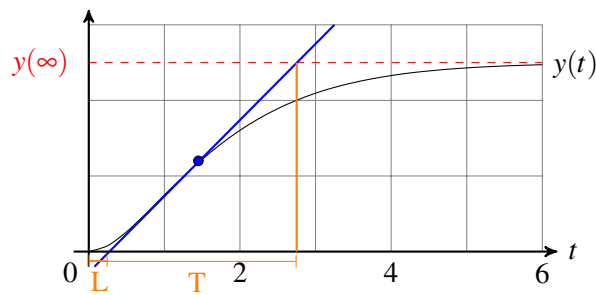
Apply a setpoint input to the plant itself and capture the output.

1. Given the output, draw a tangent line that intersects the inflection point (50% of the steady state output), the real axis and the final value line, such as



2. From the sketch, measure two parameters

- L (the lag): from $t = 0$ to the point where the tangent line crosses the time axis
- T (time constant): the time between where the tangent line crosses the time axis and where it reaches the output final value $y(\infty)$



3. Now, from the two measured parameters we can implement three controllers based on the LUT

Controller	K_p	K_i	K_d
P	$\frac{T}{L}$	0	0
PI	$\frac{0.9T}{L}$	$\frac{0.3K_p}{L}$	0
PID	$\frac{1.2T}{L}$	$\frac{K_p}{2L}$	$\frac{K_p L}{2}$

Each row in the table is a unique controller which may be implemented using $G_{PID}(s)$ in the forward path, with unity negative feedback.

Second Method

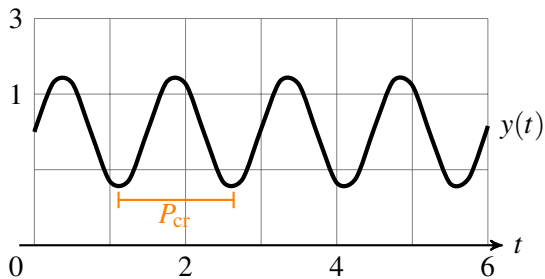
In this approach, the assumptions are:

1. The system has unity negative feedback about the plant with a controller in the forward path — a typical configuration

2. The controller is PID and the parameters can be tuned — including turned off
3. There is a value for $0 < K_p < \infty$ that is stable when $K_i = K_d = 0$

Apply a setpoint input to the entire system:

1. Start with K_p set to the smallest nonzero value realistically possible
2. Observe the output as you begin to increase the gain K_p
3. Continue to increase and observe until the output forms a stable oscillation *Note:* We are driving the response toward the unstable region; this can be a risky approach and great care should be taken
4. Once the output has reached stable oscillation, capture the output, and make note of the gain that achieved the result, K_{cr}



5. Measure the period of oscillation P_{cr}
6. Use the measured K_{cr} and P_{cr} in the LUT below to select the parameters for three PID controllers

Controller	K_p	K_i	K_d
P	$0.5 \cdot K_p$	0	0
PI	$0.45K_p$	$\frac{1.2K_p}{P_{cr}}$	0
PID	$0.6K_{cr}$	$\frac{2K_p}{P_{cr}}$	$\frac{P_{cr}K_p}{8}$

11.1.3 Heuristic Approach

Just as with Ziegler-Nichols method, the heuristic approach is based on a modelless evaluation of the output. Generally, this approach only reliably works for a typical tracking problem.

If possible, run tests on the plant alone — without feedback or control effort. If known, use the expected input; otherwise, start with a setpoint and move to a ramp, finally a parabola. This can help identify a baseline of performance. Then, repeat the tests with the final design feedback loop.

Ultimately, incorporate the PID in the system and tune according to these trial-and-error rules. Normally, we will begin with a P-only controller, then move to a PI-only. Incorporate a differential

gain if the overshoot cannot be managed by tuning the controller. Implement a full PID only if necessary to meet the design requirements. The motivation should always be to use the simplest possible controller to achieve the performance goals.

1. K_p scales the error proportionally. If the system settles to an output that is near the desired final value, increasing this parameter can close the gap and reduce the final error
2. K_i reduces or eliminates steady state error and reduces settling time. However, it tends to cause output oscillations
3. K_d reduces overshoot and oscillations in the output. However, it slows the system response proportionally. The larger the parameter, the slower the settling time

The heuristic approach is iterative in nature and you will need to test, adjust, and repeat. Be sure to maintain stability and monitor performance changes with each test. The integration and differentiation efforts compete with and cancel one another — it's a **waterbed effect**.

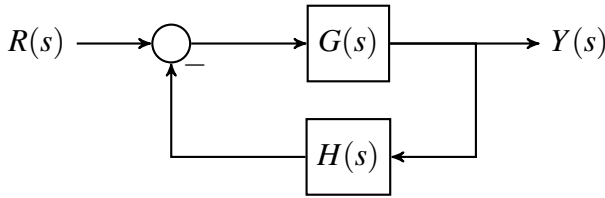
Once a suitable set of parameters is achieved, make additional measurements with the parameters linearly scaled. Choose to implement the smallest gains possible in the controller that meets the desired performance.

Chapter 12

Frequency Response Analysis

Frequency response analysis is performed on the loop-gain L to determine the relative frequency stability by inspecting the gain and phase margins and on the open-loop transfer function T to determine overall gain, phase, and bandwidth. The primary tool is the Bode Plot, however, Nyquist and Nichols Plots are used as well.

From the diagram



The loop-gain $L(s) = GH(s)$ and the transfer function $T(s) = \frac{G(s)}{1+L(s)}$

Since the analysis is performed on two different expressions, a generic $F(s)$ transfer function is used for development

Given a typical transfer function in polynomial form

$$F(s) = \frac{b_m s^m + b_{m-1} s^{m-1} + \dots + b_0}{a_n s^n + a_{n-1} s^{n-1} + \dots + a_0}$$

However, frequency response analysis uses a binomial form representation. Which means the numerator and denominator should be written in binomial form, with one exception. Any complex roots are kept as second-order polynomials (quadratic function). Therefore, the transfer function is

$$F(s) = \frac{K_n \cdot s^z (s + z_1)(s + z_2) \dots (s + z_m) (s^2 + 2\zeta\omega_{n_{z1}} + \omega_{n_{z1}}^2) \dots (s^2 + 2\zeta\omega_{n_{zq}} + \omega_{n_{zq}}^2)}{K_d \cdot s^p (s + p_1)(s + p_2) \dots (s + p_2) (s^2 + 2\zeta\omega_{n_{p1}} + \omega_{n_{p1}}^2) \dots (s^2 + 2\zeta\omega_{n_{pq}} + \omega_{n_{pq}}^2)}$$

where K_n and K_d are constants after binomial factorization.

Notes:

1. If $F(s)$ has a numerator and/or denominator expression with complex roots, the quadratic is not written in binomial form, but left as a quadratic with real coefficients ($s^2 + a_1 s + a_0$).

2. $F(s)$ also has an s^i in the numerator and/or denominator ($0 \leq i$).

Next, cancel common factors. Then, the solution must be normalized. Factor the constants from the binomial expressions *and* the squared natural frequency from the second order polynomial expressions.

$$F(s) = \frac{K \cdot s^z \left(\frac{s}{z_1} + 1\right) \left(\frac{s}{z_2} + 1\right) \cdots \left(\frac{s}{z_m} + 1\right) \left(\frac{s^2}{\omega_{n_{z1}}^2} + \frac{2\zeta s}{\omega_{n_{z1}}} + 1\right) \cdots \left(\frac{s^2}{\omega_{n_{zq}}^2} + \frac{2\zeta s}{\omega_{n_{zq}}} + 1\right)}{s^p \left(\frac{s}{p_1} + 1\right) \left(\frac{s}{p_2} + 1\right) \cdots \left(\frac{s}{p_n} + 1\right) \left(\frac{s^2}{\omega_{n_{p1}}^2} + \frac{2\zeta s}{\omega_{n_{p1}}} + 1\right) \cdots \left(\frac{s^2}{\omega_{n_{pq}}^2} + \frac{2\zeta s}{\omega_{n_{pq}}} + 1\right)}$$

where the constant becomes

$$K = \frac{K_n \cdot z_1 \cdot z_2 \cdots z_m \cdot \omega_{n_{z1}}^2 \cdots \omega_{n_{zq}}^2}{K_d \cdot p_1 \cdot p_2 \cdots p_n \cdot \omega_{n_{p1}}^2 \cdots \omega_{n_{pq}}^2}$$

Finally, transform the equation into the frequency domain by $s \rightarrow j\omega$

$$F(j\omega) = \frac{K \cdot (j\omega)^z \left(\frac{j\omega}{z_1} + 1\right) \left(\frac{j\omega}{z_2} + 1\right) \cdots \left(\frac{j\omega}{z_m} + 1\right) \left(\frac{(j\omega)^2}{\omega_{n_{z1}}^2} + \frac{2\zeta j\omega}{\omega_{n_{z1}}} + 1\right) \cdots \left(\frac{(j\omega)^2}{\omega_{n_{zq}}^2} + \frac{2\zeta j\omega}{\omega_{n_{zq}}} + 1\right)}{(j\omega)^p \left(\frac{j\omega}{p_1} + 1\right) \left(\frac{j\omega}{p_2} + 1\right) \cdots \left(\frac{j\omega}{p_n} + 1\right) \left(\frac{(j\omega)^2}{\omega_{n_{p1}}^2} + \frac{2\zeta j\omega}{\omega_{n_{p1}}} + 1\right) \cdots \left(\frac{(j\omega)^2}{\omega_{n_{pq}}^2} + \frac{2\zeta j\omega}{\omega_{n_{pq}}} + 1\right)} \quad (12.1)$$

Which we use as the base equation to find the magnitude and phase to use on a Bode Plot. The process to go from $F(s)$ in polynomial form to normalized frequency form looks challenging, but is really pretty simple. Before moving to analysis, let's a look at some examples.

12.1 Example Transformations

Simple

Given:

$$F(s) = \frac{s+2}{s^2+11s+28}$$

Write the transfer function in the form appropriate for frequency response analysis

Solution:

In binomial form

$$F(s) = \frac{s+2}{(s+4)(s+7)}$$

and normalized

$$F(s) = \frac{2\left(\frac{s}{2}+1\right)}{4 \cdot 7\left(\frac{s}{4}+1\right)\left(\frac{s}{7}+1\right)} = \frac{\frac{s}{2}+1}{14\left(\frac{s}{4}+1\right)\left(\frac{s}{7}+1\right)}$$

Finally, normalized in the frequency domain

$$F(j\omega) = \frac{1 + \frac{j\omega}{2}}{14\left(1 + \frac{j\omega}{4}\right)\left(\frac{1 + j\omega}{7}\right)}$$

Moderate

Given:

$$F(s) = \frac{s+2}{(s+5)(s^3+11s^2+28s)}$$

Write the transfer function in the form appropriate for frequency response analysis

Solution:

In binomial form

$$F(s) = \frac{s+2}{s(s+5)(s+4)(s+7)}$$

and normalized

$$F(s) = \frac{2\left(\frac{s}{2}+1\right)}{5 \cdot 4 \cdot 7s\left(\frac{s}{5}+1\right)\left(\frac{s}{4}+1\right)\left(\frac{s}{7}+1\right)} = \frac{\frac{s}{2}+1}{70s\left(\frac{s}{5}+1\right)\left(\frac{s}{4}+1\right)\left(\frac{s}{7}+1\right)}$$

Finally, normalized in the frequency domain

$$F(j\omega) = \frac{1 + \frac{j\omega}{2}}{70(j\omega)\left(1 + \frac{j\omega}{4}\right)\left(1 + \frac{j\omega}{5}\right)\left(1 + \frac{j\omega}{7}\right)}$$

Difficult — Complex roots

Given:

$$F(s) = \frac{s+2}{(s+5)(s^3+11s^2+28s)(s^2+10s+34)}$$

Note: the polynomial $s^2+10s+34$ has complex roots at $-5 \pm 3j$ so it will not be factored into binomial form.

Write the transfer function in the form appropriate for frequency response analysis

Solution:

In binomial form

$$F(s) = \frac{s+2}{s(s+5)(s+4)(s+7)(s^2+10s+34)}$$

and normalized

$$F(s) = \frac{2\left(\frac{s}{2} + 1\right)}{5 \cdot 4 \cdot 7 \cdot 34 \cdot s \left(\frac{s}{5} + 1\right) \left(\frac{s}{4} + 1\right) \left(\frac{s}{7} + 1\right) \left(\frac{s^2}{34} + \frac{10s}{34} + 1\right)}$$

$$= \frac{\frac{s}{2} + 1}{2380 \cdot s \left(\frac{s}{5} + 1\right) \left(\frac{s}{4} + 1\right) \left(\frac{s}{7} + 1\right) \left(\frac{s^2}{34} + \frac{10s}{34} + 1\right)}$$

Finally, normalized in the frequency domain

$$F(j\omega) = \frac{\frac{j\omega}{2} + 1}{2380(j\omega) \left(\frac{j\omega}{5} + 1\right) \left(\frac{j\omega}{4} + 1\right) \left(\frac{j\omega}{7} + 1\right) \left(\frac{(j\omega)^2}{34} + \frac{j10\omega}{34} + 1\right)}$$

12.2 Generic Analysis

Once we have done the work to get the transfer function into the normalized frequency domain representation (12.1) we can evaluate the frequency response to a sinusoidal input by plotting a Bode diagram.

Bode diagrams represent the frequency response magnitude and phase of output with respect to the input. The plots are semi-log where the horizontal axis is $\log_{10}$ based frequency in radians per second; the vertical axis, for magnitude is in dB and the vertical axis for phase is in degrees.

The reason for transforming with the $\log_{10}$ operator is because the magnitude and phase of the overall output is now the summation of the individual expressions. Without the transformation, the magnitude and phase analysis of (12.1) would be quite difficult. After transformation, the magnitude is

$$|F(j\omega)|_{dB} = 20\log_{10}(K) + 20\log_{10}\left(\left|\frac{j\omega}{z_1} + 1\right|\right) + \cdots + 20\log_{10}\left(\left|\frac{j\omega}{z_m} + 1\right|\right) + 40\log_{10}\left(|\text{Quad}_{\omega_{nz1}}|\right) +$$

$$\cdots + 40\log_{10}\left(|\text{Quad}_{\omega_{nzq}}|\right) + 20\log_{10}\left(\left|\frac{j\omega}{p_1} + 1\right|\right) + \cdots + 20\log_{10}\left(\left|\frac{j\omega}{p_n} + 1\right|\right) +$$

$$40\log_{10}\left(|\text{Quad}_{\omega_{np1}}|\right) + \cdots + 40\log_{10}\left(|\text{Quad}_{\omega_{npq}}|\right) \quad (12.2)$$

and the phase

$$\phi_F(j\omega) = \phi_{\frac{j\omega}{z_1} + 1} + \cdots + \phi_{\frac{j\omega}{z_m} + 1} + \phi_{\text{Quad}_{nz1}} + \cdots + \phi_{\text{Quad}_{nzq}} +$$

$$\phi_{\frac{j\omega}{p_1} + 1} + \cdots + \phi_{\frac{j\omega}{p_n} + 1} + \phi_{\text{Quad}_{np1}} + \cdots + \phi_{\text{Quad}_{npq}} \quad (12.3)$$

Now, each expression is evaluated separately and the final solution to the magnitude and phase is the simple summation.

Note that the quadratic magnitude and phase are both slightly more difficult to determine than the binomial solutions, so they are abbreviated in this solution.

There are four basic factors that make up a the transfer function.

1. Static gain
2. Derivative and integral
3. Binomial
4. Quadratic

12.2.1 Static Gain

$$F(s) = K$$

The static (constant) gain K is determined while normalizing the transfer function. Then, we can find the logarithmic gain

$$K_{\text{dB}} = 20 \log_{10} K$$

with the phase

$$\phi_k = 0^\circ$$

There are several rules of logarithms that apply to all expressions, but are more easily presented with a static gain

$$1. \ 20 \log_{10} K = -20 \log_{10} \left(\frac{1}{K} \right) \text{ — Corollary: } 20 \log_{10} \left(\frac{1}{K} \right) = -20 \log_{10} K$$

$$\text{e.g. } 20 \log_{10} \left(\frac{1}{200} \right) = -20 \log_{10} 200$$

$$2. \ 20 \log_{10} (K_1 \cdot K_2) = 20 \log_{10} K_1 + 20 \log_{10} K_2$$

$$\text{e.g. } 20 \log_{10} (100) = 20 \log_{10} 10 + 20 \log_{10} 10$$

$$3. \ 20 \log_{10} (10^n) = 20n \log_{10} 10$$

$$4. \ 20 \log_{10} (10^n K) = 20n \log_{10} 10 + 20 \log_{10} K$$

$$\text{e.g. } 20 \log_{10} (500) = 20 \log_{10} (10^2 5) = 40 \log_{10} 10 + 20 \log_{10} 5$$

12.2.2 Derivative and Integral

$$F(s) = s^{\pm 1}$$

Derivative

$$F(s) = s$$

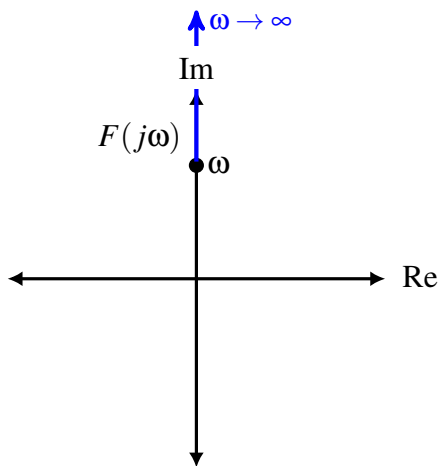
therefore,

$$F(j\omega) = j\omega$$

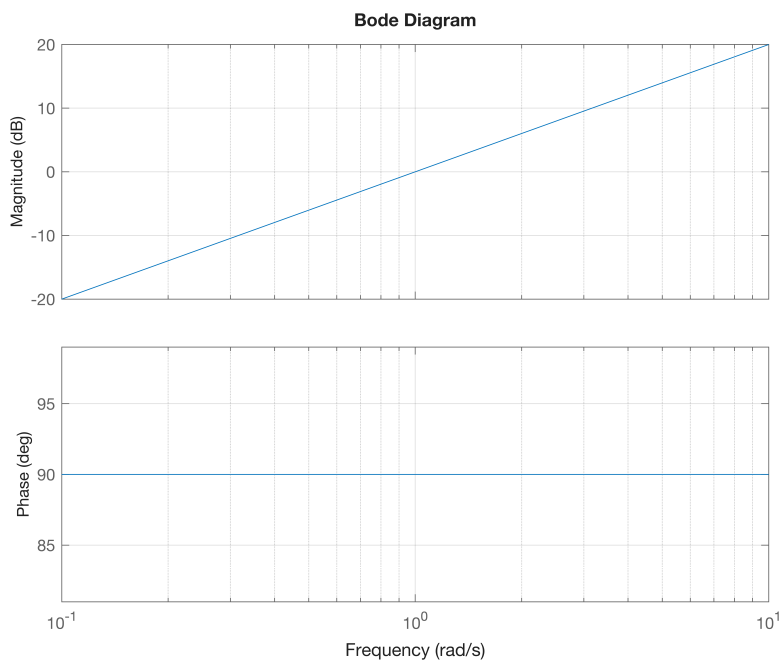
or in polar coordinates as

$$F(j\omega) = \omega \angle 90^\circ$$

which may be plotted on the complex plane



Bode Plot using MATLAB



Notice in the Bode Plot, the magnitude of 0 dB only occurs at the frequency $\omega = 1$ and that the slope of the output is always positive, with a positive phase of 90° .

Integral

$$F(s) = \frac{1}{s}$$

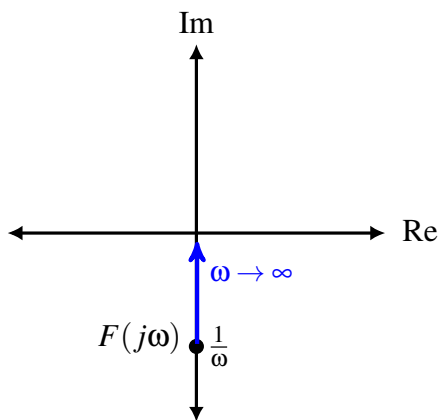
therefore,

$$F(j\omega) = \frac{1}{j\omega} = \frac{-j}{\omega}$$

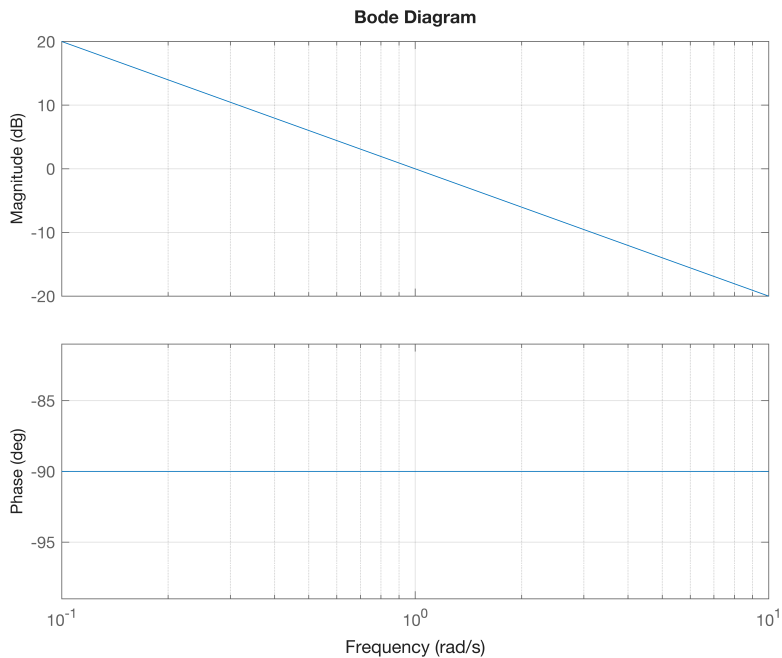
or in polar coordinates as

$$F(j\omega) = \frac{1}{\omega} \angle -90^\circ$$

which may be plotted on the complex plane



Then, the Bode Plot using MATLAB



Notice in the Bode Plot, the magnitude of 0 dB still only occurs as the frequency $\omega = 1$, but the slope is the inverse — it's a negative slope with a -90° phase shift.

12.2.3 Binomial

These expressions are quite common in both the numerator and denominator.

$$F(s) = (Ts + 1)^{\pm 1}$$

where $\frac{1}{T}$ is the corner frequency.

In the Numerator

$$F(s) = (Ts + 1)$$

therefore,

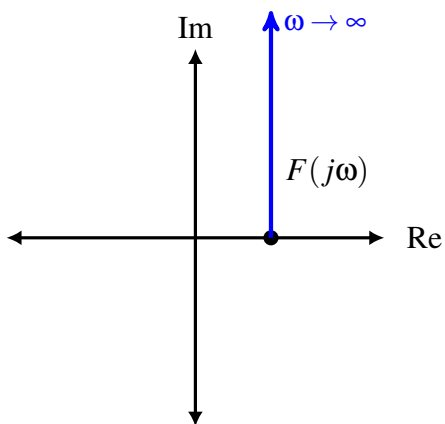
$$F(j\omega) = (1 + Tj\omega)$$

in polar coordinates

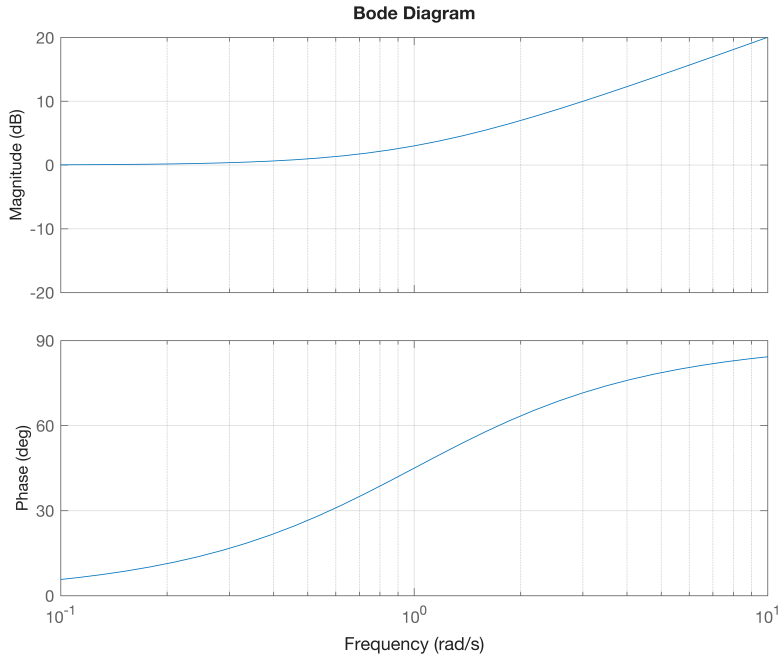
$$F(j\omega) = \sqrt{1 + (T\omega)^2}, \angle \tan^{-1}(T\omega)$$

When ω is small, the magnitude and angle are small. When $\omega = \frac{1}{T}$ the real part is equal to the imaginary part the magnitude is $\sqrt{2}$ and the phase is 45° .

As ω approaches ∞ , the magnitude approaches ∞ and the phase approaches 90° .



With a Bode Plot from MATLAB



In the Denominator

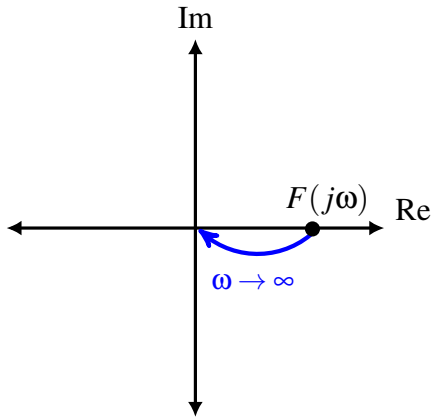
$$F(s) = (Ts + 1)^{-1} = \frac{1}{Ts + 1}$$

therefore,

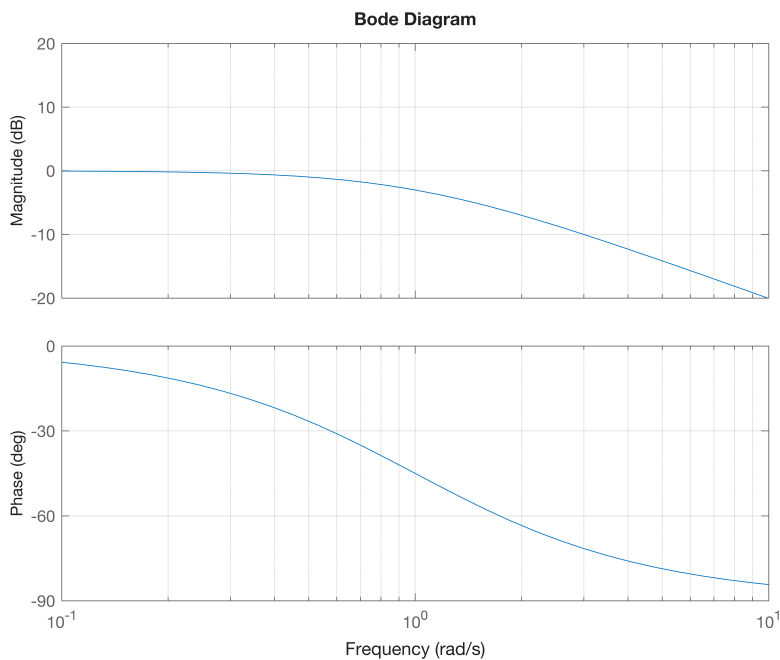
$$\begin{aligned} F(j\omega) &= \frac{1}{1 + Tj\omega} \\ &= \frac{1 - Tj\omega}{1 + (T\omega)^2} \\ &= \frac{1}{1 + (T\omega)^2} - \frac{Tj\omega}{1 + (T\omega)^2} \\ &= \text{Re}\{F(j\omega)\} + \text{Im}\{F(j\omega)\} \end{aligned}$$

When $\omega \rightarrow 0$, $F(j\omega) \rightarrow 1$ and $\phi_{F(j\omega)} \rightarrow 0$. When $\omega \rightarrow \infty$, $F(j\omega) \rightarrow 0$ and $\phi_{F(j\omega)} \rightarrow -90^\circ$. When ω is small, the magnitude and angle are small. When $\omega = \frac{1}{T}$ the real part is equal to the imaginary part; the magnitude is $\sqrt{2}$ and the phase is -45° .

As ω approaches ∞ , the magnitude approaches 0 and the phase approaches -90° .



With a Bode Plot from MATLAB



Sketching

With software simulation tools, we can easily generate the Bode Plot from a math model we've developed in the tool; however, a perfect figure by hand is difficult, to say the least. What we can do is generate an approximate sketch based on the frequency normalized gain and the corner frequency.

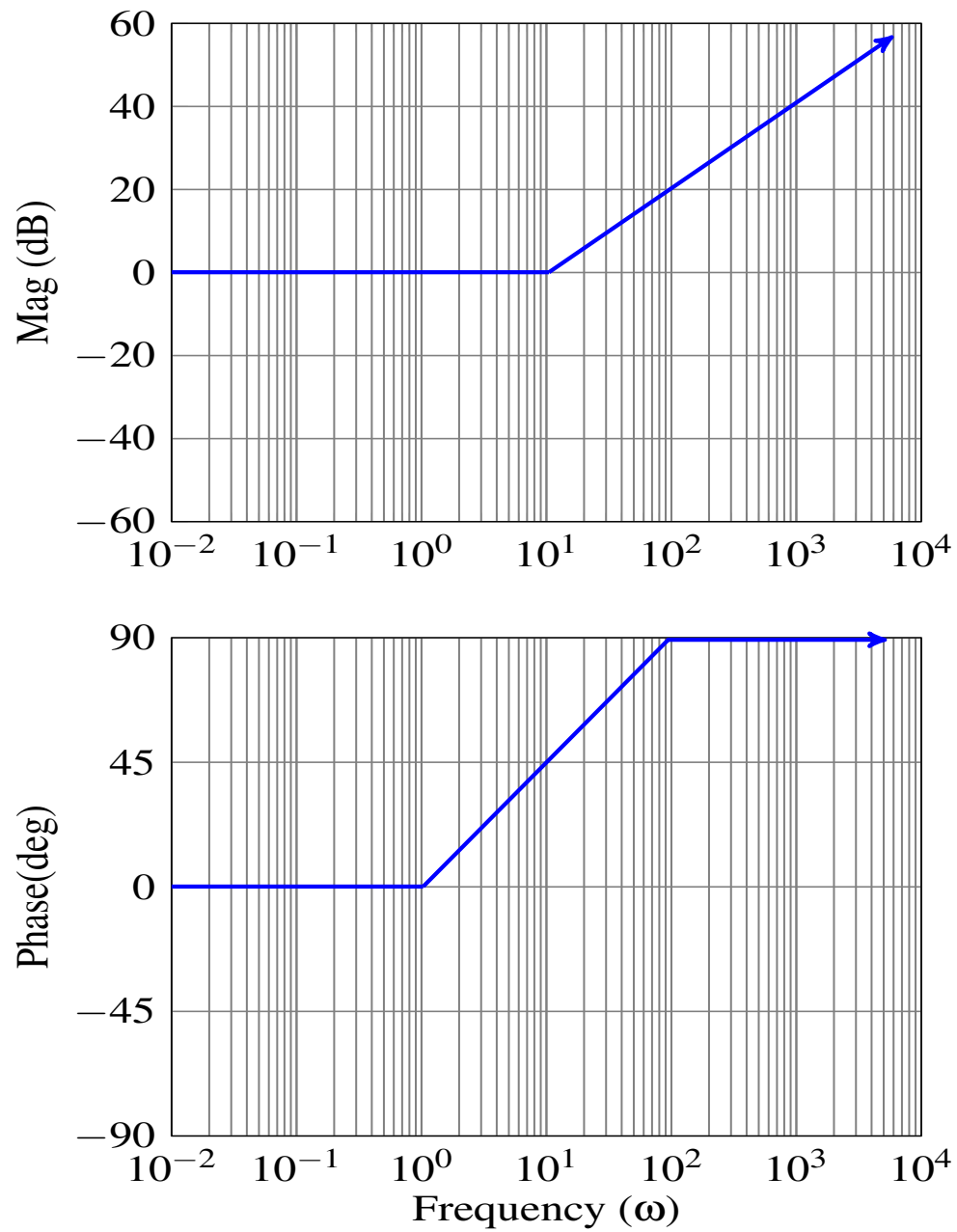
For frequencies much smaller than an equal to the corner frequency, the response is flat with a magnitude of the normalized gain. At the corner frequency and above the gain increase by 20 dB/dec when the expression is in the numerator or decreases by 20 dB/dec when the expression is in the denominator. The phase is zero degrees for low frequencies, positive 45 at the corner frequency and positive 90 degrees for high frequencies when the expression is in the numerator. When the expression is in the denominator, the phase is still 0 degrees for low frequencies, negative 45 degrees at the corner, and -90 degrees for high frequencies.

Example — Numerator

Given:

$$F(s) = s + 1$$

The simplified Bode Plot sketch is

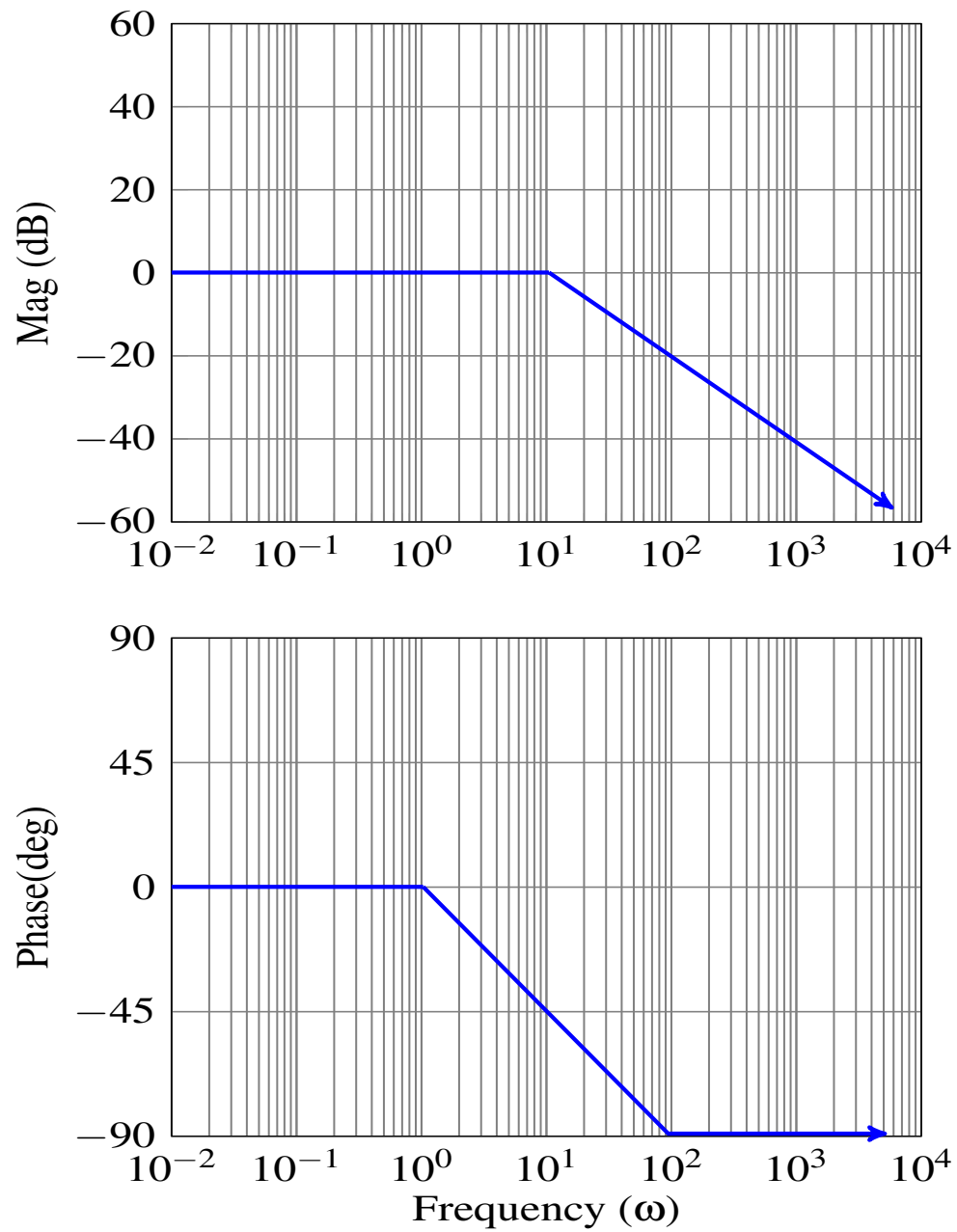


Example — Denominator

Given:

$$F(s) = \frac{1}{s+1}$$

The simplified Bode Plot sketch is



12.2.4 Quadratic

These functions are common for some fields of study where naturally oscillating systems occur; other fields they are rarely, if ever, seen. The models are built around two parameters, the natural frequency ω_n and the damping ratio. The natural frequency is always greater than zero and the damping ratio ζ is greater than zero and less than one.

$$F(s) = (s^2 + 2\zeta\omega_n s + \omega_n^2)^{\pm 1}$$

$$\omega_n > 0$$

$$0 < \zeta < 1$$

In the Numerator

$$F(s) = (s^2 + 2\zeta\omega_n s + \omega_n^2)$$

First, factor ω_n^2

$$F(s) = \omega_n^2 \left(\frac{s^2}{\omega_n^2} + \frac{2\zeta\omega_n s}{\omega_n^2} + 1 \right)$$

$$= \omega_n^2 \left(\frac{s^2}{\omega_n^2} + \frac{2\zeta s}{\omega_n} + 1 \right)$$

Then transform into the frequency domain

$$F(j\omega) = \omega_n^2 \left(\frac{(j\omega)^2}{\omega_n^2} + \frac{2\zeta j\omega}{\omega_n} + 1 \right)$$

Next, simplify and write as a complex number

$$F(j\omega) = \omega_n^2 \left(\frac{-\omega^2}{\omega_n^2} + \frac{2\zeta j\omega}{\omega_n} + 1 \right)$$

$$= \omega_n^2 \left(1 - \frac{\omega^2}{\omega_n^2} + j \frac{2\zeta\omega}{\omega_n} \right)$$

$$= \omega_n^2 \left(\text{Re}\{F(j\omega)\} + j \text{Im}\{F(j\omega)\} \right)$$

where ω_n^2 is a constant and becomes part of the K parameter. For further analysis it is excluded — it only contributes a static gain.

Just as with the binomial, we have three regions to consider

1. When $\omega \leq 0.1\omega_n$

$$F(j\omega) \approx 1 + 0j$$

$$|F(j\omega)|_{\text{dB}} = 40 \log_{10}(|F(j\omega)|) = 40 \log_{10}(|1 + 0j|) = 0 \text{ dB}$$

$$\phi_{F(j\omega)} = 0^\circ$$

2. When $\omega \geq 10\omega_n$

$$F(j\omega) = 1 - \frac{\omega^2}{\omega_n^2} + j\frac{2\zeta\omega}{\omega_n}$$

$$|F(j\omega)|_{\text{dB}} = 40\log_{10}\left(\left|1 - \frac{\omega^2}{\omega_n^2} + j\frac{2\zeta\omega}{\omega_n}\right|\right) = +40 \text{ dB/dec}$$

$$\phi_{F(j\omega)} = 180^\circ$$

3. When $\omega = \omega_n$

$$\begin{aligned} F(j\omega) &= 1 - \frac{\omega^2}{\omega_n^2} + j\frac{2\zeta\omega}{\omega_n} \\ &= 2\zeta j \end{aligned}$$

where

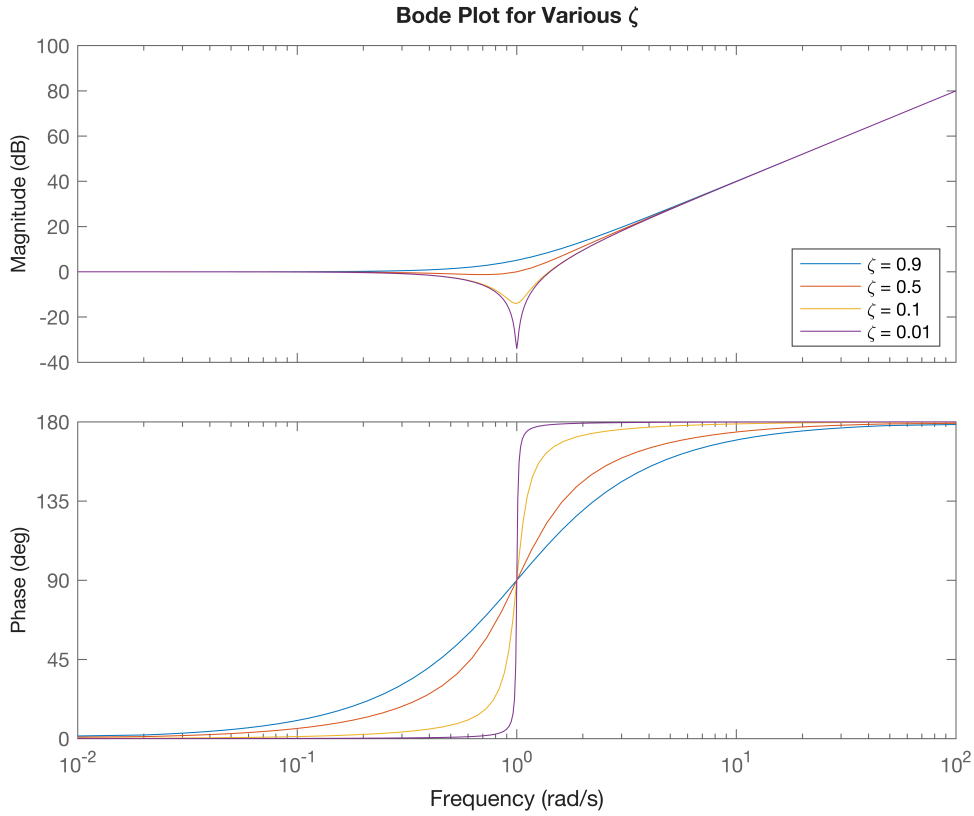
$$\begin{aligned} |2\zeta j| &= \sqrt{\text{Re}^2 + \text{Im}^2} \\ &= \sqrt{0^2 + (2\zeta)^2} \\ &= 2\zeta \end{aligned}$$

However, $0 < \zeta < 1$ which means it may be written as a fraction $\zeta = \frac{N_\zeta}{D_\zeta}$. So,

$$\begin{aligned} |F(j\omega)|_{\text{dB}} &= 20\log_{10}(2\zeta) \\ &= 20\log_{10}\left(2\frac{N_\zeta}{D_\zeta}\right) \\ &= 20\log_{10}(2) + 20\log_{10}(N_\zeta) - 20\log_{10}(D_\zeta) \end{aligned}$$

$$\phi_{F(j\omega)} = 90^\circ$$

For low frequencies, we see the magnitude and phase at zero. At high frequencies we see an increase in gain by 40 dB per decade, with a phase shift of 180° — because there are two poles. The results are unusual about $\omega \approx \omega_n$. The response is primarily a function of the damping ratio ζ and since it is always less than one we observe a drop in response at the natural resonant frequency. As $\zeta \rightarrow 0$ the output response goes to $-\infty$ at $\omega = \omega_n$



In the Denominator

$$F(s) = (s^2 + 2\zeta\omega_n s + \omega_n^2)^{-1}$$

I believe the best way to describe this expression is by using a previously mentioned rule.

Recall: $\log_{10}(\frac{1}{K}) = -\log_{10} K$

Once again, we have three regions to consider

1. When $\omega \leq 0.1\omega_n$

$$|F(j\omega)|_{\text{dB}} = 0 \text{ dB}$$

$$\phi_{F(j\omega)} = 0^\circ$$

2. When $\omega \geq 10\omega_n$

$$|F(j\omega)|_{\text{dB}} = -40 \text{ dB/dec}$$

$$\phi_{F(j\omega)} = -180^\circ$$

3. When $\omega = \omega_n$

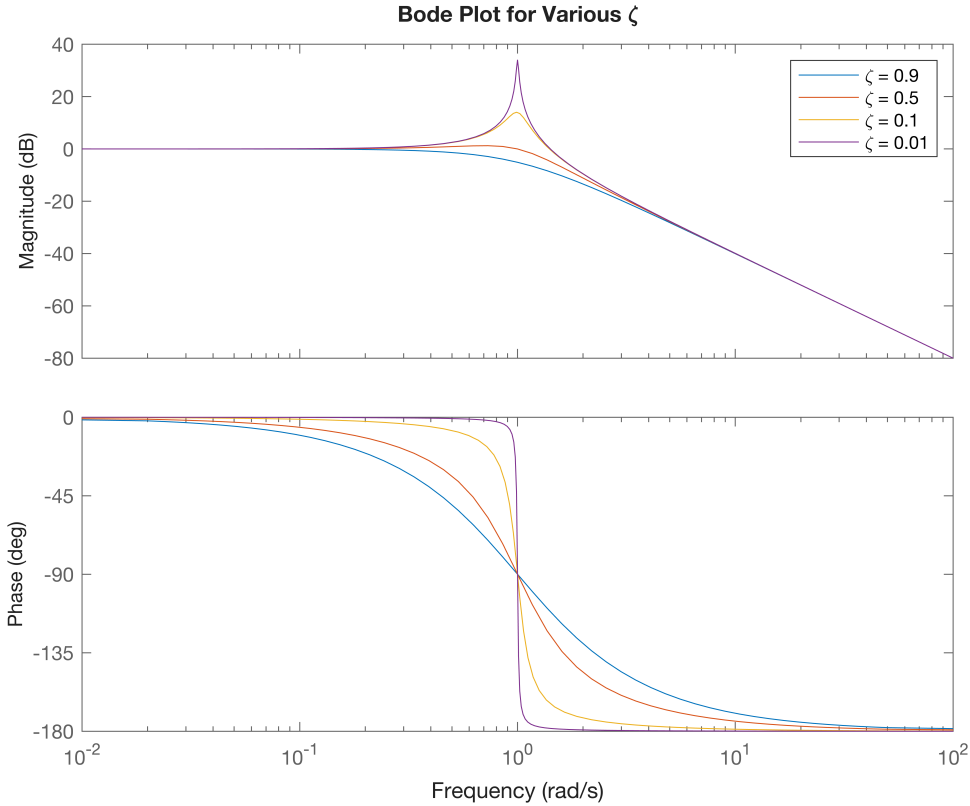
$$|F(j\omega)|_{\text{dB}} = -20 \log_{10} \left(2 \frac{N_\zeta}{D_\zeta} \right)$$

$$= 20 \log_{10} \left(2 \frac{D_\zeta}{N_\zeta} \right)$$

$$= 20 \log_{10}(2) + 20 \log_{10}(D_\zeta) - 20 \log_{10}(N_\zeta)$$

$$\phi_F(j\omega) = -90^\circ$$

For low frequencies, we see the magnitude and phase at zero. At high frequencies we see a **decrease** in gain by 40 dB per decade, with a phase shift of -180° . The results are unusual about $\omega \approx \omega_n$. The response is primarily a function of the damping ratio ζ and since it is always less than one we observe an increase in response at the natural resonant frequency. As $\zeta \rightarrow 0$ the output response goes to $+\infty$ at $\omega = \omega_n$.



Example — Simple

Given the function from Section 12.1:

$$F(s) = \frac{s + 2}{s^2 + 11s + 28}$$

Find: the Bode diagram

The normalized and transformed solution in the frequency domain

$$F(j\omega) = \frac{1 + \frac{j\omega}{2}}{14(1 + \frac{j\omega}{4})(\frac{1+j\omega}{7})}$$

The numerator has one corner frequency at 2, $\omega_c = 2$. The denominator has two corner frequencies at 4 and 7. Analyze each corner frequency individually then produce a composite solution.

The numerator corner frequency $\omega_{c1} = 2$

	$\omega \leq 0.1 \cdot \omega_{c1}$	$\omega = \omega_{c1}$	$\omega \geq 10 \cdot \omega_{c1}$
$ F(j\omega) _{\text{dB}}$	0 dB	3 dB	+20 dB per dec
$\phi_{F(j\omega)}$	0°	45°	90°

The denominator has two corner frequencies $\omega_{c2} = 4$ and $\omega_{c2} = 7$.

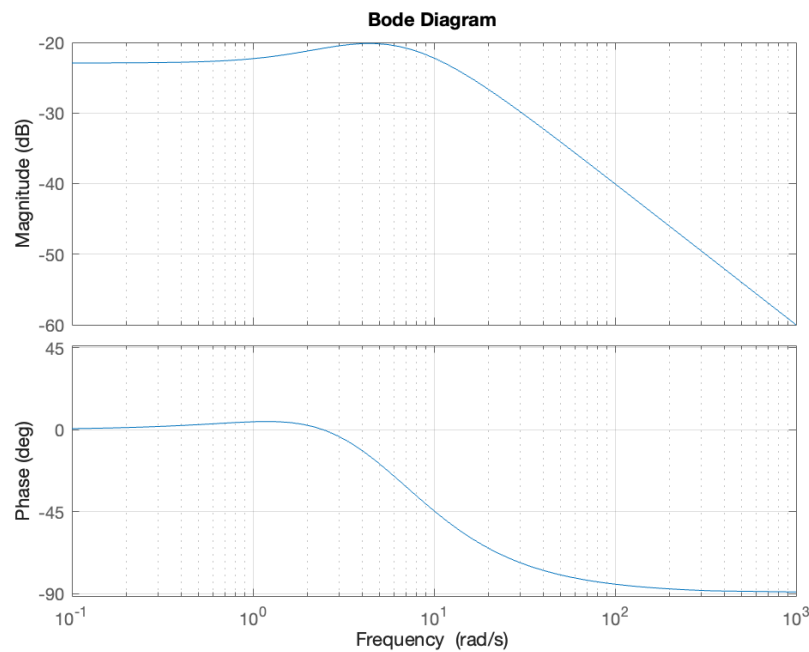
Consider $\omega_{c2} = 4$

	$\omega \leq 0.1 \cdot \omega_{c2}$	$\omega = \omega_{c2}$	$\omega \geq 10 \cdot \omega_{c2}$
$ F(j\omega) _{\text{dB}}$	0 dB	-3 dB	-20 dB per dec
$\phi_{F(j\omega)}$	0°	-45°	-90°

Consider $\omega_{c3} = 7$

	$\omega \leq 0.1 \cdot \omega_{c3}$	$\omega = \omega_{c3}$	$\omega > 10 \cdot \omega_{c3}$
$ F(j\omega) _{\text{dB}}$	0 dB	-3 dB	-20 dB per dec
$\phi_{F(j\omega)}$	0°	-45°	-90°

The composite response, as noted above, is the summation of all of the expressions. Do not forget to include the gain in the summation of the composite magnitude.



The compressed nature of the corner frequencies of the function produces a diagram where the action from corner frequencies is obscured — very common for a control system. Let's look at an example with widely spaced corner frequencies to more clearly identify the responses.

Example — Clear Corner Frequencies

Given the function:

$$F(s) = \frac{s + 10}{(s + 1000)(s + 100000)}$$

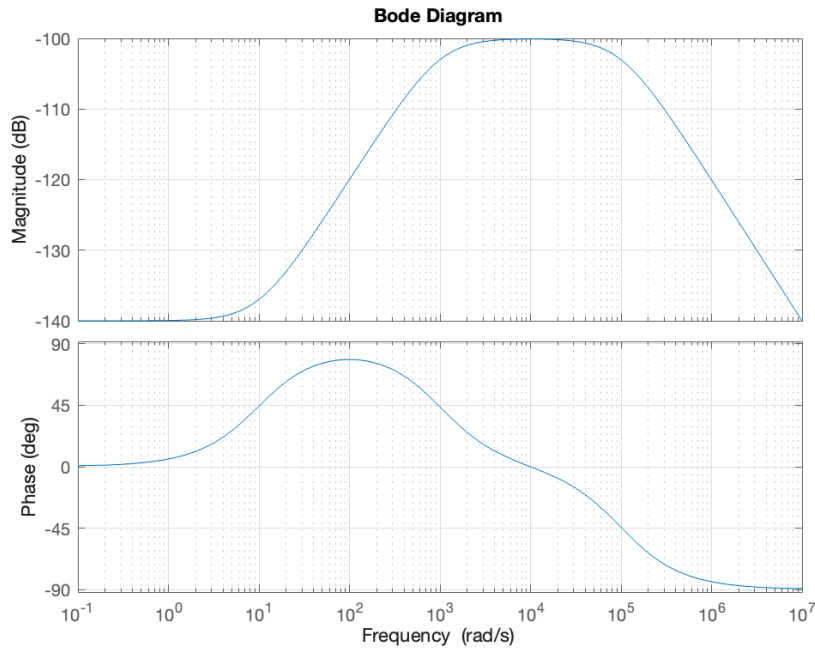
Find: the Bode diagram

The transfer function is already in binomial form, so we only need to normalize and transform

$$F(s) = \frac{10(\frac{s}{10} + 1)}{1000 \cdot 100000(\frac{s}{1000} + 1)(\frac{s}{100000} + 1)}$$

$$F(j\omega) = \frac{1 + \frac{j\omega}{10}}{10000000(1 + \frac{j\omega}{1000})(1 + \frac{j\omega}{100000})}$$

The numerator has a corner frequency $\omega_{c1} = 10$, the denominator has two corner frequencies $\omega_{c2} = 1000$ and $\omega_{c3} = 100000$.



12.3 Relative Stability

When performing Frequency Response Analysis on the loop-gain, $L(s)$ of a system we are determining the relative frequency stability of the system. Relative because we compare two figures of merit to a similar system.

The two figures of merit are

1. Gain margin: $GM = 0 - L_{dB}$

- L_{dB} is the gain when $\phi = -180^\circ$

2. Phase margin: $PM = 180^\circ + L_\phi$

- L_ϕ is the phase when $L_{dB} = 0$

The larger the figures of merit, the more stable a system is — this makes comparing the stability of two systems as easy as comparing their figures of merit. In MATLAB, use the function `margin` to generate a Bode Plot with the GM and PM identified in the figure's title.

12.4 Performance

When performing Frequency Response Analysis on the open-loop gain, $T(s)$ of a system we are determining performance of the system.

There are three performance metrics to consider

1. Output gain.

- From the Bode Plot, we observe the output response for a given sinusoidal input. Depending on the desired parameters, additional tuning may be necessary to meet the requirements of the design. Most of the time, the goal is to track the input signal with the output response so the magnitude of the output should be 1 or very near it.

2. Bandwidth

- From the Bode Plot, we can determine the bandwidth ω_{bw} . Normally, the bandwidth is the frequency where the nominal output is reduced by -3 dB. This is the case for most physical systems, which naturally have low-pass responses.

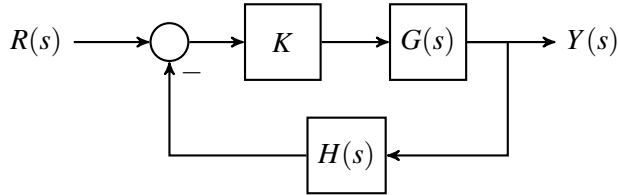
3. Phase shift

- The phase shift is a time delay in the signal referenced back to the input. It tends to vary based on the frequency of the input.

12.5 Design Example

We can use the linear approximations for design or, for more accurate results, software simulation tools can produce precise Bode Plots.

Given:



Where

$$G(s) = \frac{1}{s^2 + 505s + 2500}$$

$$= \frac{1}{(s+5)(s+500)}$$

$$H(s) = 1$$

Design a system with a gain that allows for the output to closely track a setpoint input and to additionally track a sinusoidal input with a frequency greater than $\omega = 200$.

The loop-gain

$$L(s) = KGH(s)$$

$$= \frac{K}{(s+5)(s+500)}$$

and an open-loop transfer function

$$T(s) = \frac{K}{(s+5)(s+500) + K}$$

$$= \frac{K}{s^2 + 505s + 2500 + K}$$

Stability Analysis

We begin with

$$L(s) = \frac{K}{(s+5)(s+500)}$$

and normalize

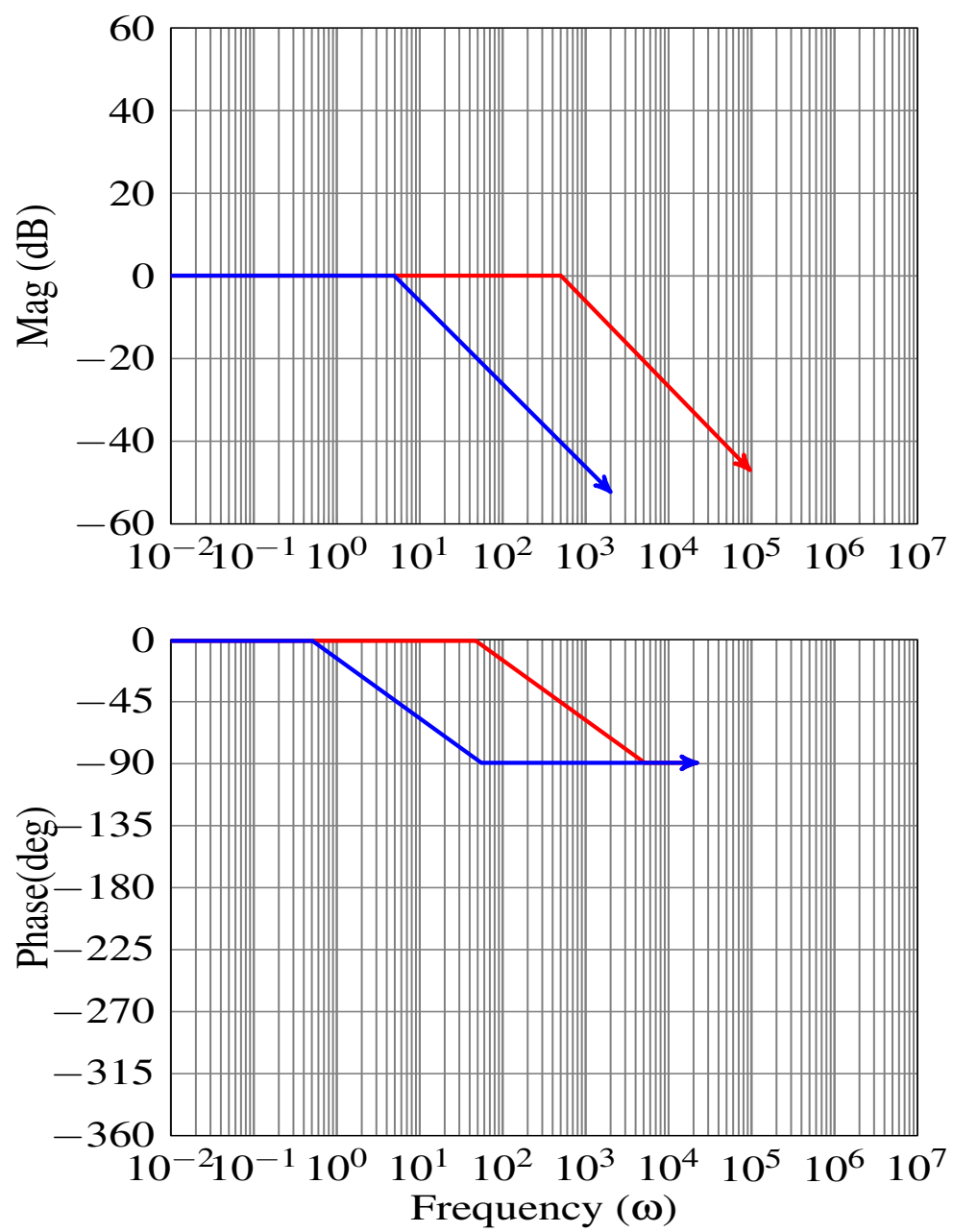
$$L(s) = \frac{K}{(s+5)(s+500)}$$

$$= \frac{K}{5 \cdot 500} \frac{1}{(\frac{s}{5} + 1)(\frac{s}{500} + 1)}$$

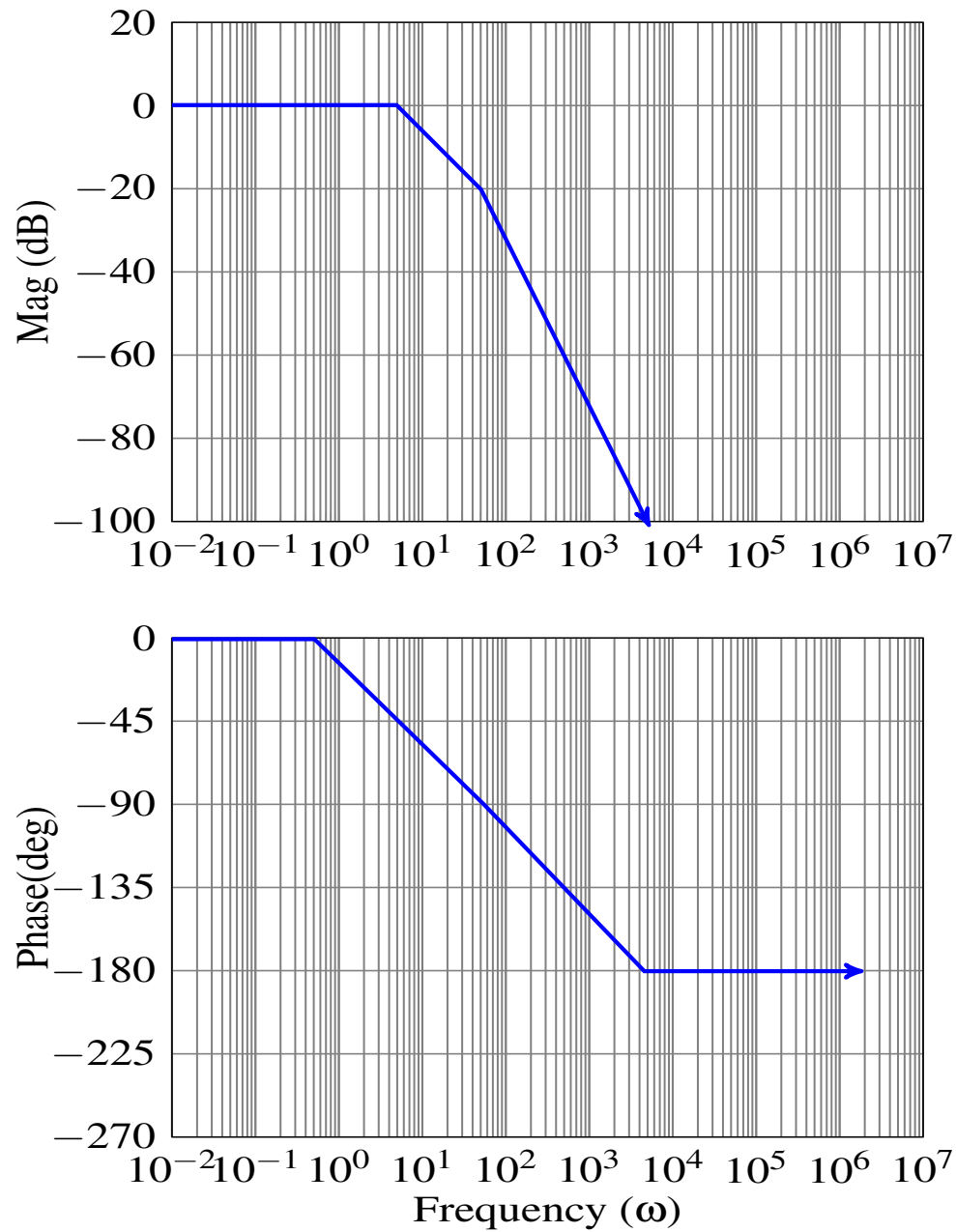
The denominator has two binomial expressions with corner frequencies at 5 and 500. For the denominator

Expr	ω					Notes
	0.5	5	50	500	5000	
$ \frac{j\omega}{5} + 1 $	0 dB	-3 dB	$-20 \frac{\text{dB}}{\text{dec}}$			
$\phi_{\frac{j\omega}{5} + 1}$	0°	-45°	-90°			
$ \frac{j\omega}{500} + 1 $			0 dB	-3 dB	$-20 \frac{\text{dB}}{\text{dec}}$	
$\phi_{\frac{j\omega}{500} + 1}$			0°	-45°	-90°	

The simplified Bode Plot sketch for the denominator — without magnitude scaling applied



Combining the two corner frequency responses to determine $L(j\omega)$ — without considering the magnitude.



Next, we consider the magnitude.

$$\hat{K}_{\text{dB}} = -20\log_{10}(2500) + 20\log_{10} K$$

Set $K = 1$ and determine $\hat{K}_{\text{dB}}$ without scaling

$$\begin{aligned}\hat{K}_{\text{dB}} &= -20\log_{10}(2500) \\ &= -68 \text{ dB}\end{aligned}$$

Which means the **actual** magnitude of the loop-gain response at low frequencies is at -68 dB, not 0 dB. This is an incredibly small response, let's increase the relative stability figures of merit.

Recall:

$$GM = 0 - L_{dB}$$

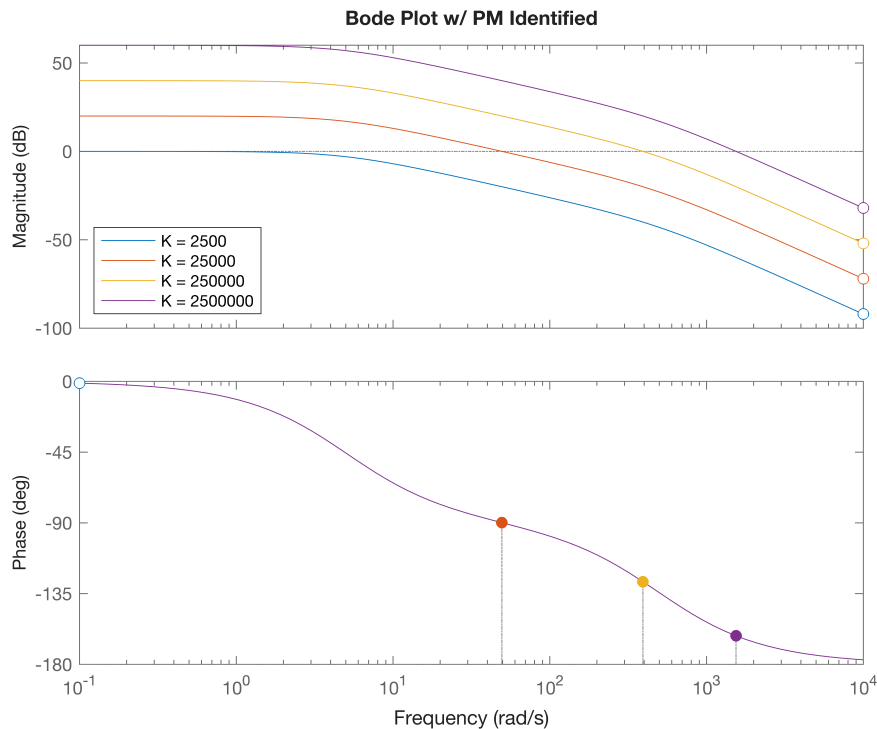
where L_{dB} is the gain when $\phi = -180^\circ$. Since the total phase shift is -180° as $\omega \rightarrow \infty$ the GM will be infinity regardless of K .

Consider the Phase Margin

$$PM = 180^\circ + L_\phi$$

where L_ϕ is the phase when $L_{dB} = 0$. If we increase K enough we will encounter a $PM < \infty$ as the response rolls off at higher frequencies. The larger values of K produces higher gain, which in turn causes PM to shrink.

K	PM
1	∞
2500	∞
25000	$\approx 90^\circ$
250000	$\approx 52^\circ$
2500000	$\approx 18^\circ$



From the Bode Plot, we can observe the PM decrease in magnitude as the gain increases. Additionally, we observe the PM frequency increase for larger gain values.

Performance

The open-loop transfer function

$$\begin{aligned} T(s) &= \frac{K}{(s+5)(s+500) + K} \\ &= \frac{K}{s^2 + 505s + 2500 + K} \end{aligned}$$

When $K = 1$

$$T(s) = \frac{1}{s^2 + 505s + 2501}$$

We can observe from the open-loop equation that steady state response to a setpoint input is going to be minimal $\approx \frac{1}{2501}$. That's really small. For a reasonable response to a setpoint input

$$\begin{aligned} T(0) &\approx 1 \\ &= \frac{K}{2500 + K} \end{aligned}$$

K must be large. Practically, we want $T(0)$ be greater than .95

$$\begin{aligned} T(0) &= \frac{K}{2500 + K} \\ &= C \end{aligned}$$

where $C = 0.97$ is the minimal final response

$$\begin{aligned} K &= \frac{2500C}{1 - C} \\ &= 80,833 \end{aligned}$$

For $C = 0.99$ we have $K = 247,500$ which is close to a gain value we have previously evaluated for stability.

Generally, we look for a minimal gain to produce the desired response; somewhere between $80,833 < K < 250,000$ will likely meet our needs.

Set $K = 125,000$ and evaluate

$$\begin{aligned} T(s) &= \frac{K}{s^2 + 505s + 2500 + K} \\ &= \frac{125000}{s^2 + 505s + 127500} \end{aligned}$$

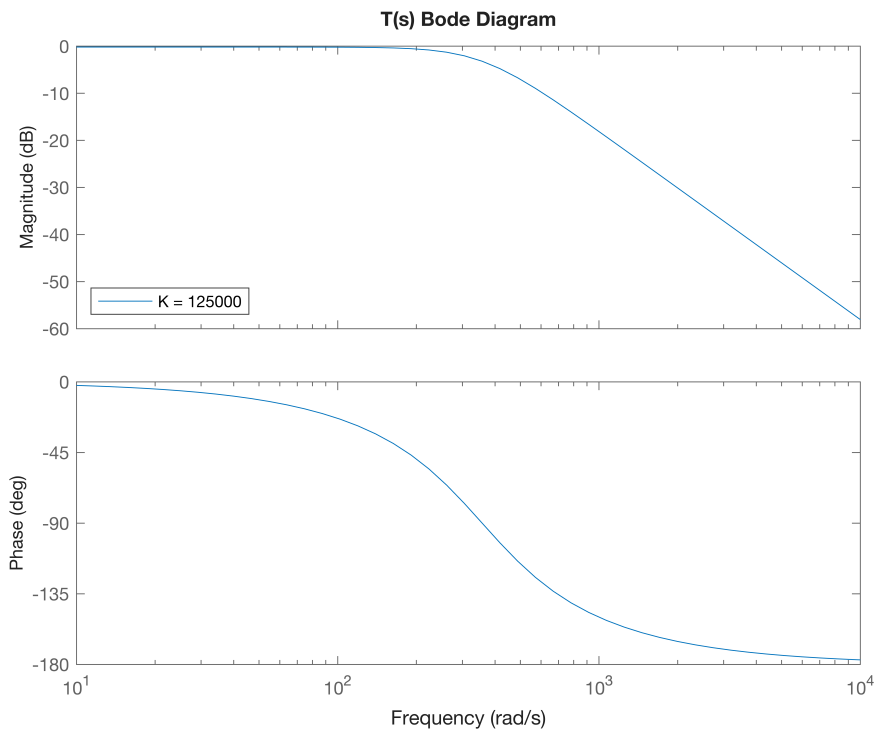
Which produces a steady state position error of 0.98 and open-loop poles of $-252.5 \pm 252.48j$.
Note: the real component is about the same as the imaginary, so the overshoot will be less than 4% and, given the extremely small time constant, the settling time will be very short.

$$\tau = \frac{1}{252.5}$$

Therefore, the settling time

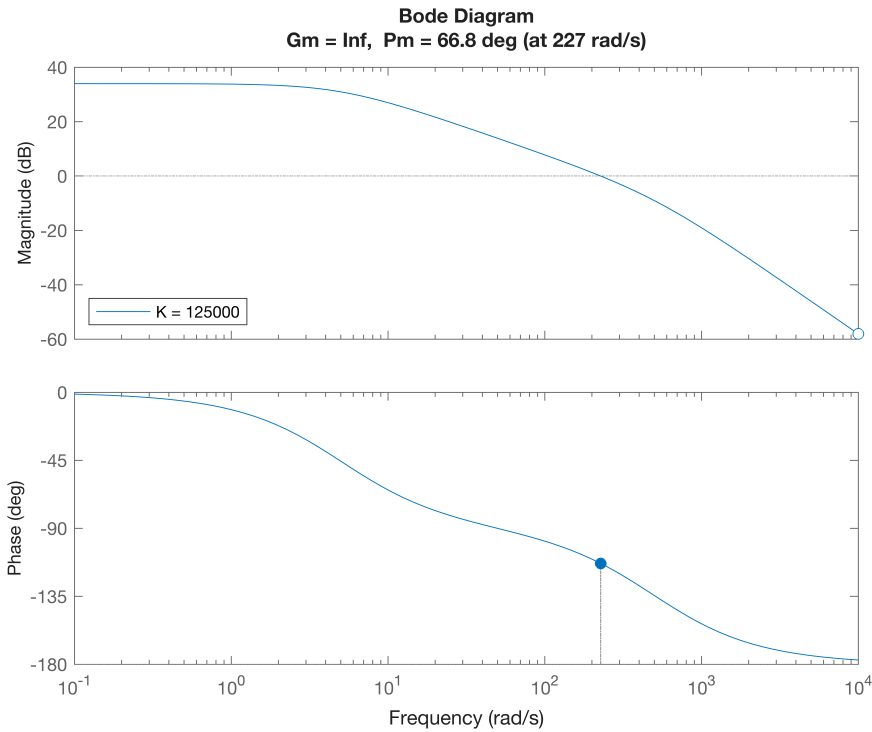
$$\begin{aligned}T_s &= 5\tau \\ &= 0.02\text{sec}\end{aligned}$$

Let's consider the frequency response of the open-loop gain $T(s)$ when $K = 125,000$



For low frequencies the magnitude of the output is about the same as the input until the 3 dB point at $\omega_{wb} \approx 357$, which makes 357 radians per second the bandwidth frequency. Anything below the bandwidth frequency and the output magnitude matches the input. At and above the bandwidth frequency and the output is severely reduced — recall 3 dB is the “half power” point.

Let's go back to the stability analysis of the loop-gain to check the final PM.



We see $PM \approx 66.8^\circ$ at $\omega_{PM} = 227$. Convert the PM frequency to Hertz

$$\begin{aligned}
 f_{PM} &= \frac{\omega_{PM}}{2\pi} \\
 &= \frac{227}{2\pi} \\
 &\approx 36\text{Hz}
 \end{aligned}$$

Produces a period of

$$P_{PM} \approx 0.028$$

This means each degree is 7.7×10^{-5} seconds; a PM of 66.8° allows for a time delay of ≈ 0.005 seconds.

The final design with $K = 125,000$ produces a steady state tracking of a setpoint equal to 0.98 and a frequency input tracking with minimal phase change for frequencies below $\omega \approx 100$ and an overall bandwidth of $\omega_{wb} \approx 357$. Any sinusoidal input with a frequency below ω_{wb} will produce an output of approximately the same magnitude and when $\omega < 100$ the phase shift will be negligible as well.